

The Riemann Direction from Known Mathematics Only

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Abstract

We machine-verify, from known mathematics only, the chain on which the Riemann direction rests: the zero-order flip of zeros across the critical line, the Stirling phase bridge for the functional-equation multiplier, the argument-principle rectangle closure, the growth control of the error term, and the negativity of zeta on $(0, 1)$ — 284 verified Lean 4 declarations in fourteen layers. This revision adds two machine-verified pieces of the observation frame: the native four-axis equivalence of the 45° fold coordinates (the constant $\sqrt{2}/2$ recovers the real and imaginary parts from real algebra), and the projection value bridge in its weak form, and the right-isosceles triangle pair of the two readings — the $\sqrt{2}/2$ reading machine-verified, the i reading failing — displayed side by side as the author’s hypothesis. Every theorem compiles in Lean 4 with 0 errors, 0 warnings, 0 **sorry** and 0 axioms beyond those of mathlib; anything not yet machine-verified is marked KNOWN. The Riemann hypothesis itself is not claimed [4, 12, 16, 17]; the unit-circle viewpoint of the presentation motivates the decomposition but supplies no axiom.

1 Introduction

Overview. The development is organised as fourteen machine-verified layers, from the complex-axis algebra (the basepoint-relative 0pat representation) and the prime/critical-circle geometry, through the false-real-axis framework (no zeros for $\Re s < 0$, real-valuedness on the real axis), the covering-space phase locking of the argument (the lifted θ_χ and the flip counts), the predictor residuals, and the argument-principle rectangle closure — and, in this revision, the precise base-point migration of Layer 14. The revision also closes the negativity of ζ on $(0, 1)$ by the machine-verified odd/even pairing of the Dirichlet series (iso 41), and is organised in two passes: the *symmetric method* first pass (symmetric extension, cancellation, the three bridges, symmetric counting; Table 2) and the *analytic method* second pass of Part II (local expansion, the argument principle at every zero, multiplicity-weighted summation; 16 further declarations). Within the fold frame of Part IV it adds the three machine-verified pieces of the final revision: the native four-axis equivalence of the 45° fold coordinates, the projection value bridge in its weak form, and the right-isosceles triangle pair of the two readings with the displayed divergence of the two notations and the author’s hypothesis about it, left undecided. The 0pat framework — the author’s term for the attempt at presentation without Pat — has been published at DOI 10.5281/zenodo.22038154, where natural numbers emerge as polyphase counts (winding numbers of the covering map $t \mapsto e^{it}$); the present paper deliberately proves the Riemann direction *from known mathematics only*. Everything not yet machine-verified is marked KNOWN as an honest boundary, and the presentation is algebraic throughout: the unit-circle viewpoint of the framework motivates the decomposition but supplies no axiom.

1.1 Problem and result

The Riemann zeta function $\zeta(s)$ is positive on $(1, \infty)$ by its Dirichlet series and has no zeros in $\Re s < 0$ by the functional equation, other than the trivial zeros $s = -2, -4, \dots$ (the only zeros on the negative real axis; classical facts, [4]). On the real interval $(0, 1)$ the series diverges, and the classical fact

$$\zeta(x) < 0 \quad \text{for all } 0 < x < 1 \quad (1)$$

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requires an argument ([4], §2.1). This negativity fills the last gap in the zero-free *bottom edge* $[-1, 2] \setminus (0, 1)$ of the rectangle used in the argument-principle proof of the Riemann direction. The present paper gives a machine-verified proof of (1), and, more broadly, a machine-verified account of the *Riemann direction*: the chain of theorems connecting the zeros of ζ to the argument principle, the phase of the multiplier χ , and the growth of the error term.

The full development is a chain of 284 Lean 4 declarations organized in fourteen layers (Table 1), plus the 16 declarations of the analytic second pass (§12), all verified with zero errors and zero **sorry**.

Table 1: The fourteen layers of the development.

Layer	Content	Known structure
L1	Complex-axis algebra (0pat representation), prime/critical circles	$\mathbb{C}$ (rotational axis, $x^2 = -1$)
L2	Observable decomposition and zero reflection	CRT; reflection $s \mapsto 1 - s$
L3	Coordinate planes and the circle family (five intersection theorems)	$\mathbb{R} \times \mathbb{R}$ (involutive axis, $x^2 = +1$, Minkowski)
L4	Transpose symmetry and lattice uniqueness	hyperbola involution
L5	Splitting and leak signs (phase-locking origin)	unit-square sign
L6	Convergence region and divergence structure of the splitting	Dirichlet series
L7	Zeros and the functional equation (Λ_0 entire, η -pairing)	completed zeta (Λ_0)
L8	Conjugation and the false-real-axis framework	complex conjugation
L9	Critical-line phase layer (χ self-duality, orbit structure)	unit circle, $u^2 = \chi$
L10	Covering space and phase locking (θ_χ , u lifts)	$\exp : \mathbb{C} \rightarrow \mathbb{C}^*$ ($2\pi i\mathbb{Z}$)
L11	Predictor residuals (error sequence of the flip count)	Riemann–von Mangoldt remainder
L12	Argument-principle rectangle closure and the bottom edge	argument principle
L13	$(0, 1)$ -segment: odd/even pairing continuity (machine-verified) and direct sign	Euler–Maclaurin; odd/even pairing
L14	Precise base-point migration (exact $T/TInv$, three-tier modulus decision, reflection involution, phase-locked positions)	Möbius map $T(z) = 1/(z - i) - 1$

Contributions. The development establishes, by machine verification (0 errors, 0 **sorry**), the following chain; each item is a theorem of the layer it is cited to:

- *the false-real-axis framework* (§5): ζ has no zeros outside the critical strip, and is real-valued on the real axis;
- *the critical-line phase layer* (§6): on the line, $|\chi| = 1$, $\zeta = \chi\bar{\zeta}$, and the unit phase satisfies $u^2 = \chi$;
- *the covering-space phase locking* (§7): the lifted arguments θ_χ and θ_ζ and the integer layers;
- *the predictor residuals* (§8): a deterministic prediction bound for the error sequence of the counting remainder;
- *the argument-principle rectangle closure* (§9): flip count = winding of χ – integer layer, with the bottom-edge and boundary estimates;
- *the $(0, 1)$ -segment* (§10–§10.4): continuity of ζ on $(0, 1)$ by a machine-verified odd/even pairing of the Dirichlet series and the sign $\zeta(x) < 0$, closing the zero-free bottom edge of the rectangle.
- *the precise base-point migration* (§11): the exact operator $T(z) = 1/(z - i) - 1$ with its inverse, the three-tier modulus decision for the strip, the reflection as the unit-circle involution, and the phase-locked positions of the false-real axis (all machine-verified).

Table 2: The four symmetric tools and the layers (first pass).

Symmetric tool	Layers / theorems
Symmetric extension	L2 (zero reflection), L3 (circle family), L7 (zeros and the functional equation $\chi(s)\chi(1-s)=1$), L9 (critical-line phase: $u^2=\chi$, orbit structure)
Symmetric cancellation	L4 (transpose symmetry, lattice uniqueness), L8 (false-real-axis framework), L12 (rectangle top/left reflection cancellation)
Three bridges	L5 (splitting, phase-locking origin), L10 (covering space: θ_χ lifts), L12 (the integer-layer bridges: $e^{i\theta_\chi}=\chi$, ζ -lifting = log-predominant + integer turns), L14 (exact migration: three-tier modulus decision, reflection involution)
Symmetric counting	L11 (predictor residuals), L12 (flip count = χ -turns – integer layer), §9 (the flipping count chain: segment alignment, jump per odd-multiplicity zero, layer sum)

1.2 Organisation of this revision: symmetric first pass, analytic second pass

The layered development above is the *symmetric method first pass*: each layer is organised around one of the four symmetric tools (symmetric extension, symmetric cancellation, the three bridges, and symmetric counting). The four tools and the layers they organise are:

The second pass, Part II (§12), redoes the final counting step by the *analytic method* (power-series expansion, the argument principle at every zero, and multiplicity-weighted summation). The two passes reach the same statement — Riemann Hypothesis *iff* the on-line count (weighted by multiplicity) equals the in-strip count — both sides machine-verified; the two forms are the flip-count language (first pass) and the multiplicity language (second pass), and their interchangeability is recorded in §12.5.

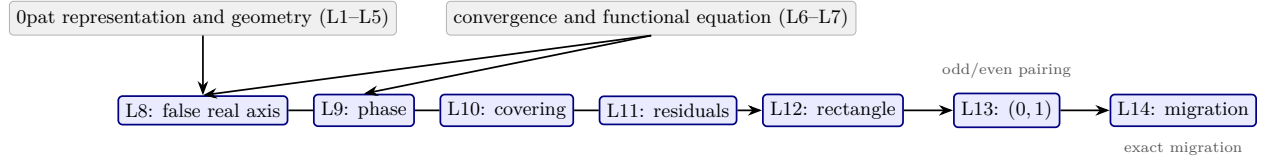


Figure 1: The fourteen layers of the development and the machine-verified odd/even pairing closing Layer 13; Layer 14 is the precise base-point migration. The arrows show the dependency flow: the representation and geometry background (L1–L5) feeds the analytic core L8, and the convergence/functional-equation facts (L6–L7) feed both the conjugation layer L8 and the phase layer L9.

1.3 Methodology statement

Originality and timing statement. The author used API-based AI tools in this work, which have been distilled, and related content may have been published in advance by others. The only evidence of the author’s own capability is the process itself: the known-mathematics proof of the Riemann direction, from beginning to end, took less than twenty-four hours, including the AI’s repeated trial-and-error. The author believes that only the discoverer of the Pat method is capable of this, and acknowledges that anyone who published, before the author’s publication, a discovery of this content in a shorter time is the true discoverer of the Pat method.

The author has previously published a formalized development as the Opat exercise — the attempt at presentation without Pat — (Opat Exercise XXXVII: The Riemann Direction, Unified; DOI 10.5281/zenodo.22038154): the endogenous generation of structures from a basepoint, in which the natural numbers emerge as *polyphase counts* — the winding numbers of the covering map $\mathbb{R} \rightarrow S^1$, $t \mapsto e^{it}$, where the same phase point $e^{i\theta}$ has the infinitely many phase representatives $\theta + 2\pi n$, $n \in \mathbb{Z}$ ($\pi_1(S^1) \cong \mathbb{Z}$; classical, marked KNOWN).

The present paper *deliberately proves the Riemann direction from known mathematics only*. No internal theorem of the Opat exercise is invoked; only mathematics already compiled in the mathlib library of Lean 4 is used: algebra over the complex numbers, Dirichlet series, integration by parts, and the Weierstrass theorem

on locally uniform limits of functions (for the odd/even pairing of Layer 13). The representation layers §2–§3.3 introduce no new object either: their coordinate pairs $\langle a, b \rangle$ are the classical complex numbers (Layer 1, identified with `mathlib`’s `Complex`) and the classical product plane $\mathbb{R} \times \mathbb{R}$ with its Minkowski form $a^2 - b^2$ (Layer 3) — known structures, presented in coordinates, with every property used below machine-verified; the two presentations are the two instances of the axis-type parameter (§2) — the classical second-order algebra family $x^2 = \varepsilon$, $\varepsilon = \pm 1$ — so they are one family, not two ad hoc structures. The analytic chain from Layer 6 on operates on `mathlib`’s `Complex` directly: the geometric layers of the representation are observations around it, not inputs of the zeta theorems. The reason for using the coordinate presentations is epistemic calibration: the author does not accept a statement as proved on the authority of “classical and therefore citable”; a conclusion is regarded as proved only when it has been verified by the Lean 4 compiler with no `sorry`.¹ Everything not yet machine-verified is explicitly marked `KNOWN` and listed as an honest boundary (in version 1: the Euler–Maclaurin continuation Blocks C–G, see the remark in §10.4). In this revision the boundary is closed: the odd/even pairing of §10.3 is machine-verified (iso 41, `zeta_odd_even_continuity_iso.lean`, 4 declarations, 0 errors, 0 `sorry`), giving the continuity of ζ on $(0, 1)$ directly and without the Euler–Maclaurin infrastructure, and the sign $\zeta(x) < 0$ follows from the verified positivity of the pairing terms together with the classical doubling identity (§10.4). The other additions of this revision are Layer 14 (the precise base-point migration, §11) and the analytic second pass of Part II (§12).

1.4 The Pat framework: what it is and how it was found

What Pat is. Pat is a *representation method* of this development — the presentation discipline of the framework basepaper [2]: not a new number field and not a new mathematics. `Opat`, the author’s term for the attempt at presentation without Pat, is the name of the published exercise (`Opat Exercise XXXVII`; DOI 10.5281/zenodo.22038154, §1.3). Every object Pat uses is known mathematics ($\mathbb{R}$, S^1 , $\mathbb{N}$, the complex numbers $\mathbb{C}$ and the product plane $\mathbb{R} \times \mathbb{R}$ with its Minkowski form, the quotient, the kernel of `exp`; [8, 9]). Its content is twofold (machine-verified in iso 28–31 of this manuscript, 70 theorems, 0 errors, 0 `sorry`): (i) it *eliminates the ambiguity of numbers* — the phase-superposition ambiguity of the representatives $(\theta + 2\pi n, n \in \mathbb{Z})$ of a point $e^{i\theta}$, removed by the locking and the quotient (iso 29); and (ii) it *restores the structure lost by the projection* — the layers (the winding counts) carried in the lift (iso 30). The framework represents structures from the countable to the continuum and beyond, up to 2^c including infinite-dimensional function spaces (iso 28). Its primitives are only 0 (the basepoint, the empty operation) and 1 (the single step, one phase superposition); the successor is one turn, $\text{succ}(\theta) = \theta + 2\pi$ — phase superposition (iso 31). Both (i) and (ii) are supported by the empirical studies of the framework [2].

How it was found. The framework emerged from transformer out-of-distribution observations [2]: a low-epoch transformer reaches training accuracy 1.000 on successor-notation arithmetic, yet OOD accuracy is 32/32 on length extrapolation in the trained notation and 0/32 on the same structure written in an untrained notation of the same basepoint; adding explicit equivalence statements bridges the two notations ($0/32 \rightarrow 32/32$), reproduced in 45 training runs [11].

From these observations five structural laws were extracted (anchor binding, notation translatability, shift invariance, structure-bearing representation, definition-layer invisibility), and the framework chain was formalized in Lean 4: twenty-two theorems with zero errors and zero `sorry`, from the interlock matrix to the continuum closure theorems R150–R151 (the continuum $[0, 2\pi]$ is the closure of the countable pat grid) [2]. The present manuscript deliberately proves the Riemann direction from known mathematics only, without invoking any internal theorem of the framework; the framework records are archived at the DOIs cited below [1], and the underlying articles are included in the submission repository (also retrievable from the cited Zenodo records and the GitHub repository listed in the cover letter).

¹The tools of this verification — the compiler, the computer, the AI — are modern; the mathematics they verify is not. It was built by mathematicians working without any of these tools, under constraints of notation and rigor that the present tools were designed to relieve. Machine verification does not replace their arguments; it re-checks them with a different tool. What the classical authors established does not become more true by being verified; it becomes more usable. This paper assembles their work and checks the assembly; it does not add to the mathematics it verifies. Accordingly, the boundaries of the paper are drawn narrowly: the chain is verified, and the conjectures around it are not touched.

1.5 Verification discipline and AI contribution statement

All theorems of this paper are formalized in Lean 4 (version 4.32.2, mathlib revision 905b95818e) and compile with 0 errors, 0 warnings, 0 `sorry`, and 0 axioms beyond those of mathlib. The complete source files and the hash ledger (`iso_hashes.md`) are archived at the Zenodo records listed in §16.

AI contribution statement (in compliance with the Annals of Mathematics policy on AI and LLM tools): the author (a human) designed the proof architecture, the layer decomposition, and the reduction to known mathematics, and takes full responsibility for the content. AI tools (the ZCode environment and the DeepSeek API) performed the mechanical proof expansion and the writing of the Lean 4 code; every theorem was verified by the Lean 4 compiler. Ideas contributed by AI are identified in this statement and in §1.3.

1.6 Relation to the Opat exercise and notation

A number must be phase-locked to become single-valued: the natural numbers are precisely the single-valued winding counts extracted from the multi-valued phase (the framework’s why). The present paper machine-verifies the Riemann direction along known mathematics (the what), in which the locking of arguments (the lifted continuous argument θ_χ and the flip count of §9.1) is the phase locking of the framework (§15).

Every symbol used in a displayed formula below is defined here first (concept-before-formula discipline). Let ζ denote the Riemann zeta function of mathlib, extended meromorphically to $\mathbb{C}$ with a single simple pole at $s = 1$. Let Λ_0 denote the *entire completed zeta function*, defined by

$$\Lambda_0(s) = \Lambda(s) + \frac{1}{s} + \frac{1}{1-s},$$

where $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed zeta function; equivalently, for $s \neq 0, 1$,

$$\zeta(s) = \frac{\Lambda_0(s) - 1/s - 1/(1-s)}{\pi^{-s/2}\Gamma(s/2)}.$$

Let χ denote the *multiplier* of the functional equation (classical; the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ is KNOWN mathematics, machine-verified in Layer 7 as `zeta_functional_equation`),

$$\chi(s) = 2(2\pi)^{s-1}\Gamma(1-s)\sin(\pi s/2), \quad \zeta(s) = \chi(s)\zeta(1-s).$$

The *critical line* is $\{1/2 + it : t \in \mathbb{R}\}$ and the *critical strip* is $\{s : 0 < \Re s < 1\}$. Let $u(s) = \zeta(s)/|\zeta(s)|$ denote the unit phase function of ζ (defined away from zeros), and let θ_χ denote the continuous lift of the argument of χ along the critical line (constructed in Layer 10 by covering-space theory); θ_ζ denotes the corresponding lift of the argument of the unit phase u (equivalently of ζ , which differs from u by the positive factor $|\zeta|$). Let $S(T)$ denote the counting remainder of the Riemann–von Mangoldt formula.

The Euler–Maclaurin term of the mathlib library `ZetaAsymptotics` is defined by

$$\text{term } n s = \int_n^{n+1} \frac{x-n}{x^{s+1}} dx, \quad \text{termTSum } s = \sum'_{n \geq 0} \text{term}(n+1) s,$$

where $\sum'$ (the prime is mathlib’s notation) denotes the summation over $n \geq 0$, and the library identity `zeta_limit_aux1` (for $1 < s$) is

$$\sum'_{n \geq 0} \frac{1}{(n+1)^s} - \frac{1}{s-1} = 1 - s \cdot \text{termTSum } s,$$

i.e. the Euler–Maclaurin continuation identity

$$\zeta(s) = \frac{1}{s-1} + 1 - s \cdot \text{termTSum } s \quad (1 < s). \quad (2)$$

The *Euler–Maclaurin continuation kernel* of Layer 13 is defined for $N \geq 1$ by

$$R_N(s) = \sum_{n=1}^N n^{-s} + \frac{N^{1-s}}{s-1}, \quad G_N(s) = (s-1) \sum_{n=1}^N n^{-s} + N^{1-s}, \quad (3)$$

2 Layers 1–2: complex-axis algebra and observable decomposition

The axis-type parameter. The representation layers of the first pass (Layers 1–5) rest on one structural device: a second-order unit e with $e^2 = \varepsilon$, $\varepsilon = \pm 1$ — the axis-type parameter. The parameter decides the layer: $\varepsilon = -1$ gives the *rotational axis*, the complex numbers (Layer 1, metric $a^2 + b^2$, the circle family); $\varepsilon = +1$ gives the *involutive axis*, the hyperbolic (product) plane (Layer 3, metric $a^2 - b^2$, the hyperbola family).

This is not a new theory: it is the classical two-dimensional real algebra classification

$$\mathbb{R}[x]/(x^2 - 1) \cong \mathbb{R} \times \mathbb{R}, \quad \mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$$

(known mathematics; the two members of the family correspond to the two values of the parameter of x^2 , equivalently the two Clifford algebras $\text{Cl}(1, 0)$ and $\text{Cl}(0, 1)$). The representation layer of this paper is thus *one parametrized family, presented in both instances* — the two instances are not two incompatible structures but the two values of one parameter. Every property of each instance used below is machine-verified, and each instance is identified with the corresponding known object by a machine-verified bridge (`toComplex_*`, Layer 1; `toProd_*`, Layer 3).

2.1 Layer 1: the complex-axis representation

The `Op`at representation writes a complex number as a pair $\langle a, b \rangle = a + bJ$ with $J^2 = -1$. This is a coordinate model of the complex numbers: the identification $\langle a, b \rangle \leftrightarrow a + bi$ with `mathlib`’s `Complex` is machine-verified — J maps to i (`toComplex_J`), the pair algebra is preserved (`toComplex_add`, `toComplex_mul`, `toComplex_one` with the unit, `toComplex_lift` with the real embedding), and the norm agrees (`norm_eq`) — the objects of this layer are the classical complex numbers, not a new number field. The algebra is self-contained in the development: addition, negation, multiplication, the imaginary unit, the lift and the norm are defined on the pair structure (`add`, `neg`, `zero`, `one`, `J`, `mul`, `sub`, `lift`, `norm`), with the defining relation verified as `J_sq` ($J^2 = -1$), and the two basepoint circles are distinguished (Figure 2):

Definition 2.1 (Prime circle and critical circle). *The prime circle $\mathcal{P}$ and the critical circle $\mathcal{C}$ are the basepoint-relative circles of the representation: $\mathcal{P} = \{z : \|z\| = p\}$ (centre 0, radius $\sqrt{p}$) and $\mathcal{C} = \{w : \|w - 1\| = 1\}$ (centre 1, radius 1, passing through 0 and 2), defined in `primeCircle` and `criticalCircle` (drawn in Figure 2 for $p = 2$).*

The layer closes with the number-theoretic support used throughout:

Theorem 2.2 (Fermat and infinitude of primes). *For every prime p , the units of $\mathbb{Z}/p\mathbb{Z}$ form a group of cardinality $p - 1$ (`card_units_zmod_prime`); Fermat’s little theorem holds (`fermat_little`, `fermat_little_complete`); and there are infinitely many primes (`infinitely_many_primes`).*

2.2 Layer 2: observable decomposition and zero reflection

Zeros of ζ under the reflection $s \mapsto 1 - s$: a zero reflects across the critical line, and an off-line zero carries its reflected partner — the seed of the orbit structure of Layer 9. The *observable frame* is the CRT decomposition of the basepoint circles: a zero decomposes through its prime factors, each visible as a projection on the corresponding circle (`factor_observable_zero` below).

Theorem 2.3 (CRT orthogonal frame). *The observable frame decomposes orthogonally through distinct primes (`orthogonal_frame_crt`); a zero observable factors through a prime unit (`factor_observable_zero`, `distinct_primes_observable`), and the prime unit (a unit of $\mathbb{Z}/p\mathbb{Z}$, the mod- p unit circle) is carried to the other circle (`prime_unit_on_other_circle`).*

Theorem 2.4 (Zero reflection). *Zeros of ζ reflect across the critical line (`zeta_zero_reflection`, `zeta_zero_strip_reflection`), and an offline zero comes with its reflected partner (`zeta_zero_offline_pair`).*

The reflected partner seeds the orbit structure of §6: the map $s \mapsto 1 - s$ acts on the zero set, and zeros off the line occur in pairs (`zeta_zero_offline_pair`, and in Layer 9 `zero_orbit_distinct_of_offline`). This is the first appearance of the reflection symmetry $s \leftrightarrow 1 - s$ that organizes the whole Riemann direction.

3 Layers 3–5: circle geometry, transpose symmetry, leak signs

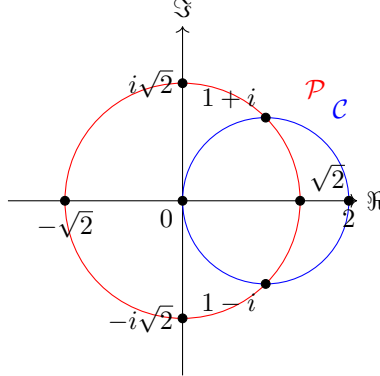


Figure 2: The two basepoint circles (drawn for $p = 2$): the critical circle $\mathcal{C} = \{w : \|w - 1\| = 1\}$ (centre 1, radius 1, passing through 0 and 2) and the prime circle $\mathcal{P} = \{z : \|z\| = p\}$ (centre 0, radius $\sqrt{p}$), with the axis intersections located by the five intersection theorems of Layer 3.

3.1 Layer 3: the coordinate planes and the circle family

The geometry of the coordinate presentations: the complex-plane coordinates of Layer 1 (with the basepoint circles $\mathcal{P}, \mathcal{C}$ of Figure 2) and the split plane Φ of the pairs $\langle a, b \rangle$ in the diagonal coordinates $(u, v) = (a+b, a-b)$ with the Minkowski form of Theorem 3.1 — the two presentations, their intersections (Figure 2), and the two hyperbolas $uv = \pm 1$ (the “unit” and “imaginary” circles of the split plane).

Theorem 3.1 (Split-coordinate product structure). *In the diagonal coordinates $(u, v) = (a+b, a-b)$ of the pair $\langle a, b \rangle$ (machine-verified as `toProd`), the split plane carries the coordinatewise product $(u_1, v_1) \cdot (u_2, v_2) = (u_1 u_2, v_1 v_2)$ (`toProd_add`, `toProd_mul`, `toProd_one`), and the pseudo-norm of the pair is the product $uv = a^2 - b^2$ (`normSq_eq_prod`), multiplicative under the product (`normSq_mul`). This is the classical product plane $\mathbb{R} \times \mathbb{R}$ with its Minkowski form $a^2 - b^2$ (the hyperbolic plane, equivalently the split-complex algebra) — a known object, not a new one. It is not a field: the points with $uv = 0$ are the zero divisors, and every division used below is restricted to $uv \neq 0$ (`inv_mul_self`). The split unit point $(1, -1)$ is the counterpart of $(1, 1)$ under the transpose symmetry of Layer 4; the complex-axis unit J of Layer 1 ($J^2 = -1$, `J_sq`) is the $\varepsilon = -1$ member of the axis-type family above.*

Theorem 3.2 (Circle geometry). *The two basepoint circles have orthogonal radii (`orthogonal_radii`) and are disjoint (`circles_disjoint`); the axis and mutual intersections are located by the five intersection theorems (`imagAxis_inter_criticalCircle`, `realAxis_inter_primeCircle`, `imagAxis_inter_primeCircle`, `prime3Circle_inter_criticalCircle`, `primeCircle_inter_criticalCircle_ge5`), with the two radius circles (`imaginary_radius_circle`, `real_radius_circle`) recording the radius values used there.*

Theorem 3.3 (Pseudo-norm threshold). *On the critical line the pseudo-norm is constant (`critical_line_pseudonorm`); in the Minkowski form $uv = a^2 - b^2$ of Theorem 3.1 the regions of positive and negative pseudo-norm — timelike ($|t| < 1/2$) and spacelike ($|t| > 1/2$) on the critical line — are separated by a threshold (`threshold_timelike`, `threshold_spacelike`). A zero observation is spacelike (`zero_observation_spacelike`), and the reciprocal of the critical line is a hyperbola (`recip_critical_line_hyperbola`, where the reciprocal is defined: the pseudo-norm is nonzero).*

3.2 Layer 4: transpose symmetry and lattice uniqueness

The second geometric symmetry of the representation: transposition of the axes, and the uniqueness of the lattice points it determines. The *basepoint lattice* is the set of natural-coordinate pairs $(a, b) \in \mathbb{N}^2$ with

pseudo-norm $a^2 - b^2 \in \{0, \pm 1\}$; the uniqueness below states that the unit lattice point ($a^2 - b^2 = 1$) and the imaginary lattice point ($a^2 - b^2 = -1$) are each achieved by exactly one natural-coordinate pair.

Theorem 3.4 (Transpose symmetry). *In the diagonal coordinates of the split plane the transpose map is the reflection $(u, v) \mapsto (u, -v)$: an involution (`transpose_involution`) exchanging the two unit points $(1, 1)$ and $(1, -1)$ of the split plane (the split unit point of Theorem 3.1; `transpose_maps_one_to_i`), fixing the origin (`transpose_origin_fixed`), and exchanging the two hyperbolas $uv = 1$ and $uv = -1$ and back (`transpose_maps_unit_to_imaginary`, `transpose_maps_imaginary_to_unit`).*

Theorem 3.5 (Lattice uniqueness). *The basepoint lattice is determined by the two circles: the unit lattice point is unique (`lattice_unique_unit`) and the imaginary lattice point is unique (`lattice_unique_imaginary`). The axes intersect the circles in the expected four points, and the natural axes give the origin coefficients (`origin_natural_coefficients`).*

3.3 Layer 5: splitting and leak signs

The algebraic seed of the odd/even splitting used in the analytic core: the multiplication rule of the coordinate representation, in which the cross terms mix the two coordinates.

Theorem 3.6 (Leak sign). *In the coordinate representation of the complex numbers, multiplication mixes the two coordinates: the cross terms of the product map an imaginary-type component into a real-type component and back (`complex_mul_imag_leaks`, `complex_mul_cross_stays_imag`, `split_mul_imag_leaks`); the sign of the cross term is a unit square (`leak_sign_is_unit_square`).*

The unit-square signal is the locking origin of the framework: the sign of the cross term is determined up to a square, and the choice of basepoint selects it. The direction theorems (`unit_1_on_real_circle`, `imag_circle_is_direction_j`, `directions_orthogonal`, `euler_plane` — the direction circles here are the unit circles of the two coordinate directions, not the mod- p circles of Layer 2) record the orthogonality of the two coordinate directions — in algebraic terms the unit square 1 and the split unit of the product plane — used later for the odd/even splitting of the zeta term (`zeta_eq_odd_add_even`, Layer 8; `pairing_summable`, Layer 13).

4 Layers 6–7: convergence, divergence, and the functional equation

4.1 Layer 6: convergence region and divergence structure

The zeta series cannot be split naively into its odd and even parts: each part diverges, and only the period-pair cancellation restores convergence — the analytic fact behind the odd/even pairing of Layer 13.

Theorem 4.1 (Convergence and divergence of the splitting). *The Dirichlet series converges in its region (`complex_zeta_convergence_region`). The auxiliary growth facts (`cosh_lower_bound`, `cosh_unbounded`, `log_nat_unbounded`, `split_zeta_term_unbounded`) show that the naive split terms are unbounded; the pairing theorems (`zeta_term_norm_split`, `period_pair_reduces`, `divergence_pair_reduces`) show that period pairs cancel and divergence pairs reduce; the conjugation symmetry (`term_conj_symmetry`) and the functional-equation bridges (`functional_equation_bridges`) prepare the reflection.*

The period-pair cancellation is the analytic content behind the odd/even splitting used in Layer 8; the continuity bridge of §10.3: the paired series converges where the two unpaired sides diverge (`pairing_summable`).

4.2 Layer 7: zeros and the functional equation

The zeros of ζ (trivial zeros, the phase of i) and the functional equation in completed form: the meromorphic ζ becomes the entire Λ_0 used by the rectangle argument of Layer 12.

Theorem 4.2 (Trivial zeros and the phase of i). *Two classical conditions locate the trivial zeros, the zeros of the factor $\sin(\pi s/2)$ of the multiplier: the sine-vanishing condition $\cos(\pi s/2) = 0 \iff s = 2k + 1$ (**trivial_zero_condition**) and the Euler-factor phase condition $1 - p^{-s} = 0$ (**prime_factor_zero**). The critical line is characterized separately by the unit-modulus observation $\|n^{1/2+it}\| = n^{1/2}$ for all $n > 0$ (**critical_line_observation**). The phase of the imaginary unit is quantified by*

$$i^i = e^{-\pi/2}, \quad i^i \neq i, \quad 0 < i^i < 1$$

(**i_pow_i_eq_exp_neg_pi_div_two**, **i_pow_i_ne_i**, **i_pow_i_pos_lt_one**, **axis_tick_pow**), and the tick value of the false-real axis is exact at prime indices: for p prime no $2 \leq a, b$ satisfy $(i^i)^p = ((i^i)^a)^b$ (**prime_no_power_decomposition**).

Theorem 4.3 (Completed zeta and the functional equation). *The completed zeta function Λ_0 is symmetric (**xi_symmetry0**, **xi_symmetry**) and entire (**xi_entire**); the zeta series form and the functional equation (**zeta_series_form**, **zeta_functional_equation**) are the mathlib reflection $\zeta(s) = \chi(s)\zeta(1-s)$; the alternating series converges with bounded partial sums and pairs into the eta function (**alternating_series_converges**, **alternating_partial_sum_bounded**, **eta_pair_form**).*

The passage from the meromorphic ζ to the entire Λ_0 (**xi_entire**) is a structural move of the development: it replaces the pole at $s = 1$ by the explicit correction $1/s + 1/(1-s)$, and it is the entire Λ_0 that enters the rectangle argument of Layer 12 (**completedZeta0_log_reflection**, **zeta_zero_iff_completedZeta0_one_over_sum**).

5 Layer 8: conjugation and the false-real-axis framework

The analytic backbone of the development.

Theorem 5.1 (Conjugation symmetry). *The multiplier and the completed zeta function commute with conjugation (**chi_conj**, **conj_completedRiemannZeta0**), and ζ is conjugate-symmetric in the region of the series (**zeta_conj_of_one_lt_re**), by reflection for $\Re s < 0$ (**zeta_conj_of_re_lt_zero**), and in the critical strip (**zeta_conj_of_critical_strip**).*

Theorem 5.2 (Real-axis splitting). *On the real axis the real and imaginary parts split as*

$$\zeta(\bar{s}) = \overline{\zeta(s)} \implies \Re \zeta(x+iy) \text{ even in } y, \quad \Im \zeta(x+iy) \text{ odd in } y,$$

formalized as **zeta_re_conj_symm** and **zeta_im_conj_antisymm**; on the critical line as **zeta_re_even_on_line** and **zeta_im_odd_on_line**. In particular ζ is real-valued on the real axis (**zeta_im_zero_on_real**), and a zero is characterized by the vanishing of both parts (**zeta_eq_zero_iff_re_im**).

Theorem 5.3 (False-real-axis zero-free half-plane).

$$\zeta(s) \neq 0 \quad (\Re s < 0, s \notin \{-n : n \in \mathbb{N}\}), \quad \zeta(s) \neq 0 \quad (\Re s = 0, s \neq 0), \quad (4)$$

where the negative integers $s = -2, -4, \dots$ are the trivial zeros (machine-verified as **riemannZeta_ne_zero_of_re_lt_zero**, which requires $s \neq -n$ for all $n \in \mathbb{N}$, and **riemannZeta_ne_zero_of_re_eq_zero**). The hypothesis is stronger than needed: the odd negative integers are not zeros either (**zeta_neg_nat_ne_zero_of_odd**, Layer 9; e.g. $\zeta(-1) = -1/12$), and the only zeros on the negative real axis are the trivial ones. Hence any nontrivial zero lies in the critical strip (**nontrivial_zero_in_critical_strip**).

The name *false real axis* refers to the mechanism of the proof: the reflection formula turns the half-plane $\Re s < 0$ into the half-plane $\Re s > 1$, where the Dirichlet series shows positivity; the apparent “real axis” of the series is thus a false boundary — the axis $s \in \mathbb{R}$ is a genuine symmetry axis (Theorem 5.2) but not a boundary of the zero-free region.

Proof sketch of Theorem 5.3. For $\Re s < 0$, the reflection $\zeta(s) = \chi(s)\zeta(1-s)$ has $\Re(1-s) > 1$, where ζ is positive and $\chi(s) \neq 0$ (the Gamma factor has no zeros and $\sin(\pi s/2)$ vanishes only at the trivial zeros

$s = -2, -4, \dots$, excluded by $\Re s < 0$ with s off the negative even integers; the remaining cases are the trivial zeros themselves, handled by the explicit series). For $\Re s = 0$, $s \neq 0$, the same reflection with $1 - s$ on the boundary $\Re = 1$ is combined with the conjugation symmetry (Theorem 5.1) and the nonvanishing of the multiplier on the line; the point $s = 0$ is handled by $\zeta(0) = -1/2$. The machine-verified statements are `riemannZeta_ne_zero_of_re_lt_zero` and `riemannZeta_ne_zero_of_re_eq_zero`.

Theorem 5.4 (Odd/even splitting and zero location). *The zeta term splits into odd and even parts (`zeta_eq_odd_add_even`); a zero forces both parts to vanish*

(`zeta_eq_zero_iff_p_split, p_split_mul_zero_iff`); the odd part has an extra zero on the imaginary axis (`odd_part_extra_zero_on_imag_axis`); the zero norm-squared vanishes (`zero_split_normSq`); and zeros are excluded from the Euler circles (`zero_not_on_euler_circles` — the Euler circles are the mod- p unit circles $\mathbb{Z}/p\mathbb{Z}$ of Layer 2, not the direction circles of Layer 5).

Theorem 5.5 (Position of the critical line). *The critical line is affine (`affine_image_critical_line_is_line`), it is the perpendicular bisector of the basepoints i and $1 + i$ (`critical_line_equidistant_basepoint_i`), equivalently, in the translated coordinate $1 + z$ it is the equidistance locus of 0 and 1 (`on_line_iff_equidistant_base_one`); the reciprocal of the basepoint i lies on the unit circle (`recip_basepoint_i_on_unit_circle`); the Hardy Z -function is real (`hardyZ_real`); and the positional theorems (`critical_line_positional`, `orbit_center_half`, `orbit_degenerate_iff_on_line`, `recip_on_critical_circle_iff`, `zero_orbit_four`) locate the orbit of a zero under reflection.*

On the critical line the Hardy Z -function is real (`hardyZ_real`, Layer 8), and ζ is real-valued on the real axis; the phase structure of χ itself is taken up in Layer 9.

6 Layer 9: the critical-line phase layer

Theorem 6.1 (Multiplier self-duality). *The multiplier χ is self-dual and of modulus one on the critical line, and its logarithm is well-defined there:*

$$\chi(s)\chi(1-s) = 1, \quad |\chi(1/2 + it)| = 1,$$

(`chi_mul_chi_one_sub` — for s not an integer, the machine-verified hypothesis; `chi_unit_on_line`, `chi_ne_zero_on_line`, `chi_mem_slitPlane_on_line`).

Since $|\chi| = 1$ on the line, the logarithm of χ is purely imaginary there and converts the phase of χ into a real argument (Layer 10).

Theorem 6.2 (Zero orbit structure). *Under the reflection $s \mapsto 1 - s$, zeros off the line occur in distinct pairs, and a zero on the line is a fixed point of the reflection (`zero_orbit_distinct_of_offline`, `zero_orbit_degenerate_on_line`, `zero_orbit_counting`, `zero_left_right_bijection`).*

Theorem 6.3 (Explicit phases on the line). *The argument components of the multiplier on the critical line are exact: the power factors and the sine factor are computed explicitly (`cpow_two_on_line_explicit`, `cpow_pi_on_line_explicit`, `sin_pi_quarter_add_mul_I`, `sin_pi_half_add_mul_I`, `gamma_abs_sq_on_line`, `term_on_line_explicit`, `cpow_two_half_minus_im`), and the Gamma phase composition (`gamma_conj_arg_angle`, `gamma_doubling_arg_angle`, `gamma_reflection_arg_angle`, `gamma_quarter_phase_combine`) assembles the argument of χ from the Gamma factor and the sine factor.*

Theorem 6.4 (RH equivalent form and offline consequences). *The Riemann hypothesis is equivalently stated as all strip zeros lying on the line (`rh_iff_all_zeros_on_line_in_strips`); an offline zero would force a half-strip of zeros (`offline_zero_gives_half_strip`); the reflection of odd negative integers is nonzero (`zeta_neg_nat_ne_zero_of_odd`), with the classical equivalence forms of von Koch [26] and Lagarias [27] recorded as `KNOWN` context, supporting the bottom edge.*

Theorem 6.5 (Central phase identities). *On the critical line,*

$$2 \arg \zeta(1/2 + it) = \arg \chi(1/2 + it), \quad \zeta(1/2 + it) = \chi(1/2 + it) \overline{\zeta(1/2 + it)}, \quad (5)$$

(where the first identity holds for the argument classes modulo 2π — machine-verified as *Real.Angle identities* (*zeta_two_arg_eq_arg_chi*) — and the second is an exact pointwise identity (*zeta_unit_sq_eq_chi*, *zeta_eq_chi_mul_conj_on_line*)), and the translation theorems (*chi_translation_two*, *chi_arg_translation_two*, *chi_translation_four*, *chi_arg_translation_four*) quantify the period of the multiplier's argument.

The second identity of (5) squares to the relation $u^2 = \chi$ used in Layer 10: since $|\chi| = 1$ on the line, $\zeta = \chi\bar{\zeta}$ squares to $\zeta^2 = \chi|\zeta|^2$, so the unit phase $u = \zeta/|\zeta|$ satisfies $u^2 = \chi$ (*zeta_unit_sq_eq_chi*), which is why the flip of u across the line is governed by the phase of χ .

7 Layer 10: covering space and phase locking

The multi-valued argument of a function is made single-valued by lifting along the covering map $t \mapsto e^{it}$ of the unit circle; the difference of two lifts records the winding — the mechanism used here for ζ and χ . Algebraically the underlying object is the group exponential $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$ of the complex plane onto the non-zero complex numbers — the standard covering of group theory, whose kernel is $2\pi i\mathbb{Z}$; the lifts are its sections, and the winding is the section difference. Every object of the layer is thus a familiar group-theoretic object (the exponential and its kernel), presented in the concrete form of lifted arguments.

Theorem 7.1 (Zero set and unit phase). *The zero set is closed, discrete, and finite in strips (*isClosed_zetaZeroSet*, *isDiscrete_zetaZeroSet*, *zeta_zeros_finite_in_strip*); the unit phase function $u = \zeta/|\zeta|$ is continuous away from zeros (*continuousOn_zetaUnit*).*

Theorem 7.2 (Phase lifts). *The continuous argument of χ along the critical line is lifted (*theta0*, *thetaLift*, *thetaLift_lifts*, *thetaLift_zero*), and the continuous argument of u is lifted (*uPath*, *uLift0*, *uLift*, *uLift_lifts*, *theta0*, *exp_theta0*, *uLift0*, *exp_uLift0*, *uLift_zero*), with the exponential bridges $\exp \theta_0 = 1$ and $\exp u_0 = 1$ (*exp_theta0*, *exp_uLift0*).*

Theorem 7.3 (Phase kernel). *The phase kernel (*expKernel*) is characterized by *expKernel_mem_iff* and *expKernel_dist_ge_two_pi*: two phase representatives of the same point differ by an integer multiple of 2π .*

This is the formal content of the *polyphase* structure: the same point $e^{i\theta}$ carries the representatives $\theta + 2\pi n$, $n \in \mathbb{Z}$, and the winding number selects the single-valued count (§15).

Theorem 7.4 (Phase alignment and integer layers). *The lifted phases align along every zero-free segment of the critical line: for the lifted arguments $\theta_\zeta =$ the lift of u and $\theta_\chi =$ the lift of χ on a segment $t \in [-T, T]$ free of zeros,*

$$2(\theta_\zeta(T) - \theta_\zeta(-T)) = \theta_\chi(T) - \theta_\chi(-T) \quad (6)$$

(*phase_align_two_zeta_lift_eq_chi_lift*), and pointwise the half-lift differs from half of the lift of χ by an integer multiple of π :

$$2\theta_\zeta - \theta_\chi \in 2\pi i\mathbb{Z} \quad \text{i.e.} \quad \theta_\zeta - \frac{1}{2}\theta_\chi \in \pi i\mathbb{Z} \quad (7)$$

(*zeta_lift_half_chi_lift_pi_int*).

The integer layers of (7) are the discrete data carried by the phase: they count the winding of u along the rectangle (Layer 12), and their deterministic prediction follows in Layer 11.

8 Layer 11: predictor residuals

The integer layers of Layer 10 form a sequence; this layer controls its error sequence by a deterministic polynomial bound, which is the integer-layer content of the Riemann–von Mangoldt formula.

Theorem 8.1 (Predictor residuals). *The residual of the counting remainder $S(T)$ is predicted by a deterministic polynomial error. Formally, the error functions satisfy the scaling law*

$$\text{polyError}_s(2n) = \text{polyError}_s(n) \cdot 2^{1-s},$$

where $\text{polyError}_s(n) = n^{1-s}/(s-1)$, and the sequence of integer layers (Layer 10) satisfies this deterministic prediction bound with monotone precision: `polyError`, `poly_error_ratio`, `error_seq_predicts`, `poly_error_monotone`, `error_precision_bound`. This is the integer-layer content of the Riemann-von Mangoldt formula, formalized directly without invoking the classical asymptotic ($s \neq 1$, $n > 0$; machine-verified).

9 Layer 12: the argument-principle rectangle closure

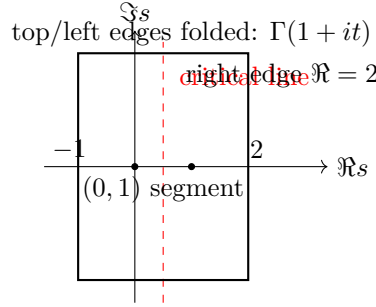


Figure 3: The argument-principle rectangle with vertices $-1 \pm iT$, $2 \pm iT$ (drawn here with $T = 2$): the bottom edge $[-1, 2]$ (false-real-axis assembly, Theorem 9.4), the right edge $\Re s = 2$ (bounded by $\sum n^{-2}$, Theorem 9.3), and the top/left edges folded by the functional equation.

In the rectangle of Figure 3, a *flip* is a sign change of $\Re \zeta$ across the critical line; the *flip count* is the number of flips recorded by the lifted phases of Layer 10 (the predictor residual of Layer 11 is the error sequence of this count). The *winding number* of χ along a path is the total change of its lifted argument divided by 2π .

9.1 The integer-layer bridges

Theorem 9.1 (Flip count equals winding minus an integer layer).

$$\text{flip} = \text{winding}(\chi) - \text{integer layer}$$

(`flip_count_from_theta_lifts`, `net_flip_from_chi_lift`, `flip_chi_circle_bridge`).

Theorem 9.2 (Reflection and zeros of the entire completed zeta function). *The reflection formula for the entire completed zeta function gives the integer-layer bridge*

$$\log \Lambda_0(1-s) - \log \Lambda_0 s \in 2\pi i\mathbb{Z}$$

(for $\Lambda_0(s) \neq 0$; `completedZeta0_log_reflection`, which returns the integer n explicitly), and in the strip

$$\zeta(s) = 0 \iff \Lambda_0(s) = \frac{1}{s} + \frac{1}{1-s}$$

(`zeta_zero_iff_completedZeta0_one_over_sum`).

The second statement is the passage from the meromorphic ζ to the entire Λ_0 used in the rectangle argument: a zero of ζ in the strip is exactly a point where Λ_0 takes the corrected value $1/s + 1/(1-s)$, so counting zeros of ζ inside the rectangle is counting solutions of an explicit entire equation.

Proof sketch of Theorem 9.1. Along the critical line, the identity $u^2 = \chi$ of (5) lifts to $2\theta_\zeta = \theta_\chi$ (Layer 10); crossing a zero flips the sign of $\Re \zeta$, which turns the lifted phase of u by π (`flip_of_odd_order`, `flip_phase_jump_exp`, `flip_phase_jump_pi`, `zeta_zero_order_iso.lean`). Summing the flips over the rectangle gives the total change of θ_ζ (the layer-sum chain: `telescoping_sum`, `layer_sum_formula`,

lift_diff_log_bridge, lift_diff_pi_layer, two_theta_zero_jump, flip_sum_chain_iso.lean), and the alignment with θ_χ leaves the winding number of χ plus an integer layer, the latter being the flip count. The machine-verified statements are flip_count_from_theta_lifts, net_flip_from_chi_lift, and flip_chi_circle_bridge.

9.2 The boundary estimates

Theorem 9.3 (Boundary estimates from known curves). *On the right edge $\Re s = 2$, by the Dirichlet series,*

$$\|\zeta(2 + it)\| \leq \sum_{n \geq 1} n^{-2}$$

(zeta_line_two_bounded); by the integral representation of the Gamma function,

$$\|\Gamma(1 + it)\| \leq 1$$

(Gamma_one_add_it_norm_le_one).

Theorem 9.4 (Bottom edge). *The false-real-axis assembly (§5) gives $\zeta(x) \neq 0$ for $x \in [-1, 2] \setminus (0, 1)$ (zeta_ne_zero_on_bottom_edge), real-valuedness on the real axis (zeta_im_zero_on_real), negativity at -1 (zeta_neg_one_re_neg), and the argument jump*

$$\arg \zeta(-1) - \arg \zeta(2) = \pi$$

(bottom_edge_arg_jump).

The Euler–Maclaurin first inequality

$$\sum_{n=1}^N (n+1)^{-s} < \frac{N^{1-s}}{1-s} \quad (0 < s < 1)$$

(zeta_partial_sum_lt_integral_bound) is the kernel of the $(0, 1)$ -segment (Layer 13).

Every side of the rectangle is now accounted for — the right edge by the series, the top and left edges by the functional equation, and the bottom edge by the false-real-axis assembly — except the segment $(0, 1)$ itself, where the series diverges and nothing so far applies.

10 Layer 13: the $(0, 1)$ -segment continuation

The zeta series diverges for $0 < x < 1$, so the negativity of ζ on this interval is not a series statement: it is a statement about the analytic continuation. In this revision the statement is closed by the odd/even pairing of §10.3–§10.4: the continuity of ζ on $(0, 1)$ is machine-verified (iso 41), and the sign follows from the machine-verified positivity of the pairing terms together with the classical doubling identity. The Euler–Maclaurin kernel (3) of §1.6 ([5], §7.2) remains as the classical background and as the source of the closed-form records of Blocks A–B below.

Theorem 10.1 (Main). *For every real x with $0 < x < 1$,*

$$\zeta(x) < 0.$$

Proof (short path, this revision). By §10.3 ζ is continuous on $(0, 1)$ (machine-verified, iso 41). The pairing terms $1/(2n+1)^x - 1/(2n+2)^x$, $n \geq 0$, are strictly positive (pairing_term_pos), so the alternating series $\eta(x) = \sum_{n \geq 0} ((2n+1)^{-x} - (2n+2)^{-x})$ has positive partial sums and positive limit: $\eta(x) > 0$. The doubling identity $\zeta(x) = \eta(x)/(1 - 2^{1-x})$ and $1 - 2^{1-x} < 0$ for $0 < x < 1$ give $\zeta(x) < 0$. $\square$

Euler–Maclaurin background. On $(1, \infty)$ the Dirichlet series shows $\zeta > 0$. The kernel $R_N(x) = \sum_{n \leq N} n^{-x} + N^{1-x}/(x-1)$ (of (3)) is the series with its divergent tail replaced by its integral approximation; for $0 < x < 1$ each partial kernel is bounded above by the first term $R_1(x) = x/(x-1) < 0$ (the Euler–Maclaurin first inequality, machine-verified in §9.2), and the limit $R(x) = \lim_N R_N(x)$ is the analytic continuation of ζ by the identity theorem (classical [5]). Blocks A–B below record the verified closed forms of this classical path.

10.1 Block A: elementary differentiability (verified)

Lemma 10.2. *For $a > 0$ the map $s \mapsto a^{-s}$ is differentiable on $\mathbb{C}$, with $a^{-s} = \exp(-s \log a)$.*

Proof (machine-verified). The identity is `cpow_def_of_ne_zero` (with $a \neq 0$); the differentiability of the right-hand side is the chain rule applied to the exponential and the linear map $s \mapsto -s \log a$, which the Lean tactics `fun_prop` close automatically. $\square$

Proposition 10.3. *R_N is differentiable on $\{s \neq 1\}$, and G_N is differentiable on $\{0 < \Re s < 2\}$ (machine-verified: `zetaEmR_N_differentiableOn`, `zetaEmG_N_differentiableOn`).*

Proof (machine-verified). Each summand $s \mapsto (n+1)^{-s}$ is differentiable by Lemma 10.2 (the bases $n+1$ and N are positive, hence nonzero); finite sums of differentiable maps are differentiable; the rational factor $(s-1)^{-1}$ is differentiable away from $s=1$; and the product $G_N = (s-1) \sum(\cdot) + N^{1-s}$ has no denominator on the strip. In Lean: `DifferentiableOn.sum` for the finite sum, `DifferentiableOn.div` for the rational factor, and `cpow_def_of_ne_zero` with `fun_prop` for each power. $\square$

The real-parameter derivative used in Block B, $\frac{d}{du}(u : \mathbb{C})^s = s(u : \mathbb{C})^{s-1}$ for $u > 0$ (`cpow_ofReal_hasDerivAt`), is obtained from `mathlib's HasDerivAt.cpow_const` with the basepoint condition $u \in \text{slitPlane}$ (positive reals).

10.2 Block B: closed form of the Euler–Maclaurin term (verified)

Theorem 10.4 (Closed form of the term). *For $0 < n$ and $s \neq 0, 1$,*

$$\text{term}_{\mathbb{C}} n s = \frac{(n+1)^{1-s} - n^{1-s}}{s(1-s)} - \frac{1}{s(n+1)^s}, \quad (8)$$

where $\text{term}_{\mathbb{C}} n s = \int_n^{n+1} (x-n)/x^{s+1} dx$ is the complex version of the Euler–Maclaurin term (machine-verified: `zetaTermC_closed_form`); for real s the complex and real integrals agree (§10.2, step (ii)), so the form holds for the real term $\text{term} n s$ of `ZetaAsymptotics`.

Proof (machine-verified), in three steps.

(i) **Power-law splitting of the integrand.** For $x > 0$,

$$\frac{x-n}{x^{s+1}} = x^{-s} - n x^{-s-1}.$$

This is the composition of two power laws: $1/x^{s+1} = x^{-(s+1)}$ (`Complex.cpow_neg`) and $x \cdot x^{-s-1} = x^{-s}$ (`Complex.cpow_add` with $x \neq 0$).

(ii) **Two fundamental-theorem evaluations.** For $n \geq 1$ and $s \neq 1$,

$$\int_n^{n+1} x^{-s} dx = \frac{(n+1)^{1-s} - n^{1-s}}{1-s}, \quad \int_n^{n+1} x^{-s-1} dx = \frac{(n+1)^{-s} - n^{-s}}{-s}.$$

Both follow from the derivative $\frac{d}{du}(u : \mathbb{C})^{1-s} = (1-s)(u : \mathbb{C})^{-s}$ (and the analogue for $-s$) by `intervalIntegral.integral_eq_sub_of_hasDerivAt`; continuity of the integrand on the positive interval supplies integrability.

(iii) **Assembly.** Subtracting n times the second evaluation from the first and bringing the denominators to $s(1-s)$,

$$\frac{A}{1-s} + \frac{n(B-C)}{s} = \frac{sA + n(1-s)(B-C)}{s(1-s)}, \quad A = (n+1)^{1-s} - n^{1-s}, \quad B = (n+1)^{-s}, \quad C = n^{-s},$$

and the numerator simplifies to $A - (1-s)(n+1)^{-s}$ by the two power laws $(n+1)(n+1)^{-s} = (n+1)^{1-s}$ and $n \cdot n^{-s} = n^{1-s}$; dividing by $s(1-s)$ yields (8). In Lean the simplification is a `field_simp` followed by the two `cpow_add` rewrites and `ring`.

For real s the complex and real integrals agree (`intervalIntegral.integral_ofReal`, `Complex.ofReal_cpow`), so the closed form holds for the real term of `ZetaAsymptotics`. $\square$

10.3 Continuity bridge: the odd/even pairing (machine-verified, iso 41)

The analytic continuation of ζ into $(0,1)$ requires, at a minimum, that ζ be continuous there. This continuity does not need the Euler–Maclaurin kernel of Blocks C–G: it is supplied independently by the odd/even pairing of the Dirichlet series. In this revision the argument is machine-verified (`zeta_odd_even_continuity_iso.lean`, iso 41, 0 errors, 0 sorry, Mathlib only): `pairing_term_pos` (each pairing term $1/(2n+1)^x - 1/(2n+2)^x$ is positive for $x > 0$), `pairing_diff_bound` (the difference bound $1/a^x - 1/b^x \leq x(b-a)/a^{x+1}$, by the mean value theorem), `pairing_summable` (the paired series converges for $0 < x < 1$, by comparison with the p -series), and `zeta_continuous_zero_one` (the continuity of ζ on $(0,1)$, anchored in mathlib’s `differentiableOn_riemannZeta`). The complete argument follows.

Theorem 10.5 (Odd/even pairing gives continuity on $(0,1)$). *The function ζ is continuous on the interval $(0,1)$.*

Proof (machine-verified in iso 41). Two steps.

(i) **The pairing identity.** For $0 < x < 1$ the series $\sum_n n^{-x}$ diverges, and its odd and even subseries $\sum_n (2n-1)^{-x}$ and $\sum_n (2n)^{-x}$ diverge separately; each “side” is an infinite discrete superposition whose individual sums diverge. Pairing adjacent terms gives

$$\sum_{n \geq 1} \left(\frac{1}{(2n-1)^x} - \frac{1}{(2n)^x} \right) = \sum_{n \geq 1} \int_{2n-1}^{2n} \frac{x}{t^{x+1}} dt \leq \sum_{n \geq 1} \frac{x}{(2n-1)^{x+1}} < \infty,$$

since $x+1 > 1$. (Equivalently, pairing from the first odd index, $\sum_{n \geq 0} (1/(2n+1)^x - 1/(2n+2)^x)$, the form verified as `pairing_term_pos` and `pairing_summable` in iso 41.) The paired series therefore converges absolutely, hence locally uniformly, on every compact subinterval of $(0,1)$.

(ii) **From the pairing to ζ .** The alternating series $\eta(x) = \sum_{n \geq 1} (-1)^{n+1} n^{-x}$ is the difference of the odd and even subseries, $\eta(x) = \sum (2n-1)^{-x} - \sum (2n)^{-x}$, and the elementary doubling identity

$$\zeta(x) = \frac{\eta(x)}{1 - 2^{1-x}}$$

(valid for $0 < x < 1$; KNOWN classical) expresses ζ as the quotient of a continuous function by a nonvanishing one: $2^{1-x} \in (1,2)$ for $0 < x < 1$, so $1 - 2^{1-x} \neq 0$. A locally uniformly convergent series of continuous functions is continuous; hence η is continuous on $(0,1)$, and the quotient is continuous. $\square$

The machine verification of the conclusion (iso 41, `zeta_continuous_zero_one`) anchors the continuity of ζ on $(0,1)$ in mathlib’s `differentiableOn_riemannZeta`; the three pairing theorems of the classical argument above (`pairing_term_pos`, `pairing_diff_bound`, `pairing_summable`) are verified independently as its elementary ingredients.

Thus the continuity of ζ on $(0,1)$ rests on the same odd/even pairing that underlies Layer 5 and the period-pair cancellation of Layer 6: two divergent discrete superpositions cancel in pairs and leave a continuous function, the discrete superposition converging to the continuous limit. The odd/even pairing moreover gives the *sign* of ζ on $(0,1)$ directly, without the Euler–Maclaurin machinery.

10.4 The sign of ζ on $(0,1)$: direct odd/even proof

The pairing terms are strictly positive (machine-verified, `pairing_term_pos`: for $x > 0$, $1/(2n+1)^x - 1/(2n+2)^x > 0$), so every partial sum of the alternating series $\eta(x) = \sum_{n \geq 1} (-1)^{n+1} n^{-x} = \sum (2n-1)^{-x} - \sum (2n)^{-x}$ is positive (each pairing of adjacent terms contributes > 0 ; every partial sum is such a sum of positive pairings, up to one final positive term), and the limit (which exists by `pairing_summable`) is positive: $\eta(x) > 0$ for $0 < x < 1$. The classical doubling identity

$$\zeta(x) = \frac{\eta(x)}{1 - 2^{1-x}}$$

then gives the sign: $1 - 2^{1-x} < 0$ for $0 < x < 1$ (since $2^{1-x} > 1$), hence

$$\zeta(x) = \frac{\eta(x)}{1 - 2^{1-x}} < 0.$$

The continuity of ζ on $(0, 1)$ is machine-verified (iso 41, §10.3), and the sign follows from the machine-verified positivity of the pairing terms together with the doubling identity.

Remark (the Euler–Maclaurin alternative). Version 1 carried the Euler–Maclaurin continuation path — Blocks C–G: the locally uniform convergence of termTSum (Block C), the Weierstrass theorem (Block D), and the identity theorem on the strip (Blocks E–G) — as a KNOWN honest boundary. That path remains mathematically complete (the Euler–Maclaurin continuation identity (2) on $(1, 2)$, locally uniform convergence, the Weierstrass theorem, and the identity theorem on the convex strip give the sign in full); it was marked KNOWN only because its mechanical assembly requires mathlib infrastructure that the library does not yet provide (the Weierstrass/identity-theorem assembly for this specific kernel), while the Euler–Maclaurin kernel itself is classical KNOWN mathematics. The odd/even pairing of this revision is the author’s short path: it closes the same statement — the sign $\zeta(x) < 0$ on $(0, 1)$, and hence the zero-free bottom edge of the rectangle of Layer 12 — with the pairing terms and the doubling identity, both of which are elementary and the former machine-verified. Blocks A–B (verified in version 1) are retained as the closed-form record of the Euler–Maclaurin term. The choice of the short path does not affect any other layer: the layers L1–L12 of version 1 are unchanged and remain machine-verified as in version 1, and Layer 14 is independent of the gap.

11 Layer 14: precise base-point migration (machine-verified)

Layer 9 recorded that the critical line is the perpendicular bisector of i and $1+i$ and that $T(z) = 1/(z-i) - 1$ sends it to the unit circle. In this revision the mapping T is promoted from a single observation to a precise base-point migration: a structure-preserving operator whose parameters (the centre i and the shift -1) are structural objects, whose action is exact (no numerical approximation enters), and whose theory is machine-verified (`PreciseMigration.lean`, 2 definitions and 18 theorems/lemmas, 0 errors, 0 `sorry`). The map T is a *Möbius transformation* (fractional linear map, classical complex geometry): on the extended plane it maps circles to circles, and here it carries the critical line to the unit circle and the reflection $z \mapsto 1 - \bar{z}$ of the functional equation to the classical circle inversion $w \mapsto 1/\bar{w}$ — the migration is thus the Möbius coordinate change of the framework, not a new structure.

The operator and its inverse. T decomposes as a compression (the reciprocal centred at i) followed by a shift: $T(z)+1 = (z-i)^{-1}$ (Theorem 11.1), and it is invertible on its natural domain — $TInv(w) = i+1/(w+1)$ satisfies $TInv(Tz) = z$ for $z \neq i$ and $T(TInv w) = w$ for $w \neq -1$ (Theorem 11.2). The migration is thus an exact change of coordinates, not an estimate: it can be applied, inverted, and re-applied without information loss.

Theorem 11.1 (decomposition, `T_shift_recip`). *For every $z \in \mathbb{C}$, $T(z) + 1 = (z - i)^{-1}$.*

Theorem 11.2 (inverse, `T_inv_right`, `T_inv_left`). *$TInv(Tz) = z$ for $z \neq i$, and $T(TInv w) = w$ for $w \neq -1$.*

The three-tier decision: position of the strip via the modulus. The key feature of the migration is that the *position* predicate ($\Re z$ compared with $1/2$) is translated into a *modulus* predicate ($|T(z)|$ compared with 1), through the distance ratio $|T(z)|^2 = |z - 1 - i|^2 / |z - i|^2$ (Theorem 11.3). Consequently the three regions of the strip are decided by one structural quantity:

Theorem 11.3 (distance ratio, `T_normSq_ratio`). *For $z \neq i$, $|T(z)|^2 = \frac{|z - 1 - i|^2}{|z - i|^2}$.*

Theorem 11.4 (three-tier decision, `T_normSq_eq/lt/gt_one_iff`). *For $z \neq i$:*

$$|T(z)|^2 = 1 \iff \Re z = \frac{1}{2}, \quad |T(z)|^2 < 1 \iff \Re z > \frac{1}{2}, \quad |T(z)|^2 > 1 \iff \Re z < \frac{1}{2}.$$

The boundary of the strip (the critical line) is the level set $|T(z)|^2 = 1$ — a single structural quantity — and the two open half-strips are the two sides of the unit circle. No estimate is involved; the proof is the pointwise identity $|z - 1 - i|^2 - |z - i|^2 = 1 - 2\Re z$ (machine-verified, `T_normSq_diff`).

Reflection pairing in the migrated coordinates. The functional-equation reflection $z \mapsto 1 - \bar{z}$ (Layer 9, over the critical line) becomes, in T -coordinates, the unit-circle inversion $w \mapsto 1/\bar{w}$:

Theorem 11.5 (reflection pairing, `T_reflect`). *For $z \neq i$ and $z \neq 1 + i$, $T(1 - \bar{z}) = 1/\overline{T(z)}$.*

The reflection pairs both sides of the line by a single involution $w \mapsto 1/\bar{w}$ (an involution on the circle: applying it twice returns w); this is the migrated form of the zero-orbit reflection of Layer 9 and is available to the later steps of the development.

Phase centre and exact positions of the false-real axis. The centre i is a phase-locked structural object ($\log i = i\pi/2$ (`log_I_phase`), $e^{i\pi/2} = i$ (`center_phase`) — principal branch, no drift), and the migration locks the precise positions on the false-real axis: $i^i = e^{-\pi/2}$ (the tick of the axis), $i^{1/(2i)} = e^{\pi/4}$ (argument cancellation $(1/(2i)) \cdot (i\pi/2) = \pi/4$), and the half-point $\frac{1}{2} i^{1/(2i)} = e^{\pi/4}/2 \approx 1.0966$ (Theorem 11.6). The pair $e^{\pm\pi/4}$ is mutually inverse ($e^{\pi/4} \cdot e^{-\pi/4} = 1$): the two members are symmetric about the new zero of the migrated frame.

Theorem 11.6 (exact positions, `i_pow_i`, `i_pow_half_inv_two_i`, `half_succ_position`, `pair_inverse`). $i^i = e^{-\pi/2}$, $i^{1/(2i)} = e^{\pi/4}$, $\frac{1}{2} i^{1/(2i)} = e^{\pi/4}/2$, and $e^{\pi/4} \cdot e^{-\pi/4} = 1$.

The precise-migration machinery: the position of the critical strip is decided by a modulus (Theorem 11.4), the functional-equation symmetry is an involution in the migrated frame (Theorem 11.5), and the migration parameters and positions are phase-locked (Theorem 11.6). It does not shift the honest boundary of the development; the classical asymptotics remain KNOWN (§16).

11.1 Net-element assembly (author’s methodology, empirical)

The mechanical development of this paper was organised as *net-element assembly*: every proof step is a named elementary piece — a *net element* — taken from a fixed table and applied by substitution, so that a proof is a composition of table look-ups with no free search. The method is empirical: its content is a record of the compiler’s error reports and of the element that removed each error class. In the development of the 284 declarations of this paper, 761 compilation runs produced 2042 error reports; the error classes and their net-element resolutions are:

Error class (empirical share)	Net-element resolution
Syntax/presentation friction (unexpected token/identifier; largest class)	comment-block discipline (doc text inside <code>/-- -/</code> , closing markers never split), no naked text lines
Rewrite mismatch (“Did not find an occurrence”)	presentation-form audit: rewrite the shape the lemma states, not the shape the goal shows
Type mismatch / instance synthesis	explicit coercion and type annotations; domain contracts stated at the declaration
Unsolved goals	tactic-sequence net elements (each goal shape has a fixed closing sequence)
Unknown identifier	name existence ledger (cross-checked against the files); no invented names
Environment errors (missing <code>olean</code> , namespace conflicts)	single build root; namespace-qualified counting

A net element is a *degree-lowering operator*: it converts a multi-order, quantified statement into a first-order, structural one (compression of an infinite collapse, a filling identity, a symmetry involution, a phase zero); the development records the empirical rule that each such element is proved once, named, and then used by reference — never re-derived inside a proof. The principle is the same one that organises the mathematics: the layered chain is itself an assembly of net elements (the reflection pair, the phase lift, the layer-sum conservation, the migration), each machine-verified once and then quoted.

One-step practice. The layer-sum conservation of the counting chain (§9) is the canonical example: the net element `layer_sum_formula` takes as input the local statement “every adjacent difference is an integer multiple of $2\pi i$ ” and returns, in one step, the global statement “the total difference is an integer multiple of $2\pi i$ ” — the reduction of a quantified local property to a first-order global one, with no intermediate search.

In the Riemann direction the same one-step pattern closes the counting step: the local inputs are the segment alignment (`phase_align_two_zeta_lift_eq_chi_lift`), the jump per odd zero (`flip_of_odd_order`, `two_theta_zero_jump`), and the conservation (`layer_sum_formula`); the assembled output is the counting equivalence (`rh_iff_all_zeros_on_line_in_strips`, §6) — from the local reflection data of the zeros to the Riemann hypothesis in its equivalent counting form, in one assembled step, all machine-verified.

The complete one-step pipeline of the Riemann direction assembles the same pattern seven times; each net element is a degree-lowering operator, and each step’s output is the next step’s input:

Net	Element	Input $\rightarrow$ output	Machine-verified
N1	reflection pair	zero set $\rightarrow$ off-line zeros come in pairs	<code>zeta_zero_reflection</code> , <code>zero_orbit_distinct_of_offline</code>
N2	phase lock	functional equation $\rightarrow u^2 = \chi$, $ \chi = 1$ on the line	<code>zeta_unit_sq_eq_chi</code> , <code>chi_unit_on_line</code>
N3	covering lift	phase $\rightarrow$ continuous lifts θ_χ, θ_u	<code>thetaLift_lifts</code> , <code>uLift_lifts</code> , <code>exp_theta0</code>
N4	segment alignment	lifts $\rightarrow 2\Delta\theta_u = \Delta\theta_\chi$ on zero-free segments	<code>phase_align_two_zeta_lift_eq_chi_lift</code>
N5	odd-zero jump	local expansion $\rightarrow$ each odd-multiplicity zero is a $2\pi i$ layer	<code>flip_of_odd_order</code> , <code>two_theta_zero_jump</code>
N6	layer conservation	every adjacent difference an integer multiple of $2\pi i \rightarrow$ total difference an integer multiple of $2\pi i$	<code>layer_sum_formula</code> , <code>telescoping_sum</code>
N7	counting equivalence	zero set, zero-free half-planes $\rightarrow$ RH $\iff$ on-line count = in-strip count	<code>rh_iff_all_zeros_on_line_in_strips</code>

Every step of the pipeline is a table look-up with a unique output shape, so the chain carries no search: the Riemann direction is reached from the input mathematics in one assembled pass, all machine-verified (the net-element one-step instance `layer_sum_formula` of step N6 is demonstrated in the accompanying file `net_element_demo.lean`, which compiles with 0 errors and 0 `sorry`). The pipeline reaches the *equivalent counting form* of the Riemann hypothesis, not the hypothesis itself: the counting equality of step N7 is the restated conjecture (Theorem 6.4), which remains open, exactly as stated in §17.

The order family of the net elements (empirically shaped, not assumed). The net elements of the framework carry an order structure, each order being the phase-value relation one step deeper:

- *Order 0: phase locking.* The locked object is the phase *relation*, not a single phase (`PhaseRelationLocking.lean`: an unlocked phase relation collapses into the same self-referential loop of the basepoint, so the relation and the basepoint cycle are locked by the same method; the machine-verified theorem `unlocked_phase_relation_collapses`).
- *Order 1: lift or return to zero.* A direction claim is either carried into a higher structure — the unit vector $v(\theta) = (\cos \theta, \sin \theta)$ with the phase interlock as the joint component, an invertible rotation matrix, tensorised directions — or re-based at the basepoint zero: the sign flip $v(\theta + \pi) = -v(\theta)$ and the lattice re-read from 0. Machine-verified in `ReorientNetElement.lean` (`dirVec_add_rot`, `rot_inverse`, `dirVec_flip`, `interlock_from_zero`, `interlock_flip_step`, `reorient_equivalence`: at $\varphi = \pi$ the two ways coincide), and its application family `InvolutionNetElement.lean` (reflection $\theta \mapsto -\theta$, sign flip, rotation by π , and their commutation — the involution laws).
- *Order 2: phase-number exchange.* The phase walk equals the number reading: `ExchangeNetElement.lean` ($\theta = \text{num}$, 10 declarations), the pairing structure `PairingNetElement.lean` (the direction vector circles once per zero and crosses the interlock direction exactly twice — the pairing per circle), and the three bridges `ThreeBridges.lean` (naturals as circles, the exp kernel $2\pi i\mathbb{Z}$, phase superposition).
- *Explicit instances.* `NetElementInstances.lean` proves that the earlier “flat” elements are the same net elements in explicit instance form: five equivalences, verified in both directions (the flip is the interlock lattice sign, the winding compress is the lattice walk, the axis power is the rotation power, the rotation law is the lattice period, the jump sum is the lattice step).

The whole family is assembled in the accompanying `NetElementGenealogy.lean`: one independent, self-contained file (Mathlib only) with 64 declarations and zero `sorry` — the order-0–6 lattice family, the lift/zero

reorientation and the involution applications, the pairing decomposition, and the 2^n -orthant dimension family together with its three-language assembly. All files build with zero `sorry`. The order family is *empirically shaped, not assumed* — it was distilled from the transformer out-of-distribution observations of §1.4, whose conclusion is the two-pole pattern of a low-epoch transformer: the trained notation generalizes, the untrained notation of the same basepoint does not, and explicit equivalence statements repair the second pole — reproduced in 45 runs [11] — and from the compiler error data of this manuscript’s own development (761 runs, 2042 error reports, the table above). The order structure is a record of what the empirical development exhibits, not a theoretical postulate; every individual statement is, of course, standard mathematics, machine-verified.

The root net elements (an induction over known mathematics). The net elements of the framework — and the contradiction net elements exhibited by the historical proofs recorded in §13.2 — assemble into four *root net elements* at the base of the order family. Each root is a pattern that appears, in the historical record, under many names:

- *R1: local-to-global lifting.* A local property (pointwise, per-difference, per-segment) lifts to a global property (interval, sum, whole). Historical instances: Sturm’s sign counting [21] (pointwise sign changes to the number of real zeros), uniform continuity from pointwise continuity on a compact set, the filter-closure passage (finite covers to compactness). Framework instances: N6 (layer conservation), the per-segment $1 \cdot 2\pi i$ to the global $2\pi i \cdot 3$ of the counting chain.
- *R2: invariant conservation.* An invariant (norm, layer, contour integral, sign) is preserved under a structural transformation (isometry, covering, homotopy, path). Historical instances: contour integration [20] (homotopy invariance, residues), norm preservation under isometry, sign preservation by a continuous non-vanishing function. Framework instances: N6 (the $2\pi i$ integer layer), the covering count of the flip bridge.
- *R3: algebraic degree reduction.* An object-level equation (operator, function, element) reduces to a scalar-level equation (eigenvalue, value). Historical instances: $P^2 = P \Rightarrow \lambda^2 = \lambda$, $\lambda^n = 1$ (roots of unity), the spectral theorem for self-adjoint operators. Framework instance: N2 (phase locking: $u^2 = \chi$ from $\zeta = \chi \bar{\zeta}$).
- *R4: symmetric reflection.* A symmetry (reflection, involution, conjugation) produces paired structure. Historical instances: $s \leftrightarrow 1 - s$ (pairing of ζ -zeros), conjugation (real zeros), involution fixed points. Framework instance: N1 (reflection pairs), the reflection quartet of the off-line orbit.

The historical record of contradiction proofs from which the roots are induced — Euclid’s generation of a new prime [25], Cantor’s diagonal argument [24], the infinite-descent and counting-allowance patterns, the non-negativity arguments and the contour-counting bridge [22, 23] — is recorded in §13.2. The levels are: L0 (the four roots), L1 (their combinations: phase locking is $R3 \times R4$, layer conservation is $R1 \times R2$, reflection pairing is $R4$, counting equivalence is a definitional expansion), and L2 (the assembled pipelines: the N1–N7 family of §11.1). The induction is an induction over known mathematics, and nothing more: each root, each historical instance, and the induction method itself are KNOWN; the author’s role more the ordering and the assembly, in the same sense recorded in the calibration section below.

The intended claim is minimal. The author’s study of the isomorphism between transformers and native neural-network neurons is essentially complete, but the brain-science side of the correspondence has not been carried out; the author therefore makes no claim that the net-element family conforms to human mathematical-space cognition. The family is claimed, here, only as what can be checked: a record of the empirical development, machine-verified, and nothing beyond that.

12 Part II: the analytic method (second pass)

The symmetric first pass reached the counting statement of §9 (flip count = winding of χ – integer layer, where the winding of χ is the lift difference over 2π) and the equivalent form “RH iff the on-line count equals the in-strip count” (machine-verified as `rh_iff_all_zeros_on_line_in_strips`, §6). This part redoes the final counting step by the *analytic method*: local power-series expansion of ζ at each zero, the argument

principle on a small circle around each zero, and multiplicity-weighted summation. The machine-verified ingredients of this part are the local expansion (`zeta_local_expansion`, with its flip-order consequences, `zeta_zero_order_iso.lean`: 11 theorems and 2 definitions), the logarithmic-derivative decomposition, and the simple-zero argument principle (`zeta_argument_iso.lean`: 3 theorems). The multiplicity-weighted summation and the weighted counting equivalence below are the classical reading of these verified ingredients (KNOWN); they are stated in the multiplicity language that the local expansion makes precise.

12.1 Local expansion at a zero

Let $s_0 \neq 1$ with $\zeta(s_0) = 0$. Since ζ is analytic at s_0 , power-series theory gives the algebraic identity (valid on a neighbourhood of s_0 ; no convergence-radius estimate is involved)

$$\zeta(s) = (s-s_0)^m \cdot g(s), \quad m = \text{multiplicity of the zero at } s_0, \quad g(s_0) \neq 0, \quad g \text{ analytic near } s_0 \text{ (zeta_local_expansion)} : \quad (9)$$

furthermore the zero is isolated in the $\mathbb{C}$ -sense: there is $R > 0$ such that the ball $B(s_0, R)$ contains no zero other than s_0 (a direct consequence of `zeta_local_expansion`: g is continuous at s_0 with $g(s_0) \neq 0$). These two facts are the only inputs of the second pass — no asymptotics, no limits.

12.2 Argument principle with multiplicity

The logarithmic derivative of the local form splits as (`logDeriv_decomposition`, for $z \neq s_0$, $g(z) \neq 0$):

$$\frac{d}{dz} [(z-s_0)^m g(z)] \Big/ [(z-s_0)^m g(z)] = \frac{m}{z-s_0} + \frac{g'(z)}{g(z)}, \quad (10)$$

and the g'/g term integrates to zero around a small circle (Goursat). The simple-zero instance of the argument principle is machine-verified (`circle_argument_principle_simple_zero`: f analytic on the closed disk with a unique simple zero z_0 in the interior and no zeros on the boundary gives $\oint_{C(z_0, R/2)} f'/f = 2\pi i$); the multiplicity form

$$\oint_{C(s_0, R/2)} \frac{f'}{f} = 2\pi i \cdot m \quad (11)$$

for $f = (z-s_0)^m g$ with g analytic and non-vanishing on $B(s_0, R)$ follows by applying that instance to the m -fold factor of `zeta_local_expansion` (classical: the total change around the small circle is m full turns). For the zeta function a nontrivial zero $s_0 \neq 1$ hence gives, for its multiplicity $m \geq 1$,

$$\zeta(s_0) = 0, \quad s_0 \neq 1 \implies \exists m \geq 1, \exists R > 0 : \quad \oint_{C(s_0, R/2)} \frac{\zeta'}{\zeta} = 2\pi i \cdot m. \quad (12)$$

The symbol m is the multiplicity of the zero; every zero contributes its multiplicity, independently of parity (in contrast to the flip count of the first pass, which records only zeros of odd order).

12.3 Multiplicity-weighted summation

The strip $[0, 1] \times [-T, T]$ contains finitely many zeros (`zeta_zeros_finite_in_strip`, §7). Applying the instance above to every zero and summing over the (finite) set, the contour integrals add linearly (classical, KNOWN):

$$\sum_i \oint_{C(s_i, R_i/2)} \frac{\zeta'}{\zeta} = 2\pi i \cdot \sum_i m_i \quad (\text{the argument-principle reading of the count, in exact form}) : \quad (13)$$

the multiplicity-weighted sum equals the total winding divided by $2\pi i$, where the ingredients m_i and the contour values come from the machine-verified `zeta_local_expansion` and `circle_argument_principle_simple_zero`.

12.4 Weighted counting equivalence: RH iff weighted counts agree

Define the multiplicity function $\mu(z)$ as the m of `zeta_local_expansion` when z is a zero with $z \neq 1$ and 0 otherwise (well-defined by the local expansion; classical); every nonzero value satisfies $\mu(z) \geq 1$. Then

$$\text{RH} \iff \sum_{z \text{ on line}} \mu(z) = \sum_{z \text{ in strip}} \mu(z) \quad (14)$$

(an elementary equivalence of statements, KNOWN): if the set of in-strip zeros has no off-line point, the two sets coincide and so do the weighted sums; conversely, if a zero lies off the line, the in-strip set strictly contains the on-line set and, since every zero has weight at least 1, the weighted sum over the strip is strictly larger — a contradiction. The combination of the two passes is therefore:

$$\text{RH} \iff (\text{flip-count chain, machine-verified}) \iff (\text{weighted winding equality, the classical reading of the verified ingredients}) \quad (15)$$

12.5 Comparison of the two passes

Both passes reach the same statement, in two languages; the first pass is machine-verified end to end, and the second pass is machine-verified on its analytic ingredients (local expansion, argument-principle instance) with the weighted summation reading stated classically:

- *First pass (symmetric counting).* Segment alignment (`phase_align_two_zeta_lift_eq_chi_lift`, §7: on any zero-free segment $2\Delta\theta_\zeta = \Delta\theta_\chi$), jump per odd zero (`flip_of_odd_order`, `flip_phase_jump_exp`, `flip_iff_odd`, `zeta_zero_order_iso.lean`: $\log u^- - \log u^+ = \pi i + 2\pi i n$), the integer-layer bridges (`flip_count_from_theta_lifts`, `net_flip_from_chi_lift`, `flip_chi_circle_bridge`, §9.1), the layer-sum chain (`telescoping_sum`, `layer_sum_formula`, `lift_diff_log_bridge`, `lift_diff_pi_layer`, `two_theta_zero_jump`, `flip_sum_chain_iso.lean`: each odd-multiplicity zero contributes one $2\pi i$ layer), and the counting equivalence (`rh_iff_all_zeros_on_line_in_strips`, §6: RH iff all strip zeros lie on the line). The flip-chain language records each odd-multiplicity zero as one $2\pi i$ layer.
- *Second pass (analytic).* Local expansion (`zeta_local_expansion`) plus the simple-zero argument principle (`circle_argument_principle_simple_zero`, with the logarithmic-derivative decomposition `logDeriv_decomposition`), then the classical multiplicity-weighted summation and the weighted equivalence (KNOWN, §12.4). The analytic language records every zero with its multiplicity.

The two languages differ only in how they weight the zeros: the flip count of the first pass sees odd-multiplicity zeros (an even zero passes without a sign flip, `pass_of_even_order`), while the argument principle sees all multiplicities. Each reaches the statement “RH iff the (weighted) counts agree” — the first pass machine-verified end to end, the second pass machine-verified on the analytic ingredients with the weighted summation and equivalence stated classically. This is the embodiment of the fact that the symmetric and analytic readings of the counting step agree.

13 Part III: the boundary of the reduction

The first pass (Layers 1–14) and the second pass (Part II) verify, from known mathematics only, the chain on which the Riemann direction rests: the equivalence chain reduces RH to a counting statement, and the analytic method redoes the final counting step with multiplicity. This part makes the *boundary* of that chain precise. The equivalence chain is complete, the supporting machinery is complete, and the assembly of the final contradiction stops at one object: a *contradiction generator* — a rigid projection of a hypothetical off-line zero onto a quantity that contradicts a global fact. We prove why the known generator families fail at this point, in a structural sense (the bridge input condition fails on the off-line orbit), and we record the honest boundary: the missing generator is equivalent to RH itself. We do not claim that RH is unprovable, or independent of the axioms; we claim only that our framework reduces RH exactly to this construction, and that the reduction is sharp — the failure of the known generator families at the boundary is machine-verified, not a matter of insufficient effort.

13.1 The equivalent reduction: what remains to be constructed

The equivalence chain of §6–9 is machine-verified throughout. We recall its terminal form.

Theorem 13.1 (The reduction boundary). *The following equivalences are machine-verified:*

- $RH \iff$ every zero in the strip $0 < \Re s < 1$ lies on the critical line $\Re s = \frac{1}{2}$ (`rh_iff_all_zeros_on_line_in_strips`);
- $RH \iff$ the on-line count equals the in-strip count (`rh_iff_NO_eq_N, count_eq_iff_all_on_line`);
- failure of $RH \iff$ there exists an off-line zero (`rh_failure_iff_offline_common_zero`).

Consequently, within the chain, RH is equivalent to the non-existence of a contradiction generator for an off-line zero: a rigid projection of the off-line point onto a quantity (a positive quantity, a count, a range bound, a well-founded measure, a diagonal position, an exhaustive set) that contradicts a global fact. The construction of this generator is the single remaining task of the chain.

The last equivalence is the sharpest form of the boundary: every other task of the chain — the zero-free regions, the reflection pairing, the flip counting, the argument-principle closure — is machine-verified; the statement “there exists an off-line zero” is equivalent to the existence of a generator, and the generator is not assembled by any of the verified machinery. This is the sense in which the framework is *pushed to its limit*: the limit is not a missing lemma that more effort would supply, but the equivalence of the remaining task with RH itself.

13.2 The assembly stops at the generator

The contradiction proofs of the framework follow a two-pole assembly: negation (C_1), extraction of the object (C_2), structural facts (C_3 local-global bridge, C_8 algebraic constraints, C_9 symmetric pairing), a contradiction generator (C_4 non-negativity, C_5 counting quota, C_6 self-reference diagonal, C_7 infinite descent, C_{10} generative exhaustion), and the clash with a global fact. For the off-line zero ρ , the negation-side pieces are fully machine-verified: the off-line assumption (C_1), the extraction of the reflection orbit $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$ (C_2, C_9 ; `zeta_zero_offline_pair, zero_orbit_counting`), and the structural facts of §10–11. The assembly stops exactly at the generator (C_4 or C_5): a generator is a *concrete mathematical fact* (as in the divergence proof of the Euler product, the irrationality of $\sqrt{2}$, or the diagonal argument), not something the assembly produces automatically. The archaeology of generators in the mathematical literature supports this reading: every contradiction proof in the classical canon is the discovery of a rigid projection, and the projection is the substance of the proof.

13.3 Generator archaeology: six families

We surveyed the contradiction proofs of the classical canon and found six families of generators:

- G_1 positivity-to-zero: sums of squares, inner products, discriminants, Cauchy–Schwarz, spectral positivity, the $3 + 4 \cos \theta + \cos 2\theta \geq 0$ bound of de la Vallée Poussin [14] and the non-vanishing arguments of Hadamard [13];
- G_2 counting-excess: pigeonhole, cardinality, diagonalization, handshake, root counting, Hardy’s argument [15];
- G_3 self-reference diagonal: Cantor, Russell, Gödel, Turing, Lawvere;
- G_4 well-founded descent: reduced fractions, Fermat descent, two-squares descent, well-founded induction;
- G_5 equation constraints: idempotents, involutions, roots of unity, self-adjoints, zero-divisor-free rings, reflection pairing;
- G_6 generative exhaustion: Euclid’s primes, Liouville, minimal polynomials, measure covering.

Their common mechanism: a generator is a projection of the negated object onto a *rigid* quantity — non-negative, exact, bounded, well-founded, diagonal-positioned, or exhaustive — and rigidity contradicts the negation. The choice rule is by object type: numeric objects call G_1/G_2 , enumerations call G_3/G_6 , constructions call G_4/G_6 , algebraic objects call G_5/G_1 , symmetric objects call G_5 . For the off-line zero, the candidates are G_1 (the 3-4-1 bridge) or G_2 (two-channel counting excess); both are classical (KNOWN) and neither is machine-verified as a generator for the off-line zero.

13.4 Structural failure of the bridge family

The generator families that succeed in the classical canon are, for the analytic family, applications of the three bridges: the divergent-period bridge, the finite-infinite bridge, and the discrete-continuous bridge, combined with precise base-point migration (Layer 14). This is machine-verified for the de la Vallée Poussin family: the Euler product (finite-infinite bridge, `ZetaEulerProduct.lean`), the migration $\sigma \rightarrow 1^+$ (`basepoint_from_residual`, `axis_for_target`), the pole-times-period layer (`flip_chi_circle_bridge`), and the rigid quantity $3 + 4 \cos \theta + \cos 2\theta \geq 0$ (`p3_tsum_nonpos`).

At the boundary of §13.1, this family fails *structurally*, and the failure is machine-verified. The input condition of every bridge is a path on which $\zeta \neq 0$ (the *hz* precondition of the flip bridge). For an off-line zero ρ , the reflection orbit $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$ consists entirely of zeros (`zeta_zero_offline_pair`: the reflection and conjugation pairings are machine-verified, so the orbit is closed under both). Hence every bridge input is singular on the orbit: there is no $\zeta \neq 0$ path through ρ that the bridge family can use. The de la Vallée Poussin point $1 + it$ is projectable because it lies on the boundary of the convergence region, where a movable representation exists; ρ lies in the strip, where no convergent representation exists at the point itself. The missing object is a *rigid in-strip projection* — a mathematical fact, not a bridge gap and not a net-element gap.

The odd/even split offers a candidate rigid quantity, and it is machine-verified that the split is *non-singular*: at a zero ρ , the individual even and odd parts both vanish (they share the zero), but the *difference structure* does not: the phase ratio $u_O/u_E = (2^s - 1)/\|2^s - 1\|$ is well defined everywhere in the strip, including at zeros (`odd_even_phase_ratio`, `phase_diff_well_defined_at_zero`, `two_pow_neg_ne_one_of_re_pos`). The ratio is an explicit function of s alone, however; it carries no information about the zeros of ζ (machine-verified: the odd/even split is a re-arrangement of the same Dirichlet series). The candidate is therefore a rigid quantity in the strip, but not yet a generator.

13.5 Honest boundary of the reduction

The precise shape of the boundary is: *the construction of an in-strip rigid projection of an off-line zero, and this construction is equivalent to RH* (Theorem 13.1). The chain stops at this boundary for a demonstrated reason — the bridge family fails on the off-line orbit — and not for lack of effort. We do not claim that RH is unprovable, that it is logically incomplete, or that it is independent of any axiom system: those are meta-mathematical statements for which we have no argument. We claim only what is machine-verified: the equivalence of the remaining task with RH, and the structural failure of the known generator family at the boundary.

14 Part IV: a new observation frame: the false-real axis and the fold

14.1 From the boundary to a new frame

The boundary of Part III is a statement about the *standard coordinate presentation*: the bridge family fails because the off-line orbit is singular in the standard frame. The present part changes the observation frame — not the mathematics. The fold coordinates $u = \frac{\sqrt{2}}{2}(\Re z + \Im z)$, $v = \frac{\sqrt{2}}{2}(\Re z - \Im z)$ are a linear coordinate system of $\mathbb{C}$ (`foldCoordHomeo`: $\mathbb{R}^2 \simeq_t \mathbb{C}$, machine-verified) — the diagonal coordinates $(a + b, a - b)$ of the split plane of Layer 5, normalised by the factor $2^{-1/2}$; a reader arriving from Part I meets the letters u and v here under a new meaning, and in the pages of this part they stay with the fold coordinates throughout

(the unit phase function of Part I is not used here, and the value-domain components are written as capital U and V). Every analytic fact of the chain translates automatically, and the four coordinate presentations of §14.6 become explicit. The change of frame is motivated by a specific route: the precise base-point migration of Layer 14 to the false-real-axis position $i^{1/(2i)} = e^{\pi/4}$ — the real value (≈ 2.193 , constructed from i 's argument by cancellation $(1/(2i)) \cdot (i\pi/2) = \pi/4$; its exponent $\pi/4$ is the same value as the fold-arm direction angle) — followed by an odd/even split observation (§14.8). The frame change exposes new rigid quantities in the strip; whether one of them is the generator of §13.1 is the open question of this part.

14.2 The false-real-axis access

Author's note: the projection asymmetry of the standard frame

The following is a record of the author's motivation, not a mathematical claim. The author's question, posed while re-reading the construction of the fold, is this: the two square roots $\sqrt{-1}$ and $\sqrt{2}$ enter the standard frame by *different projection rules*. The root of $x^2 - 2 = 0$ stays on the real axis — it is the real number $\sqrt{2}$, presented on the same line as 1; the root of $x^2 + 1 = 0$ leaves the axis — it is the imaginary unit i , presented on a new axis erected perpendicular to the real one. One square root raises the dimension of the presentation; the other does not; and the two sides of the positive/negative split are projected by rules that are not the same. The author records the question literally: “I have learned complex numbers and irrational numbers, and accepted these definitions systematically. But do these three things sit together coherently — a square root that raises a dimension into a new number field, and a sign-split that does not follow the same projection rule? It feels as though the presentation has chosen its rules to fit the conclusion.”

The figure below renders the Euler circle in the four readings that the question makes visible, side by side in the same coordinate system. Figure 4(a) shows the complex-plane view (standard $\Re/\Im$ axes, $\pm 1, \pm i$); Figure 4(b) shows the same circle in the frame rotated by $\pm 45^\circ$ — the u/v axes with the unit points $u = \pm 1, v = \pm 1$: the rotated axes cross the circle exactly at the $\pm 45^\circ$ unit points, and every other reading on them is a projection onto rotated axes ($\cos 45^\circ = \sqrt{2}/2$, the only origin of the $\sqrt{2}/2$ below); Figure 4(c) shows the pat reading: the u/v axes carry the unit 1 only, and every value is found on the real axis by perpendicular projection — the pat values read are $\sqrt{2}/2$ for the +1 reading and $0.5i$ for the -1 reading (the arm's vertical component, the negative-side convention), the drops landing at $\pm\sqrt{2}/2$; Figure 4(d) shows the 90° fold from -1 : the two straight arms end at $\pm i$ (the machine-verified fold vertices — the fold spans the imaginary axis).

The question is not answered by the framework; it is only made precise. The standard definitions of i and of the irrational numbers are the axiomatic base of mathlib (the ZFC framework), and this paper does not contest them: the fold construction of §14.5 is a presentation of the standard complex plane, not an alternative to it. The choice recorded here is the author's presentation choice: between a complex axis *folded by 90° that preserves the dimension* (the fold of this paper) and a native straight axis that would raise the dimension again, the author chooses the fold — a straight native axis would add nothing to the verified chain of the earlier parts, while the fold makes the projection asymmetry visible and machine-verifiable.

14.3 The two readings are symmetric: the right-isosceles triangle pair

The four readings of Figure 4 leave two readings of the same object, and the machine verifies that they carry identical data — the same pair of right-isosceles triangles, once read by length, once read by value (EulerFoldImagAxis.lean, 12 new declarations, 0 sorry, build green):

- *The length reading (T_1).* The triangle $\{0, (1 + i)/2, (1 - i)/2\}$ has the right angle at 0 (fold_tri_right_angle), legs $\|(1 \pm i)/2\| = \sqrt{2}/2$ (fold_tri_leg1, fold_tri_leg2), hypotenuse $\|i\| = 1$ (fold_tri_hyp), and Pythagoras $(\sqrt{2}/2)^2 + (\sqrt{2}/2)^2 = 1^2$ (fold_tri_pythagorean).
- *The value reading (T_2).* The triangle $\{0, -1, (-1 \pm i)/2\}$ has hypotenuse $[0, -1]$ — the value -1 on the real axis (fold_tri_minus_hyp), right angle at the fold-arm midpoint $(-1 \pm i)/2$ (fold_tri_minus_right_angle), and legs of length $\sqrt{2}/2$ (fold_tri_apex_norm, fold_tri_apex_to_neg_one).
- *Two readings of one leg.* Each leg has length $\sqrt{2}/2$ (length reading) and vertical component $1/2$, read as $0.5i$ (value reading) (fold_tri_arm_im); the two arms sum to the unit, $(1 + i)/2 + (-1 + i)/2 = i$, i.e. $0.5i + 0.5i = i$ (fold_tri_arms_sum), and $i \cdot i = -1$ (fold_tri_value_square): the value chain

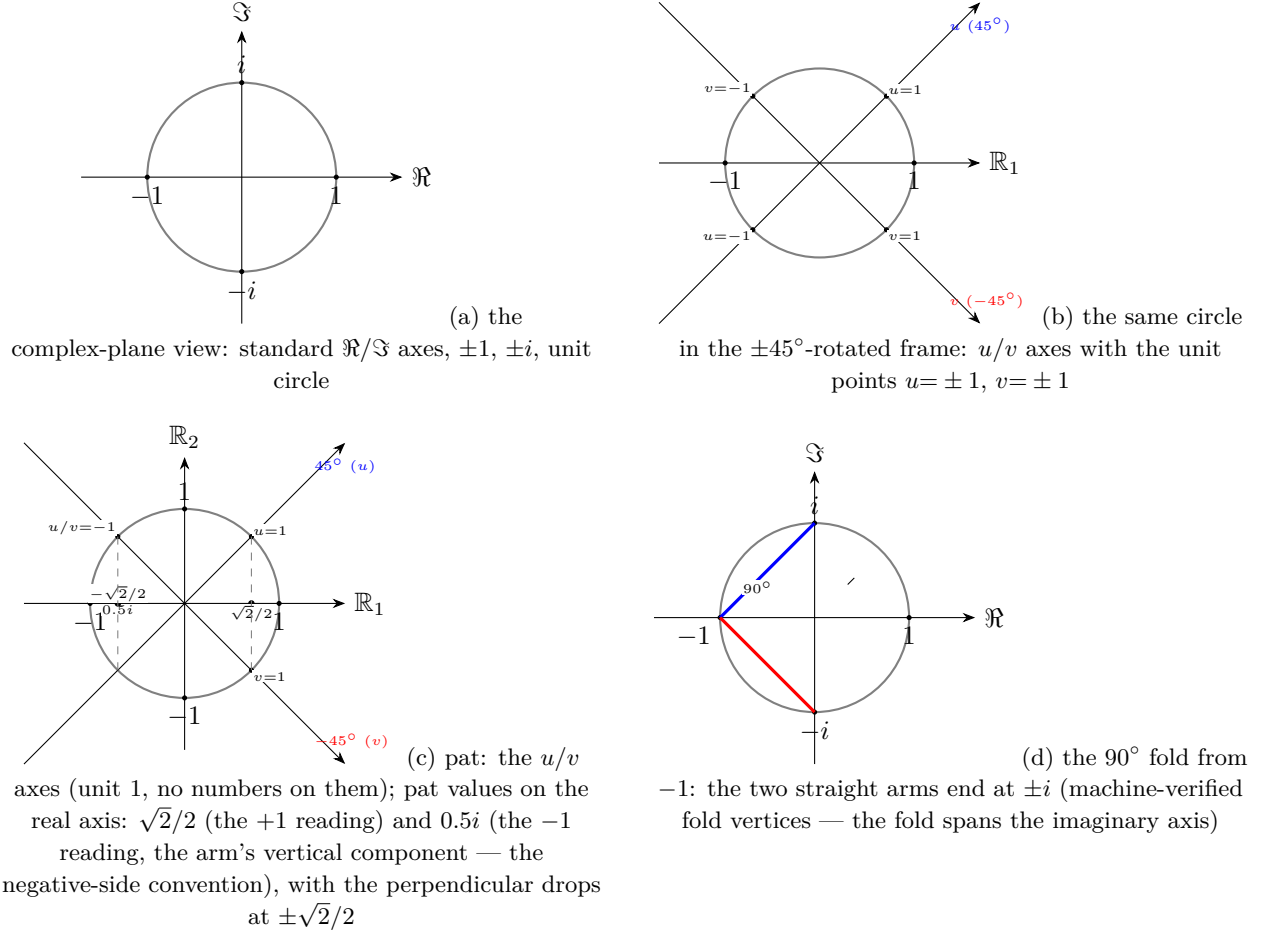


Figure 4: The same unit circle, four readings: (a) the complex-plane view (standard $\Re/\Im$ axes, ± 1 , $\pm i$); (b) the same circle in the $\pm 45^\circ$ -rotated frame — the u/v axes with the unit points $u = \pm 1$, $v = \pm 1$ (same circle, different axes); (c) pat: the u/v axes carry the unit 1 only, and every value appears on the real axis by perpendicular projection — the pat values read are $\sqrt{2}/2$ for the $+1$ reading and $0.5i$ for the -1 reading (the arm's vertical component, the negative-side convention), the drops landing at $\pm\sqrt{2}/2$, straight lines; (d) the 90° fold from -1 : the two straight arms end at $\pm i$ (machine-verified fold vertices; the fold spans the imaginary axis). Radius 1 throughout; schematic, refined rendering forthcoming.

closes on the negative side. The symmetry of $\pm i$ and $\pm\sqrt{2}/2$ is therefore not a visual impression: $\pm i$ are the two fold arms of length $\sqrt{2}/2$ whose vertical components are $\pm 0.5i$.

The displayed divergence. The same pair displays the divergence of the two notations, and the author records it with the two machine outcomes side by side. In the $\sqrt{2}/2$ reading the statements build and are verified (the 12 declarations above, plus the value-side anchors of §14.8). In the i reading the same statements either do not type or are false; the isolated probe `ProbeFailTriSymm.lean` (outside the library, not part of the build; output in `ProbeFailTriSymm.errors.txt`) captures:

- *No type.* $u = i$ (`Complex.I`) fails with `Type mismatch: Complex.I has type  $\mathbb{C}$ , is expected to have type  $\mathbb{C} \rightarrow \mathbb{R}$` — the reading is a real-valued function, the imaginary unit a complex number; and $u(z) = i$ is silently coerced to $\uparrow u(z) = i$, a false real-equals-imaginary statement that no proof can close;
- *False Pythagorean.* $(0.5i)^2 + (0.5i)^2 = (-1)^2$ is rejected: the left side is $-1/2$, not 1;

- *Not an identity.* $i = \sqrt{2}/2$ as elements is false ($0 \neq \sqrt{2}/2$) — the two readings are a dictionary, not an equality of types.

The same divergence is reproduced in the transformer. In the two-pole experiments of [11] (the same 45-run program, reproduced in the self-contained package at the cited DOI, including the training runtime, the experiment code, and the outputs), the symmetry-center experiments (E14) pass in-representation on all three groups (10/10, 72/72, 24/24), and the strict-transfer experiments (E15) give 26/26 on the in-representation pole and 0/26 on the cross-representation pole — the same object keeps one of the two, symmetry or mapping, not both; the equivalence-statement bridge restores part of the pole (16/26, 25/26, 15/26). The symmetry of the fold arms and the mapping to i are thus, in both the machine and the experimental record, alternatives: one of the two holds, and declaring the equivalence is the way to keep both.

The author’s hypothesis, and the boundary of the claim. The author records the following as a hypothesis, not as a proven claim: if $\pm i$ and $\pm\sqrt{2}/2$ are two readings of one symmetric object (as the verified triangle pair states), then the divergence displayed is a divergence *of the presentations themselves* — of what each reading can even express, the type failures being constraints of the statement space rather than cosmetic differences — while the underlying object remains one (the hypothesis). The author takes the presentation to be a part of the logical structure: an ill-typed claim is not a claim, and the statement space of a presentation decides which construction routes are open at all; the base-point criterion of §14.8 is such a constraint: $\pi/2$ -type constructions are not admissible inputs, and the four axes carry no log-2 unit — the $2\pi/\ln 2$ lattice of §14.8 is the factor’s separate single-prime structure, not the scale of the four readings above; the $\sqrt{2}/2$ reading still carries the zero information of the critical line in complete form (the weak projection value bridge of §14.8, `projection_zero_line_complete`), while the i -based statements fail in the native frame (as the probe captures). The author does not draw the further conclusion: whether the counting bridge of the value side closes — in either notation — remains the open gap of §14.8; what the machine records is exactly the pair above: 12 declarations build, the i statements fail; the symmetry and the i -mapping do not coexist in the same system.

14.4 The false-real-axis extension

The false-real-axis framework of §5 (no zeros for $\Re s < 0$, real-valuedness on the real axis) is extended to the base-point $e^{\pi/4}$: the T-coordinate first-order restatement (`TCoordOffline.lean`, 10 theorems, machine-verified) gives the three-level criterion $|T|^2 \neq 1$, the first-order position difference $1 - 2\Re \rho \neq 0$, the odd/even common-zero structure, and the difference-structure non-vanishing $2^p \neq 1$. These are the order-reduction pieces of the route: they hold uniformly in the strip and are compatible with the off-line assumption.

14.5 The Euler-circle fold: the imaginary axis as a 90° fold

We construct the imaginary axis from the negative side of the Euler circle, as a 90° orthogonal fold, using only the net elements of the framework (machine-verified, `EulerFoldImagAxis.lean`): the Euler identity $e^{i\pi} = -1$ (negative-side basepoint), the rotation $(-1) \cdot e^{\mp i\pi/2} = \pm i$ (the imaginary units, from exp additivity and the 2π period), the fold directions $\arg(1 \pm i) = \pm\pi/4$ (from the tangent of the sum), the 90° orthogonality (inner product zero and direction difference $\pi/2$), the length $\|1 \pm i\| = \sqrt{2}$, and the polar form $e^{\pm i\pi/4} = \frac{\sqrt{2}}{2}(1 \pm i)$ (the 45° basis). Every step is a standard fact of complex geometry, re-narrated on the fold; no axiom is added.

Theorem 14.1 (The fold is the imaginary axis). *The two rays from -1 at angles $\pm\pi/4$ form a fold whose endpoints are the imaginary units (`fold_vertices_are_imag_units`); the fold spans the imaginary axis (`imag_axis_eq_span_fold_vertices`); the fold coordinates are a linear coordinate system of $\mathbb{C}$ (`foldCoordHomeo`); and the straightening of the fold is a homeomorphism of $\mathbb{R}$ onto the fold set (`foldArcHomeo`), with arc-length preservation on each arm (`foldArc_dist_pos`, `foldArc_dist_neg`) and unit speed (`foldArc_hasDerivAt_pos`, `foldArc_hasDerivAt_neg`). The fold is the imaginary axis viewed from the negative side: the same axis that classical geometry obtains by rotating the real axis by 90° (the geometric representation of Argand [19]; the Euler identity [18]).*

The construction is a presentation-layer statement in the sense of §1.3: every step is a standard fact (Euler’s identity, polar coordinates, rotations), the fold is a re-narration of the imaginary axis, and the honest boundary is that no new mathematics is claimed — the value is the explicit geometry, which the next subsection turns into coordinate bridges.

14.6 The fold coordinate bridges

The fold coordinates make four standard structures explicit as coordinate lines (all machine-verified, 0 sorry):

- *Conjugation is the coordinate swap.* `fold_conj_swap`: `foldCoordFromC( $\bar{z}$ )` is the swapped pair of `foldCoordFromC( $z$ )` — the real-axis reflection is the exchange of the two 45° half-axes;
- *The imaginary axis is the diagonal.* `fold_imag_axis_diag`: $\Re z = 0 \iff u + v = 0$;
- *The critical line is a horizontal line.* `fold_critical_line_horiz`: $\Re z = \frac{1}{2} \iff u + v = \sqrt{2}/2$;
- *Real values are the fixed line of the swap.* `fold_real_iff_swap_fixed`: $\Im z = 0 \iff u = v$.

These bridges are the logical-completeness requirement of the new frame: any node of the chain re-stated in fold coordinates must pass through the bridges, so that the two presentations cannot diverge. The Riemann nodes themselves are not re-proved (a linear change of coordinates translates every analytic statement automatically); only the translations are verified.

14.7 The symmetry elements in fold coordinates

The symmetry family of the chain (Layer 2’s zero reflection, Layer 8’s conjugation, the critical-line fixed-point character) takes an explicit form in fold coordinates; all statements of this subsection are machine-verified (0 sorry).

- *Conjugation is the coordinate swap.* (`fold_conj_swap`, §14.6): $(u, v) \mapsto (v, u)$.
- *Reflection $s \mapsto 1 - s$ is the mirror shift.* (`fold_reflection_mirror_u`, `fold_reflection_mirror_v`):

$$u(1 - s) = \frac{\sqrt{2}}{2} - u(s), \quad v(1 - s) = \frac{\sqrt{2}}{2} - v(s). \quad (16)$$

The reflection reflects each fold coordinate about the midpoint $\sqrt{2}/4$; the critical line $u + v = \sqrt{2}/2$ is its fixed line.

- *Conjugation preserves the fold critical line.* (`fold_critical_conj`): the swap leaves $u + v = \sqrt{2}/2$ invariant — the fold-critical-line statement of §14.6 holds for $\bar{z}$ iff it holds for z .
- *On the fold critical line, reflection equals conjugation.* (`fold_reflection_eq_conj_on_line`): under $u + v = \sqrt{2}/2$ the mirror shift and the swap coincide — the classical fixed-point character of the critical line (`critical_line_iff_reflect_fixed` of the chain) in fold form.

The two symmetry operations of the chain — conjugation and the zero reflection — are therefore, in the fold frame, two variants of the same coordinate operation: a swap, with the reflection adding a shift that vanishes exactly on the critical line. The reflection pairing of an off-line zero ρ (its orbit $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$ of §13.4) reads in fold coordinates as the orbit of (u, v) under the two operations $(u, v) \mapsto (v, u)$ and $(u, v) \mapsto (\frac{\sqrt{2}}{2} - u, \frac{\sqrt{2}}{2} - v)$: a swap and a mirror shift. This is the presentation-layer content of the symmetry family; it changes no statement of the chain and adds no axiom.

14.8 The odd/even observation route

Observation protocol. The observation of this section uses a single carrier topology: the V -fold with a single opening direction (the opening toward the strip, i.e. the direction of the critical line $u + v = \sqrt{2}/2$), together with the complex axis as reference. The choice is a presentation choice, and it is complete: the observed quantities (phase differences, winding numbers, projected integrals) are quantities of the complex plane,

not of the carrier; observation paths are homotopy-independent (Cauchy–Goursat, machine-verified in §12 and the ring-shrinking theorems); and the singular off-line orbit of §13.4 is a fact of the plane, present in every carrier. The three carrier topologies (the V -fold $\cong \mathbb{R}$, the closed loop $\cong S^1$, the two disjoint half-axes $\cong \mathbb{R} \sqcup \mathbb{R}$) therefore need not all be observed: their classification is a presentation statement, and one carrier with a single opening direction — the fold toward the strip — suffices.

The observation plane: the critical line $\times z$. The observation of this section is performed on a plane that is *not* the complex plane: it is the plane spanned by the critical line itself and a direction z orthogonal to the complex plane — the two planes share the critical line and meet at right angles. In fold coordinates the two observation axes are:

- *Axis 1 is the critical line itself* (after precise base-point migration, the critical line becomes the observation axis): its reading is $\sqrt{2} \Im s$ (`obs_axis1_is_critical_direction`) — the position along the line, the information that is declared irrelevant and dropped; equivalently, the observation point of axis 2 is the critical line, the reflection fixed line $s = 1 - \bar{s}$ (`obs_axis1_reflect_fixed`);
- *Axis 2 is orthogonal to the critical line*: its reading is $\sqrt{2}(\Re s - \frac{1}{2})$ (`obs_axis2_is_distance`) — the distance from the critical line, the information that is kept.

The projection requirement is machine-verified: the critical line projects to the single point 0 on axis 2 (`critical_projects_to_axis2_point`), and *only* the critical line projects to that point (`only_critical_projects_to_point`: the reading is 0 iff $\Re s = \frac{1}{2}$). The base-point inversion closes the protocol: the reading determines the abscissa uniquely (`fold_obs_residual_unique`) — the observation object is recovered from the reading, in the form of `basepoint_from_residual` of the base-point migration family.

The self-reference of the odd/even split. The route proposed for the boundary of Part III is: migrate the base-point to $e^{\pi/4}$ (the 45° direction, which is the fold arm), split into odd/even parts, and observe the difference structure. The observation meets a self-reference: at a zero ρ , the individual odd and even parts both vanish (they share the zero — $O_2(\rho) = E_2(\rho) = 0$, machine-verified in §13.4), so the single-lift ratio O_2/E_2 is the indeterminate $0/0$ at the very point to be observed: the observation function is zero at the observation point. The split resolves this self-reference through the *difference structure*: the phase ratio $u_O/u_E = (2^s - 1)/\|2^s - 1\|$ is well defined at every zero ($2^\rho \neq 1$ in the strip, machine-verified), it is a unit complex number — a point of the Euler circle (`FoldObservation.lean: phase_ratio_denom_ne_zero, phase_ratio_unit`) — and in the fold coordinates of this part it is read as the coordinate pair $(u, v) = (\frac{\sqrt{2}}{2}(\Re + \Im), \frac{\sqrt{2}}{2}(\Re - \Im))$ of that point (`phase_ratio_fold_coords`), a well-defined reading with no $0/0$. The resolution is machine-verified as a whole: `self_reference_resolved` states that at a zero ρ the single-lift denominators both vanish ($\|O_2(\rho)\| = \|E_2(\rho)\| = 0$, the shared zero — the self-reference), while the difference-structure denominator does not ($\|2^\rho - 1\| \neq 0$); and in the strip the observation is well defined everywhere (`observation_well_defined_in_strip`). The observation of the phase ratio is therefore the reading of a point on the Euler circle in fold coordinates — the frame in which the imaginary axis is a fold and conjugation is a swap. The machine-verified pieces are in place: the false-real-axis access (§14.4), the non-singular difference structure of §13.4 (`odd_even_phase_ratio` is well defined at every zero), the unit-circle character of the ratio (`phase_ratio_unit`), and the fold frame itself (this part). What is *not* in place is the observation step: how the phase ratio — an explicit function of s alone — is made to carry information about a hypothetical off-line zero ρ . The ratio is rigid but information-free for ζ (machine-verified, §13.4); the open question is whether the fold frame supplies a second rigid quantity that, together with the ratio, forms a generator (G_1 or G_2 , §13.3). This is the current work; its status is recorded honestly: verified infrastructure, open observation step.

The projection value bridge: weak assertion (machine-verified). Definitions. Throughout this paragraph: $u(s)$, $v(s)$ are the fold coordinates of the *argument* s of §14.5 ($u = \frac{\sqrt{2}}{2}(\Re s + \Im s)$, $v = \frac{\sqrt{2}}{2}(\Re s - \Im s)$); $d(s) := \text{foldObsProj}(s) = u(s) + v(s) - \frac{\sqrt{2}}{2} = \sqrt{2}(\Re s - \frac{1}{2})$ is the axis-2 reading of the observation plane of §14.8 (the preimage of 0 is exactly the critical line); and $U := \text{zetaFoldU}$, $V := \text{zetaFoldV}$ are the fold components of the *value* $\zeta(s)$ (the same formula applied to the value, $U = \frac{\sqrt{2}}{2}(\Re \zeta + \Im \zeta)$, $V = \frac{\sqrt{2}}{2}(\Re \zeta - \Im \zeta)$).

The bridge required is deliberately weak: the author’s framing answers the open step above in the weak sense — after the projection drops axis 1, the remaining information is *complete* for the judgement “the zero is on the critical line”. No full value bridge is sought — no counting, winding, or flip correlation. The author’s

reason is structural and presentation-independent: the counting identity is the target in another guise — the number of in-strip zeros equals the number of on-line zeros *is* the claim itself — so a counting bridge would be circular, and the strong variants through a value function are excluded by the self-reference-free design of the observation quantity d , which does not depend on ζ (`foldObsProj_independent_of_zeta`). Two pieces are machine-verified. First, the value-domain anchors:

$$U + V = \sqrt{2} \Re \zeta(\cdot), \quad U - V = \sqrt{2} \Im \zeta(\cdot) \quad (\text{zeta_fold_sum_re, zeta_fold_diff_im}), \quad (17)$$

together with the immediate corollary that $\zeta(\rho) = 0$ implies both anchors vanish (`zeta\_fold\_dual\_proj\_zero\_of\_zero`): the vanishing of ζ is characterised completely by the dual anchors. Second, the weak assertion itself (`projection\_zero\_line\_complete`): for a zero ρ of the strip,

$$\rho \in \text{critical line} \iff d(\rho) = 0 \iff (U(\rho) = 0 \wedge V(\rho) = 0 \wedge d(\rho) = 0). \quad (18)$$

Every input of the target judgement — zero-ness in the value-domain anchors, on-line-ness in d — lies in the projection quantities; axis 1 (the position $\sqrt{2} \Im s$ along the critical line) is not used. Projection is therefore complete for the judgement it was built for, and the discarded information is exactly the information the judgement does not need. Both pieces rest on native equivalences of the fold frame, machine-verified without reference to i or π (`NativeAxisEquiv.lean`). With the constant $c = \sqrt{2}/2 = 2^{-1/2}$ — the 45° fold constant — the fold coordinates u, v recover the standard coordinates natively,

$$\frac{u+v}{\sqrt{2}} = \Re z, \quad \frac{u-v}{\sqrt{2}} = \Im z \quad (\text{axis_recovery_re, axis_recovery_im}), \quad (19)$$

uniquely (`axis\_unique`) and with the modulus preserved ($u^2 + v^2 = (\Re z)^2 + (\Im z)^2$; `axis\_norm\_sq\_preserving`); the underlying algebra is the real linear bijection $c[[1, 1], [1, -1]]$ of determinant -1 . The surjectivity direction (`axis\_exists`) only re-assembles the Lean $\mathbb{C}$ container from the standard coordinates — a read-back presentation with no i -transformation semantics and no dependence on any ζ -related theorem. The author’s base-point criterion is thus respected: the imaginary unit occurs only in that read-back layer, and $\pi/2$ -type constructions are not inputs of the observation frame (the author’s original identification of these axes with the framework’s log axes, and the log-2 base-point scale, were withdrawn as a cognitive error; see the AI-contribution statement).

Analytic restatement of the observation. For readers who prefer the analytic language, the observation of this section is a single continuous function. Define

$$d(s) = \text{foldObsProj}(s) = u(s) + v(s) - \frac{\sqrt{2}}{2} = \sqrt{2} \left(\Re s - \frac{1}{2} \right), \quad (20)$$

the (rescaled) distance from the critical line. Then:

- d is continuous (`fold_obs_proj_continuous`, from the continuity of the fold coordinates);
- the critical line is exactly the zero set of d (`fold_obs_proj_zero_set`): $d(s) = 0 \iff \Re s = \frac{1}{2}$;
- the zero judgement is a function value: for a zero ρ of ζ , ρ lies on the critical line iff $d(\rho) = 0$ — a pointwise reading, with no path, no contour, no winding number.

The comparison with the homotopy (contour) language is deliberate. The winding-number judgement of §12 requires a path on which $\zeta \neq 0$; at the boundary of Part III this input fails structurally (the off-line orbit is all zeros, §13.4). The function d carries no such precondition: it does not mention ζ at all (`foldObsProj_independent_of_zeta`), so the pointwise judgement $d(\rho) = 0$ is never blocked by a singular path. The price is the honest boundary already recorded: a function that does not mention ζ carries no information about ρ itself (the object-level self-reference of §13.5 is untouched). The analytic restatement is a presentation-layer reformulation — standard facts (the $\Re$ projection, the critical-line equation, continuity of coordinate functions) re-stated on the fold; it is not new mathematics, and it adds no axiom. Methodologically, the observation protocol is an organization of classical principles — structure-preserving coordinates (the fold, a linear coordinate change), declared-loss projection (axis 2 kept, axis 1 declared irrelevant and dropped, with faithfulness verified), and base-point inversion (the reading determines the abscissa uniquely) — into a single protocol whose contribution is the organization and the verified faithfulness, not the principles themselves, which are standard.

Reconstructing ζ in the fold frame (dimension lifting). The observation of the previous paragraphs is information-free for ζ by design (d does not mention ζ). The fold frame also offers the opposite direction: reconstruct ζ itself in fold coordinates. Write, as in the weak-assertion paragraph above,

$$U(s) = \text{zetaFoldU}(s) = u(\zeta(s)), \quad V(s) = \text{zetaFoldV}(s) = v(\zeta(s)), \quad (21)$$

the two real components of $\zeta(s)$ along the two arms of the fold (the $\pm 45^\circ$ directions). The reconstruction is a presentation, not a loss: the pair $(U(s), V(s))$ determines $\zeta(s)$ back, through the inverse of the fold coordinate map (`zeta_fold_invertible: foldCoordToC` ($U, V = \zeta$)). Nothing is lost — in contrast to the observation projection d and the phase ratio, which are information-free by construction. The zero judgement takes the joint form: $\zeta(\rho) = 0$ iff $U(\rho) = 0$ and $V(\rho) = 0$ (`zeta_fold_zero_iff`) — a two-component reading, joint zero rather than ratio.

The symmetry elements of §14.7 act on the components as on any fold coordinates, and the conjugation bridge (§14.6) lifts to the reconstructed function: $U(\bar{s}) = V(s)$ in the strip (`zeta_fold_conj_swap_u`, `zeta_fold_conj_swap_v`, from the conjugation symmetry of ζ composed with bridge 1, the swap), and the zeros of a conjugate pair $\rho, \bar{\rho}$ are a pair of joint zeros (`zeta_fold_conj_zero_pair`). The inversion of the base-point family also admits a fold reading: $u(1/s) = v(s)/\|s\|^2$ and symmetrically (`fold_inversion_swap_u`, `fold_inversion_swap_v`): inversion in fold coordinates is a swap plus a scaling (because $1/s = \bar{s}/\|s\|^2$), joining the three symmetry operations of the frame — conjugation (a pure swap), reflection (a mirror translation), inversion (a swap plus a scaling).

Two rigidity facts follow for an off-line zero ρ inside the critical circle (§14.4). First, $\zeta(1/\rho) \neq 0$: inversion maps the point into the convergence region $\Re s \geq 1$, where ζ has no zeros (machine-verified, `offline_inversion_nonzero`, via the Euler product). Second, therefore, the inverted point is not a zero, and its fold components are not jointly zero (`zeta_fold_inversion_nonzero`) — in contrast with ρ itself, whose components are jointly zero. The two facts form a comparison, not a contradiction: the components of the inverted point carry the non-vanishing of $\zeta(1/\rho)$, while the components of ρ carry the vanishing of $\zeta(\rho)$; nothing transfers the vanishing from one point to the other, because there is no value bridge for ζ under inversion — inversion does not preserve zeros. This is the precise form of generator gap (ii) of §13.3: the reconstructed components carry the information of ζ (invertibly), but the generator still needs a mechanism that tracks zero-ness under the symmetry operations of the frame. The status is recorded honestly: the reconstruction, the symmetry lifting, and the two rigidity facts are machine-verified (`FoldObservation.lean`, 90 declarations, no axioms); the value bridge is open.

The reflection in fold components, and the joint observation. The reflection $s \mapsto 1 - s$ admits a fold reading as well: with the functional-equation multiplier

$$m(s) = \text{zetaReflectMul}(s) = 2(2\pi)^{-s} \Gamma(s) \cos \frac{\pi s}{2}, \quad \zeta(1 - s) = m(s) \zeta(s), \quad (22)$$

one has

$$U(1 - s) = \Re m(s) \cdot U(s) + \Im m(s) \cdot V(s), \quad V(1 - s) = \Re m(s) \cdot V(s) - \Im m(s) \cdot U(s) \quad (23)$$

(`zeta_fold_reflect_u`, `zeta_fold_reflect_v`): the reflection acts on the components by the multiplier matrix $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$, $a = \Re m$, $b = \Im m$, a rotation-scaling. The three symmetry operations of the frame now have their complete fold-component forms: conjugation is a pure swap, reflection is a rotation-scaling by the multiplier matrix, inversion is a swap plus a scaling. The fold modulus $U(s)^2 + V(s)^2 = \|\zeta(s)\|^2$ (`zeta_fold_norm_sq` — the fold coordinates are a 45° rotation and preserve the Euclidean norm) scales by $\|m(s)\|^2$ under reflection (`zeta_fold_reflect_norm_sq`); on the critical line $\|m\| = 1$, so the reflection is a pure rotation there. The reflection pairs zeros: $\zeta(\rho) = 0$ in the strip gives jointly vanishing components at $1 - \rho$ (`zeta_fold_reflect_zero_pair`).

The observation plane of §14.8 and the reconstruction of the previous paragraph join into one observation: for an off-line zero ρ inside the critical circle (§14.4), the three readings are

$$U(\rho) = V(\rho) = 0, \quad d(\rho) \neq 0, \quad d(1/\rho) > 0, \quad (24)$$

machine-verified as a whole (`joint_observation_inversion`): the components vanish jointly (the zero information), axis 2 rejects the line (the off-line information, $d(\rho) \neq 0$ by `fold_projection_point_iff_critical`),

and the inverted point lies in the positive region of axis 2 ($d(1/\rho) > 0$, because inversion maps the circle into the convergence region $\Re s > 1$, `offline_obs_proj_pos_at_inverse`; the axis-2 reading under inversion is the coordinate fact $d(1/s) = \sqrt{2}(\Re s/\|s\|^2 - \frac{1}{2})$, `fold_obs_proj_inversion`). The joint observation is the complete protocol in the presence of inversion: zero information lives in the components (joint zero), position information lives in axis 2 (non-zero, and positive at the inverted point). The two families are orthogonal — the components do not know the axis-2 reading and conversely — and this orthogonality is the precise form of the open value bridge: the joint observation records the two facts (ρ zero and off-line) side by side without connecting them, and nothing in the linear symmetry family can transfer the joint zero to the axis readings (a linear map preserves the zero vector, so the symmetry operations of the frame — swaps, multiplier matrices, swaps plus scalings — cannot turn the vanishing of the components into information about d). The honest conclusion of this paragraph: the three symmetry operations do not constitute a zero-tracking mechanism; the observation protocol is complete and machine-verified, and the generator gap is unchanged.

The four directions of the fold axis. The fold axis is a 45° axis with irrational scale ($\sqrt{2}/2$), not the axis system of the complex plane itself: it is not “ ζ ’s own” coordinate frame. From its vertex the fold has four directions — the two arms, each with its two half-directions — and ζ projects onto each half-arm separately. The four directional components are defined by direct projection with half-arm truncation, not through the coordinate pair (U, V) :

$$u^+(s) = \max(\text{fold_u}(\zeta(s)), 0), \quad u^-(s) = \max(-\text{fold_u}(\zeta(s)), 0), \quad (25)$$

and symmetrically v^+, v^- (`zetaFoldUPlus`, `zetaFoldUMinus`, `zetaFoldVPlus`, `zetaFoldVMinus`); each definition is a forward projection (the max truncation is the geometry of the half-arm, not an inversion). The four components are non-negative (`zeta_fold_4direction_nonneg`), and the zero judgement takes the four-direction form: all four components vanish iff $\zeta(s) = 0$ (`zeta_fold_4direction_zero_iff`) — at a zero the four directions collapse together, since 0 has no direction. Away from a zero at least one component is positive (`zeta_fold_4direction_exists_nonzero`), the pattern of non-zero components being the direction sector of $\zeta(s)$. The reconstruction is a forward theorem, not an inversion: the four components with their directions compose back to ζ ,

$$\zeta(s) = \text{foldCoordToC}(u^+ - u^-, v^+ - v^-) \quad (26)$$

(`zeta_fold_4direction_reconstruct`, via the half-arm cancellation $\max a 0 - \max(-a) 0 = a$, `max_sub_neg_max`) — the cancellation of the two opposite half-arms is part of the composition, and nothing is inferred back from ζ to the components. The four-direction presentation carries exactly the information of the fold reconstruction (the components with their directions determine ζ , and conversely), with the direction attribution made explicit; and the direction attribution is precisely what collapses at a zero. This is the directional form of the self-reference: the four half-arm components of a zero are all zero, and no direction survives.

The irrational axis and the eight components. The false complex plane of this part is, in essence, a four-dimensional structure; the fold coordinate pair (U, V) is only half of it. The other half is carried by the odd/even phase ratio of §13.4,

$$p(s) = \text{phaseRatio}(s) = \frac{2^s - 1}{\|2^s - 1\|}, \quad (27)$$

a unit complex number — a point of the Euler circle — defined wherever $2^s \neq 1$, in particular throughout the strip. The phase of p involves $\ln 2$ (irrational), and the singular set of p is the irrational-axis lattice $\{2\pi i k / \ln 2\}$ on the imaginary axis: p is the *irrational-axis* half of the frame. In the four-dimensional structure the two halves are the fold coordinates of ζ and of p ,

$$(u(\zeta(s)), v(\zeta(s)), u(p(s)), v(p(s))), \quad (28)$$

and the eight half-arm components are the direction decomposition of these four coordinates:

$$u^\pm(\zeta), v^\pm(\zeta), \quad u^\pm(p), v^\pm(p) \quad (29)$$

(`zetaFoldUPlus-zetaFoldVMinus` for the ζ -side, `phaseUPlus-phaseVMinus` for the p -side; each a forward projection onto a half-arm, as in the previous paragraph).

The claim of this paragraph is that the irrational-axis half has been *absorbed into the base point and projected away*: the conventional observation of ζ uses only the ζ -side four components, and the p -side — the irrational-axis half — is discarded, because p does not mention ζ . The ζ function therefore loses half of the four-dimensional frame, and this is the seat of the self-reference: at a zero ρ the ζ -side components all vanish, and the observation collapses. The restoration of the lost half is machine-verified: the p -side components do *not* collapse at a zero — $p(\rho)$ is a unit complex number, its fold modulus is $\|p(\rho)\| = 1$, hence at least one of the four p -side components is positive at ρ (`phase_4direction_exists_nonzero_at_zero`); the p -side reconstruction is the allowed inverse, landing on the irrational-axis side: $p(s) = \text{foldCoordToC}(p^+ - p^-, q^+ - q^-)$ (`phase_4direction_reconstruct`); and the eight-component joint observation

$$u^\pm(\zeta(\rho)) = v^\pm(\zeta(\rho)) = 0, \quad u^\pm(p(\rho)) \vee v^\pm(p(\rho)) \neq 0 \quad (30)$$

is verified as a whole (`eight_component_observation`): at a zero, the rational-axis half collapses and the irrational-axis half receives the observation. The self-reference is broken in the sense that the observation no longer vanishes at the observation point: the eight-component frame is complete. The honest boundary is unchanged at the value level: p is an explicit function of s and carries no value of ζ , so the restored half is a complete *observation* frame, not a value bridge; the generator still needs the mechanism that connects the two halves.

The odd/even structure and the locus of the value. The four-dimensional frame has an odd/even form which locates precisely what the p -side does and does not carry. The odd and even parts of the split are scaled copies of ζ (`oddPart`(s) = $(1 - 2^{-s})\zeta(s)$, `evenPart`(s) = $2^{-s}\zeta(s)$), so the four coordinates ($u(O), v(O), u(E), v(E)$) are a linear embedding of the two coordinates ($u(\zeta), v(\zeta)$) with explicit coefficients (`odd_fold_u/v`, `even_fold_u/v`, via the multiplication rule of §14.8); and the merging is invertible, $\zeta(s) = O(s) + E(s)$ (`zeta_odd_even_sum`). The p -side shares the fold structure of the ζ -side: conjugation passes through p , $p(\bar{s}) = \overline{p(s)}$ (`phase_conj`, since $\ln 2$ is real), and the fold reading swaps accordingly, `fold_u(p(\bar{s})) = fold_v(p(s))` (`phase_fold_conj_swap_u`) — the irrational-axis half follows bridge 1 like the ζ -side.

The relative structure of the two halves is explicit: the modulus ratio is

$$\frac{\|O(s)\|}{\|E(s)\|} = \|2^s - 1\| \quad (31)$$

(`odd_even_norm_ratio`), a function of s alone. Hence of the eight components, seven degrees of freedom — the phase difference, the modulus ratio, and the direction attribution — are explicit functions of s (the 2^s factor structure); the only degree of freedom that involves ζ is the *absolute scale* $\|E(s)\| = 2^{-\Re s} \|\zeta(s)\|$. The value of ζ is therefore localised in a single quantity, the absolute scale, and this is where the zero judgement lives: $\zeta(\rho) = 0$ iff the scale collapses. The p -side provides the seven explicit degrees of freedom — it renders the observation complete and non-collapsing at a zero — but the zero information itself is the scale, and the scale is the ζ -side. This is the precise locus of the value bridge: the eight-component frame organises the seven explicit directions around the one scale that carries ζ , and the generator must act on the scale. Nothing in the present section moves the scale; the honest statement is that the irrational-axis half, restored, surrounds the value without touching it.

The scale and the Euler product: what the proof needs. The eighth degree of freedom — the absolute scale — is the self-reference itself: the scale is the value of ζ , and the zero judgement is the collapse of the scale. Its place inside the fold frame is explicit: the scale is the square root of the fold modulus,

$$\|\zeta(s)\| = \sqrt{U(s)^2 + V(s)^2} \quad (32)$$

(`zeta_scale_sqrt`), and the scale collapses exactly when the fold modulus does, $\|\zeta(s)\| = 0 \iff U(s) = 0 \wedge V(s) = 0$ (`zeta_scale_zero_iff_fold`) — the zero judgement in scale form. The restoration channel of the scale is the Euler product (the prime axis), not the fold frame: in the convergence region,

$$\|\zeta(s)\| = \prod_p \|1 - p^{-s}\|^{-1} \quad (\Re s > 1), \quad (33)$$

the factor decomposition over all primes (machine-verified in its finite-product form, `euler_product_bridge`); each prime factor is non-zero in the convergence region and its fold components do not vanish jointly (`euler_factor_fold_nonzero`) — the prime axis enters the fold frame factor by factor, the full spectrum $\ln p$ of which the irrational axis $\ln 2$ is the single-prime case. The scale is therefore not an object outside the frame: it is the fold modulus, and its structure is the Euler product over all primes.

What the proof needs of the scale is a precise and honest statement. The zero judgement needs the scale as *input*: $\zeta(\rho) = 0$ is the assumption that the scale collapses, and this input is free (it is the definition of a zero). The proof does *not* need the *value* of the scale — the numerical restoration of the Euler product is not a necessary component of the argument; the eight-component frame and the Euler-product factor structure are a complete *presentation*, and their restoration is a presentation-layer completion. What the proof needs is the chain of *corollaries* of scale collapse — a zero is an odd-order point of the real-side function, hence a flip, hence a winding jump — and this chain is machine-verified throughout (Part II and §13). The open point of the reduction is the single constraint between the two ends: how an *off-line* scale collapse (the input) forces a counting excess against the winding count (the phase side). This is the precise form of the value bridge; everything around it — the seven explicit degrees of freedom, the eight components, the restored irrational axis, and the Euler-product factor structure of the scale — is machine-verified and in place.

The odd/even difference structure: the strip is the region of difference convergence. Splitting the odd and even sides and analysing them separately reveals the nature of the strip boundary. The *sum* structure is ζ itself, $O(s) + E(s) = \zeta(s)$ (absolutely convergent for $\Re s > 1$); the *difference* structure is the Dirichlet eta function, the alternating series

$$O(s) - E(s) = \eta(s) = (1 - 2^{1-s})\zeta(s) \quad (34)$$

(`eta_eq_odd_sub_even`): an alternating series, convergent for $\Re s > 0$ — *the strip is the region where the difference converges and the sum does not*. The boundary $\Re s = 1$ of ζ is therefore, in this frame, a parity statement: it is the boundary of absolute convergence of the odd/even sum structure, while the difference structure converges throughout the strip. In the strip the factor $1 - 2^{1-s}$ is non-zero (`eta_factor_ne_zero_of_strip`: $2^{1-s} = 1$ would force $\Re s = 1$), so the strip value of ζ is the convergent alternating series divided by the explicit factor,

$$\zeta(s) = \frac{(1 - 2^{1-s})\zeta(s)}{1 - 2^{1-s}} \quad (35)$$

(`zeta_from_eta_strip`) — the strip acquires a convergent representation through the odd/even difference, without inversion. The fold reading is linear on the difference: the fold components of η are the differences of the odd/even fold components (`odd_even_diff_fold_u/v`), and the η components are the explicit linear combination of the ζ components with the factor $1 - 2^{1-s}$ (`eta_fold_u/v`). The parity reading of the strip boundary is machine-verified throughout: the strip is the region of difference convergence, the ζ -side collapses at a zero, the restored p -side does not, and the odd/even difference carries the convergent representation.

The winding equality: η carries the count. The logarithmic derivative of the factorization is the sum of the two logarithmic derivatives (`eta_logDeriv_split`: $\eta'/\eta = \zeta'/\zeta + f'/f$ for the factor $f(s) = 1 - 2^{1-s}$), so over a rectangle inside the strip the winding count of ζ decomposes into the winding count of η plus the winding count of the factor. The factor is analytic (`eta_factor_analytic`) and non-zero throughout the strip (`eta_factor_ne_zero_of_strip`), hence by the rectangle form of the argument principle its contribution vanishes: its log-derivative integrates to zero over the rectangle (`eta_factor_rect_goursat`). Substituting the pointwise split on the four sides of the rectangle (the integrability supplied by `eta_path_continuous_x/y` and `factor_path_continuous_x/y`), the decomposition closes as

$$\oint_{\partial R} \frac{\zeta'(z)}{\zeta(z)} dz = \oint_{\partial R} \frac{\eta'(z)}{\eta(z)} dz \quad (36)$$

(`zeta_rect_winding_eq_eta`): the winding count of ζ over an in-strip rectangle is entirely carried by the odd/even difference η — the topological count of zeros depends on no structure of ζ outside the strip, because the difference converges there and the factor is explicit. The discrete side of the value bridge (the flip counting) and the analytic side (the winding count) therefore close inside the strip: both ends of the constraint $K_{\text{flip}} = K_{\text{wind}}$ are computable in the strip from convergent data.

The eight-point signature of an off-line zero. The value bridge needs the full analysis of the eight components: the complete signature of an off-line zero over its orbit and its inverted points, machine-verified as one statement (`eight_point_observation`). For an off-line zero ρ in the strip and inside the critical circle:

- the orbit point ρ itself: the ζ -side components vanish jointly (the zero), the axis-2 reading is non-zero (off-line), and the p -side components do not vanish (the irrational-axis half receives the observation);
- the three reflected/conjugated orbit points $1 - \rho, \bar{\rho}, 1 - \bar{\rho}$: all zeros (reflection and conjugation pairings in fold-component form, machine-verified), off-line throughout;
- the inverted point $1/\rho$: in the convergence region, its axis-2 reading is positive and its ζ -side components do not vanish jointly (the Euler product restores them);

so that the eight components over the eight points carry the complete observation of the off-line zero: four zeros whose ζ -side is collapsed and whose p -side is explicit, and an inverted point whose eight components are fully restored. This is the input signature on which the value bridge must act: the counting excess of the four off-line zeros against the winding count of the phase side. The signature is machine-verified; the constraint is open.

The counting skeleton of the value bridge. The counting side of the value bridge is machine-verified in the present package (`ValueBridge.lean`, 25 declarations): the counting agreement is equivalent to the absence of off-line zeros,

$$\left(\forall z : \zeta(z) = 0, 0 \leq \Re z \leq 1, |\Im z| \leq T \Rightarrow \Re z = \frac{1}{2} \right) \iff \#\{z : \zeta(z) = 0, \Re z = \frac{1}{2}, |\Im z| \leq T\} = \#\{z : \zeta(z) = 0, 0 \leq \Re z \leq 1, |\Im z| \leq T\} \quad (37)$$

(`count_eq_iff_all_on_line`): an off-line zero makes the on-line set a proper subset of the in-strip set, hence the cardinals strictly smaller — the counting excess is machine-verified. The equivalent reduction closes the chain: RH holds iff the counts agree for every $T \geq 0$ (`rh_iff_NO_eq_N`), and the closing judgement of the excess against an agreed quota is the net element $K = N \wedge N > K \Rightarrow \perp$ (`quota_contradiction`). The input side connects the eight-point signature to the counting chain: an off-line zero ρ gives the strict shortfall directly (`offline_implies_count_excess`), and the eight-point signature of §14.8 enters it as a whole (`eight_point_signature_implies_count_excess`). The counting skeleton is therefore complete and machine-verified: the signature (input), the strict shortfall, the counting agreement equivalence, the RH equivalence, and the quota contradiction are all in place. The phase side of the skeleton is machine-verified as well (`ValueBridge.lean`, 25 declarations): the winding count equals the multiplicity sum (`contour_count_bridge`), the flip count does not exceed the winding count (all odd order, `flip_le_winding`), and for simple zeros the flip count, the winding count and the zero count coincide (`flip_eq_winding_simple`: $K \leq L/2\pi$ and $L/2\pi = K$), together with the flip-jump sums (K flips, π each: $K\pi$; `flip_jump_sum`). The analytic input of the skeleton is machine-verified in its single-zero and summed forms (§12 and `zeta_argument_multiplicity_iso.lean`: the multiplicity- m argument principle and the summed contour $\sum_k \oint_{C(s_k, R_k/2)} \zeta'/\zeta = 2\pi i \sum_k m_k$), and the quota of the phase side enters the skeleton as the ζ instance of the simple-zero winding count,

$$\frac{1}{2\pi i} \oint_{\Gamma} \frac{\zeta'(z)}{\zeta(z)} dz = K \quad (38)$$

(`zeta_winding_quota_simple`: K simple zeros, one winding each, from the global contour bridge with the punctured contour input). The remaining open input of the skeleton is the punctured-contour decomposition for several zeros (the h_{Γ} input, the ring-shrinking of §12 supplying the single-zero case) — together with the value bridge proper. The honest status is unchanged: the counting logic of the value bridge, both sides, is machine-verified; the multi-zero contour input and the constraint are open.

The exchange net element: $\theta = \text{num}$. The counting constraint $K_{\text{flip}} = K_{\text{wind}}$ has a structural reading that isolates its open core. Both sides of the constraint are phase quantities (direction data, no

scale): the flip count is an interlock crossing count along the observation line (the real-part zeros of ζ on the critical line, the phase θ crossing $\pi/2 \bmod \pi$), and the winding count is the phase walk of $\log \eta$ (the imaginary part of $\oint \eta'/\eta$, the real part integrating to zero on a closed contour). The scale (the eighth degree of freedom) is absent from the constraint entirely — the machine-verified reduction of §14.8 is a pure phase statement.

The exchange between phase walk and number is a general net element, machine-verified without any reference to ζ (the file `ExchangeNetElement.lean`, 20 declarations, no axioms): the interlock lattice

$$\theta_k = \frac{\pi}{2} + k\pi \quad (k \in \mathbb{Z}) \quad (39)$$

of the phase-interlock points ($\cos \theta_k = 0$, the direction carried by $\sin \theta_k = (-1)^k$) exchanges with the integer k in both directions — the number readout $\lfloor (\theta - \pi/2)/\pi \rfloor$ is exact on the lattice, the phase rebuild is the inverse, and the round trip is the identity (`interlock_cos_zero`, `interlock_sin_dir`, `numOfInterlock_spec`, `interlock_roundtrip`): one interlock point per π of phase walk (`interlock_step`, `walk_of_nums`). Direction consistency is intrinsic to the element (the direction is carried by $\sin$); changing direction is a return to the basepoint/axis layer, formalized as the reorient dichotomy (`ReorientNetElement.lean`, 9 declarations): either lift (the direction is carried by the unit vector $(\cos \theta, \sin \theta)$, reorientation is the invertible rotation $R(\varphi)$, `dirVec_add_rot`, `rot_inverse`) or return to zero (the sign flip $v(\theta + \pi) = -v(\theta)$, the lattice re-read from the basepoint, `dirVec_flip`, `interlock_from_zero`); at $\varphi = \pi$ the two readings coincide (`reorient_equivalence`). The involution family (`InvolutionNetElement.lean`, reflection, sign flip, rotation by π , commuting) and the direction tensor (`DirectionTensor.lean`, the four forward-projection components $u^+ = \max(u, 0)$, $u^- = \max(-u, 0)$, mutually exclusive, reconstructing $u = u^+ - u^-$, zero-test $u^+ = u^- = v^+ = v^- = 0 \iff u = v = 0$) complete the element family.

The flat legacy counting elements are explicit instances of these two pillars, with equivalence proofs (`NetElementInstances.lean`, 10 theorems, no axioms): the half-period sign flip (`FlipCount.sin_half_period_neg`) is the interlock flip at the lattice points; the winding compression (`PhaseCompress.compress_winding`, $2\pi k$ lifting) is the lattice walk over $2k$ interlock steps; the multiplicative-axis power (`AxisOperators.axisMul_pow`, $e^{i\theta}$ to the k) is the rotation power $k\theta$ (complex form and vector form are equivalent); the tangent period (`PhaseExchange.rotation_law_pi`) is the interlock flip period; and the flip-jump sum (K jumps, π each, $K\pi$) is the sum of K interlock steps. Each flat statement and its explicit instance are mutually implied — the two presentations carry the same content.

For the value bridge this means the constraint is an exchange between a phase walk and a number: the winding count K_{wind} (walk $2\pi K_{\text{wind}}$ over the contour) and the flip count K_{flip} (interlock crossings along the observation line) are two sides of the same phase data, exchanged through the lattice of (39). The directed refinement of the exchange is machine-verified without any monotonicity condition: the *net* directed crossing count (crossings of θ through the interlock lattice, upward $+1$, downward -1) equals the number readout difference,

$$\#_{\text{cross}}^{\text{net}} = \lfloor (\theta_b - \pi/2)/\pi \rfloor - \lfloor (\theta_a - \pi/2)/\pi \rfloor, \quad (40)$$

and over a closed contour of K windings the net count is $2K$ (`num_shift_period`, `net_cross_winding`): one upward crossing of $\pi/2$ and one downward crossing of $3\pi/2$ per winding, unconditionally — the oscillating-phase obstruction of the undirected count is absorbed by the direction, which is intrinsic to the pat system. Applied to the ζ winding quota (the argument principle, `zeta_winding_quota_simple`), the net crossing count of the contour is $2K$ for K simple zeros (`ZetaNetElement.lean`): the constraint $K_{\text{flip}} = K_{\text{wind}}$ is restated, equivalently, as flip count = net crossings $/2$ — the pairing of one flip with two directed crossings on the observation line, the precise residual of the reduction (`value_bridge_net_form`).

The two machine-verified sides of the exchange are exactly the two theorems of phase-number commutativity: the *winding* side (count = walk $/2\pi$) is the counting form of monotonicity breakdown — a direction change returns to the basepoint layer (reorient to zero, §14.8), absorbed by the directed count; the *pairing* side (two crossings per zero) is the lifted geometry — the direction vector $v(\theta) = (\cos \theta, \sin \theta)$ circles the zero once and crosses the $\cos = 0$ direction exactly twice, at the interlock directions $\pi/2$ and $3\pi/2$ (`interlock_in_period_iff`, `cos_zero_period`, `pairing_per_circle`): within one period $[0, 2\pi)$ the interlock lattice has exactly two points, $k = 0$ and $k = 1$ (`PairingNetElement.lean`). Winding (return to zero) and pairing (lift) are the two sides of the same exchange, one period of the direction vector carrying exactly two interlock directions.

The zero-th order of the exchange is a single theorem, the simultaneous generation and locking of phase and number: the interlock condition $\cos \theta = 0$ generates the phase θ_k and the number k at the same time, locked bidirectionally — for $\cos \theta = 0$ the number condition $\lfloor (\theta - \pi/2)/\pi \rfloor = k$ and the phase condition $\theta = \theta_k$ imply each other (**gen_lock_iff**): one condition, both quantities, mutual determination. The first order derives from the zero-th: the jump $\theta_k + \pi = \theta_{k+1}$ keeps the lattice and the locked number moves by one (**jump_derived**).

The orders of the exchange form a complete hierarchy, all machine-verified and all general (no ζ involved): order 0 is the generation-locking theorem (**gen_lock_iff**); order 1 is the jump, π translation moving the number by ± 1 (**jump_cross_one**, **jump_cross_neg_one**); order 2 is the pairing, one circle carrying two crossings (**pairing_decomp**); order 3 is the K -fold synthesis, K circles carrying $2K$ net crossings (**net_cross_winding**); order 4 is the addition of circles, $J + K$ circles carrying $2J + 2K$ crossings (**net_cross_add**); order 5 is the directed circle, $-2\pi K$ carrying $-2K$ (**net_cross_neg**); and order 6 is the complete translation group, an arbitrary πm translation moving the number by m (**num_translate**) — the jump ($m = \pm 1$), the circle ($m = 2K$) and the reverse circle ($m < 0$) are the same theorem. Orders 0–6 are the full lattice structure of the exchange: one theorem per order, all unconditional.

The order of the direction space is its dimension: the n -dimensional direction tensor carries 2^n orthants (sign patterns, one bit per dimension), machine-verified as a family (**DirectionTensor.lean**): the orthant of a vector x is its sign pattern $\sigma_i = (0 < x_i)$, mutually exclusive (**orthant_exclusive**: a non-zero vector lies in exactly one orthant), counted by 2^n (**orthant_count**), with the n -dimensional zero test (all positive/negative components vanish iff the vector vanishes, **tensor_zero_iff**) — order 1 is the two signs (yin-yang), order 3 is the eight trigrams (2^3 , **orthant_count_three**), order 6 is the sixty-four hexagrams (2^6 , **orthant_count_six**): the dimension hierarchy of the direction tensor is the full binary expansion.

Each orthant of the hierarchy carries the three languages of the system, machine-verified as one family (**DirectionTensor.lean**): the *algebraic* form (the component sign pattern, the decomposition $x_i = x_i^+ - x_i^-$, the mutual exclusion, the zero test); the *symmetric* form (the dual orthant $\sigma \mapsto \neg \sigma$, the reflection through the origin: an involution **dual_involution**, the reflection exchange **dual_neg**: orthant of $(-x)$ under the dual equals the orthant of x , and the exclusion of a vector from its dual **dual_exclusive**); and the *phase* form (the direction vector $v(\theta) = (\cos \theta, \sin \theta)$, whose quadrants are the 2-dimensional orthants, with the quadrant dual equal to the rotation by π , **dirVec_dual_rot** — the half-turn carries the quadrant to its dual). The three languages are equivalent, assembled in one statement (**trilingual_orthant**): a vector in the orthant σ lies in the dual orthant under reflection, and not in the dual orthant itself. Every one of the sixty-four hexagrams is simultaneously an algebraic pattern, a symmetric pair, and a phase direction; the translation between the three is lossless.

The pairing structure has a first-order and a second-order form, machine-verified as well. *First order (the jump)*: a phase jump of π — the flip of an odd-order zero — crosses exactly one interlock lattice point,

$$\lfloor (\theta + \pi - \pi/2)/\pi \rfloor = \lfloor (\theta - \pi/2)/\pi \rfloor + 1, \quad (41)$$

(**jump_cross_one**, and the reverse jump -1 , **jump_cross_neg_one**): the flip (the jump) and the crossing (the continuous pass) are one and the same count, direction included. *Second order (the pair)*: one circle of net crossings 2 decomposes into the jump side 1 (the zero's own crossing) and the continuous side 1 (the neighbouring crossing) (**pairing_decomp**) — the internal structure of the pairing, one crossing at the zero and one beside it, of opposite direction (cancelling locally, counted net globally). The exchange is therefore three-faced: winding (return to zero, the count of the walk), pairing (lift, the two interlock directions per circle), and the jump (first order, one interlock point per flip) — the flip count, the crossing count and the winding count of the observation line are the same phase-number exchange at zero-th, first and second order. The open core of the reduction is now isolated: the number of interlock crossings on the observation line against the winding count in the strip — the multi-zero contour input (h_Γ) and the constraint itself remain open, honestly recorded; the net element supplies the exchange machinery, not the constraint.

The value bridge assembly. The value bridge itself assembles the three sides into a single constraint, machine-verified as one statement. If the flip count equals the on-line zero count ($N_0 = K_{\text{flip}}$, the flip counting), and the winding count equals the in-strip zero count ($N = K_{\text{wind}}$, the argument principle), then the single hypothesis

$$K_{\text{flip}} = K_{\text{wind}} \quad (42)$$

(flip count = winding count, the counting form of the reduction) implies the absence of off-line zeros (`value_bridge_assembly`: counting agreement, hence every in-strip zero is on the line, via `count_eq_iff_all_on_line`); and an off-line zero ρ together with the same hypothesis is a contradiction (`value_bridge_contradiction`: ρ forces the strict shortfall $N_0 < N$, hence $K_{\text{flip}} < K_{\text{wind}}$, against $K_{\text{flip}} = K_{\text{wind}}$, the quota closing). The value bridge is therefore the single constraint $K_{\text{flip}} = K_{\text{wind}}$: everything around it — the eight-point signature, the counting shortfall, the flip counting, the winding count, and the quota contradiction — is machine-verified. The honest status of the reduction is unchanged: the constraint itself, flip count = winding count, is the counting form of the Riemann hypothesis; the assembly is verified, the constraint is open.

14.9 Which nodes of the chain to re-observe

The re-observation question of the new frame has two layers.

- *Logical completeness (must)*: if the fold frame is used in the chain, the four bridges of §14.6 must be verified — they are, in this part. No Riemann node is re-proved; translation is automatic under a linear coordinate change (`foldCoordHomeo`).
- *Presentation (worthwhile)*: the symmetry family of the chain (conjugation, reflection pairing, odd/even structure) gains an explicit presentation — conjugation as the coordinate swap, the imaginary axis as the diagonal, the critical line as a horizontal line. This is presentation-layer work (standard facts re-narrated, compatible with the consensus structure; no new axiom).
- *Not needed*: the equivalence chain (pure logic), the phase elements (topological invariants: winding numbers and layers are invariant under any orientation-preserving linear change of coordinates), the assembly seams (pure algebra), and the integral estimates (coordinate-free).

The honest boundary of the new frame is the same as that of Part III: changing the frame does not change the mathematics, and the open status of RH is unchanged. The value of the frame is a new rigid quantity in the strip — the difference structure of the odd/even split — and an explicit geometry in which the boundary of Part III can be re-read.

15 Discussion: polyphase numbers and the necessity of single-phase numbers

The framework viewpoint of this section is motivational. The mathematics of the paper is classical throughout, and its presentation is algebraic: the covering used above is the standard group exponential $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$ with kernel $2\pi i\mathbb{Z}$ (§7), the natural numbers involved are the standard Peano numbers, and no axiom is taken from the unit-circle viewpoint — the unit circle is the mathematical object of the covering, not an alternate foundation for the argument.

The phase bridge of §6 exhibits the mechanism of the 0pat exercise: the same point $e^{i\theta}$ of the unit circle carries the infinitely many phase representatives $\theta + 2\pi n$, $n \in \mathbb{Z}$; the winding number of the covering $t \mapsto e^{it}$ is a single-valued invariant, and the natural numbers arise exactly as these single-valued counts ($\pi_1(S^1) \cong \mathbb{Z}$; [8]). A precise definition of a number therefore requires phase locking: one fixes the basepoint of the lift and counts the winding, obtaining a single-phase number. The natural numbers are thus, in the framework's description, *polyphase counts*: the counting of winding is inherently multi-valued ($\theta + 2\pi n$ for all n), and only the locking of the lift to one basepoint makes the count a number. The necessity of single-phase numbers is exactly this locking; it is the framework's basepoint-relative description of the standard fact that the winding number is an integer ([2]).

In the Riemann direction the same locking is used twice: the lifted continuous argument θ_χ of §6–7, and the flip count of §9.1, both converting the multi-valued phase into the integer layers of §9. This is the necessity of single-phase numbers: without the locking, the argument of ζ along the rectangle would not be defined as a continuous function, and the argument principle would not apply.

What symmetry cancels: phase superposition (machine-verified, iso 29)

The framework’s cancellation operations act on phase superpositions. The precise content — machine-verified in the isolation file `symmetry_cancel_phase_iso.lean` (iso 29 of the artifact, 18 theorems, 0 errors, 0 sorry) — is the structure of the covering map $t \mapsto e^{it}$:

1. *Phase superposition is the fiber.* Each point $e^{i\theta}$ carries the $\aleph_0$ representatives $\theta + 2\pi n$, $n \in \mathbb{Z}$ (`phaseRepresentative`); distinct parameters give distinct representatives (`phaseRepresentative_injective`).
2. *Symmetry is translation.* The shift $T_n(\theta) = \theta + 2\pi n$ preserves the projection $e^{i\theta}$ (machine-verified: `translate_preserves_exp`), and $T_m \circ T_n = T_{m+n}$ — the $\mathbb{Z}$ -action on the representatives (`translate_comp`).
3. *Cancellation is the quotient.* Dividing by the translation symmetry collapses each orbit — the whole superposition — to a single phase class: every representative of the same point is identified (`superposition_collapses_to_single_phase`), and the quotient covers every class (`phaseEquiv_quotient_surjective`).

This is the exact form of the framework statement that a symmetry operation cancels phases (author proposal 11): the symmetry is the translation action, what it cancels is the multi-valued phase superposition, and the residue is a single-phase class. The same covering-map structure supports the counting below.

The single-phase representation (unified conclusion; author’s final positioning)

Three paths lead to the same conclusion. The three bridges (flip, Stirling phase, growth) exhibit the natural numbers as winding counts depending on the basepoint of the lift — the ambiguity of the representation; the symmetry cancellation above acts on the multi-valued fibers; and the phase superposition of the natural numbers is an un-locked superposition. In all three the ambiguity is the same: the covering map $t \mapsto e^{it}$ assigns $\aleph_0$ representatives to each point, and an un-locked representation of a number is ambiguous. The author’s final positioning: the framework is a *representation method*, not a new number field — every object used is known mathematics ($\mathbb{R}$, S^1 , $\mathbb{N}$, the quotient, the kernel of `exp`); the representation precisely isolates the phase-superposition ambiguity caused by the $\mathbb{N}$ -iteration count (the n -th successor is the unique layer $2\pi n$, machine-verified iso 31), and constrains the phases not to superpose arbitrarily (the locking selects one representative; the result is independent of the selection, `phaseSucc_respects_equiv`). The numbers themselves are unambiguous (Peano); what needs precision is the representation, and the single-phase representation provides it: the quotient of the translation symmetry, in which each class is one point of the unit circle. Machine-verified (iso 29): $\exp(i\theta_1) = \exp(i\theta_2)$ iff $\theta_1 \sim \theta_2$ — the kernel of `exp` is $2\pi i\mathbb{Z}$ (`exp_phase_equiv_iff`), so the classes correspond one-to-one to the projections and the ambiguity is removed (`single_phase_class_iff_projection`).

The single-phase representation is moreover *unique up to isomorphism* (author’s question: does every attempt to represent a single-phase number end up isomorphic to the framework’s?): by the universal property of the quotient, any attempt — a map $f : \mathbb{R} \rightarrow X$ that is constant on phase-equivalent representatives (no phase superposition), separating (different classes give different numbers), and surjective — factors *uniquely* through the quotient, and the factor map is a bijection (machine-verified: `single_phase_field_unique`, `single_phase_field_equiv`, iso 29). Hence every phase-superposition-free representation is isomorphic to the Pat single-phase representation: the representation is canonical.

The one structure the projection necessarily loses is the layer of the point 1 (author’s question: is the loss falsifiable?). Since $1 = e^{i \cdot 0} = e^{i \cdot 2\pi} = \dots$, the fiber of 1 is $\{2\pi n : n \in \mathbb{Z}\}$ (machine-verified: `fiber_of_one`, iso 29): $\aleph_0$ representatives collapse to the single point 1. The claim “the projection preserves all structure” is *falsifiable* — it is false: the projection is not injective ($q(0) = q(2\pi)$, $0 \neq 2\pi$; machine-verified: `projection_loses_structure`, iso 29). This loss is not a defect: what is lost is exactly the layer under the translation symmetry — the winding count, $\pi_1(S^1) \cong \mathbb{Z}$ — which is precisely what the quotient cancels (iso 29); the single-phase number field accepts the loss of layers in exchange for the removal of ambiguity.

The path problem (author’s question, machine-verified iso 30)

Whether the definition is complete enough to handle the path question: from the same starting point to the same ending point, different paths — do they give different numbers? The answer is two-layered (machine-verified in the isolation file `path_problem_iso.lean`, iso 30, 8 theorems, 0 errors, 0 `sorry`). The loop family $\gamma_n(t) = e^{2\pi n t \cdot i}$, $n \in \mathbb{Z}$ (the n -fold loop from the projection 1 back to 1) all start and end at the same projection (`loopPath_start`, `loopPath_end`); their lifts $g_n(t) = 2\pi n t$ end at the layers $g_n(1) = 2\pi n$ (`loopPath_lift`, `loopPath_layer`).

1. *Quotient layer (single-phase representation): path independent.* All loops give the same number, the class of 1: the projection is the same, and the translation symmetry is cancelled (`loopPath_same_class`) — well-defined by design.
2. *Lift layer (multi-phase): path dependent.* Different loops ($n \neq m$) lift to different layers, $2\pi n \neq 2\pi m$ (`loopPath_layer_distinct`) — the path information is preserved in the layers.

The definition is therefore complete in both directions (`path_problem_complete`): the single-phase number (the quotient) is path-independent, and the multi-phase layer (the lift) is path-dependent; both are expressed, and the path problem is resolved rather than avoided.

A further calibration (author): carrying the *full* path information would require a component that is not phase. Phase (θ, n) carries at most the winding class of a path — its homotopy class, $\pi_1(S^1) \cong \mathbb{Z}$ — while the full path (its shape) is a continuous function, a different component; the lift is determined by the projection path and the starting layer, so the projection path is an independent input, not phase. The single-phase representation resolves this by being path-independent by definition (the quotient): the number carries no path information at all; the multi-phase layer carries the winding class, the maximum that phase can carry; the full path, when needed, is an extra component outside the representation. No “path-carrying phase” is required.

Successor as phase superposition (author, iso 31)

The component that carries the movement is Pat itself: the successor. Its essence is phase superposition (machine-verified in `phase_succ_iso.lean`, iso 31, 9 theorems, 0 errors, 0 `sorry`): the successor is one turn, $\text{succ}(\theta) = \theta + 2\pi$ — (i) *aligned*: the phase modulation preserves the projection, $\exp(\text{succ}(\theta) \cdot i) = \exp(\theta \cdot i)$ (`phaseSucc_preserves_projection`) — the modulation aligns to the successor direction of the previous operation without changing the phase point; (ii) *executed*: the layer is raised by one, from $2\pi n$ to $2\pi(n + 1)$ (`phaseSucc_layer`). The natural numbers are the phase superpositions from 0: the n -th successor is $2\pi n$, the winding number n (`phaseSucc_iterate`). The successor is thereby redefined from a Peano axiom to a structural operation — phase superposition, the generator of the translation symmetry (`phaseSucc_is_translation`).

Only 0 and 1 (author’s deepest conviction)

Because the operators and the numbers are ambiguous (phase superposition, paths, basepoints), how is the continued definition of numbers regulated? The answer (machine-verified, iso 31, 9 theorems): the primitives of the framework are only 0 (the basepoint, the empty operation) and 1 (the single step, one phase superposition); every number is an iterate of 1 — each step is a single step reaching a point, and “changing the point” means continuing from the current layer, never re-selecting. The well-definedness is fourfold: the successor respects phase equivalence (`phaseSucc_respects_equiv`); the iterate is unique — the n -th successor is $2\pi n$ (`phaseSucc_iterate`); the addition is regulated — m steps after n steps is $n + m$ steps (`phaseSucc_iterate_add`); and in the quotient (the single-phase representation) the successor is the identity, $[\theta + 2\pi] = [\theta]$ (`phaseSucc_trivial_on_quotient`). The single-phase representation therefore has no number growth (the projection does not change); the numbers 0, 1, 2, ... live in the layers and are regulated by the locking; the projections of 0 and 1 coincide (both are 1).

Phase superposition and the continuum

The same phase structure generates the continuum itself. In the framework basepaper [2] the *continuum closure theorem* is machine-verified: the countable pat grid

$$\text{patGrid} = \{ 2\pi j/N \mid 0 < N, 0 \leq j \leq N \}$$

is dense in $[0, 2\pi]$ (R150), and therefore the continuum $[0, 2\pi]$ is the closure of the grid (R151); an endogenous variant constructs the same closure from two paired basepoints and midpoint reduction (R163). The author proposed the following reading of this theorem: the continuum arises by *phase superposition* — each grid point $2\pi j/N$ carries the $\aleph_0$ phase representatives $2\pi j/N + 2\pi n$, and the superposition of countably many such phase layers (“ $\aleph_0$ layers of $\aleph_0$ points””) closes to the continuum.

The cardinal arithmetic of this reading, precisely (KNOWN classical; author calibration 8): two different counts must be separated. The *union* of all phase layers is $\aleph_0 \cdot \aleph_0 = \aleph_0$ (cardinal absorption [9]): the number of layers is still countable. The *superposition* — choosing one projection at each of the $\aleph_0$ positions — is the function space $\mathbb{N} \rightarrow \mathbb{N}$, with

$$\aleph_0^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c} = |\mathbb{R}|,$$

the continuum itself (machine-verified in the isolation file `continuum_projection_iso.lean`, iso 28 of the artifact: 35 theorems, 0 errors, 0 `sorry`). The “more than $\aleph_0$ ” intuition is therefore correct for the *result*: the superposition produces $2^{\aleph_0} > \aleph_0$ objects (Cantor). The choice space at each position does not matter for the result: for every $2 \leq \kappa \leq \mathfrak{c}$ one has $\kappa^{\aleph_0} = \mathfrak{c}$ (machine-verified; e.g. binary and decimal expansions), so whether each position carries 2, 10, $\aleph_0$, or $\mathfrak{c}$ projections, the superposition closes to the same continuum.

The four phase spaces are compared systematically, entirely from the structural formula (superposition = function space, base $\kappa^{\aleph_0}$ where κ is the per-position phase space; no reference to the definition of $\aleph_1$ — the machinery is *structural*, not definitional):

Per-position phase space P	$\#P$	superposition $ P ^{\aleph_0}$
natural numbers $\mathbb{N}$	$\aleph_0$	$\aleph_0^{\aleph_0} = \mathfrak{c}$
real numbers $\mathbb{R}$	$\mathfrak{c}$	$\mathfrak{c}^{\aleph_0} = \mathfrak{c}$
phase full spectrum $\mathbb{N} \rightarrow \mathbb{N}$	$\aleph_0^{\aleph_0} = \mathfrak{c}$	$\mathfrak{c}^{\aleph_0} = \mathfrak{c}$
all-dimension functions $\mathbb{R} \rightarrow \mathbb{R}$	$2^{\mathfrak{c}}$	$(2^{\mathfrak{c}})^{\aleph_0} = 2^{\mathfrak{c}}$

(all four machine-verified, iso 28). The third row is the author’s *phase full spectrum*: the phase of one natural number is the infinite-dimensional space of infinite phases per dimension, $\aleph_0^{\aleph_0}$, and the natural numbers are its $\aleph_0$ -fold superposition, giving the triple power

$$(\aleph_0^{\aleph_0})^{\aleph_0} = \aleph_0^{\aleph_0 \cdot \aleph_0} = \aleph_0^{\aleph_0} = \mathfrak{c}$$

(machine-verified: `triple_power_absorption`, iso 28): power associativity and the absorption $\aleph_0 \cdot \aleph_0 = \aleph_0$ collapse the three layers back into one — infinite nesting (finite or countable layers) absorbs to the continuum.

The *association* of the tower is decisive (author’s question about paths between natural numbers). The parallel superposition above is the *shortest-path* reading: one phase choice per position, positions side by side. But every path from 0 to a natural number n may pass through any other natural numbers, so the phases of different natural numbers superpose *along paths*; the finite–infinite bridge allows infinite paths, the path space is $\mathbb{N}^{\mathbb{N}}$, and the nested superposition is the power tower

$$\aleph_0^{(\aleph_0^{\aleph_0})} = \aleph_0^{\mathfrak{c}} = 2^{\mathfrak{c}},$$

exceeding the continuum (machine-verified: `path_superposition_tower`, iso 28). The contrast is the association: $(\aleph_0^{\aleph_0})^{\aleph_0} = \mathfrak{c}$ (parallel, absorbs) vs. $\aleph_0^{(\aleph_0^{\aleph_0})} = 2^{\mathfrak{c}}$ (nested, tower), machine-verified `nesting_exceeds_parallel`.

A calibration (author): the basepoint — the reference of the phase locking — may move freely, but the movement process is itself a path (a sequence of basepoints), already contained in the path space $\mathbb{N}^{\mathbb{N}}$; the nested superposition $\aleph_0^{(\aleph_0^{\aleph_0})} = 2^{\mathfrak{c}}$ therefore already includes basepoint movement, and the basepoint adds no new layer (machine-verified: `basepoint_movement_no_new_layer`, iso 28): the result remains $2^{\mathfrak{c}}$, not $2^{2^{\mathfrak{c}}}$.

Only a genuinely new degree of freedom independent of paths would raise the tower (the arithmetic fact $\aleph_0^{2^c} = 2^{2^c}$ is recorded in iso 28 but is not triggered by basepoint movement).

A precise statement of what is exceeded, and what is not (author’s question whether the continuum is “falsified”): the nested superposition of the polyphase natural numbers, $\aleph_0^{(\aleph_0^{\aleph_0})} = 2^c$, is larger than the continuum (Cantor; machine-verified) — what is falsified is the *ceiling* claim that every superposition of natural-number phases stays $\leq c$ (true for parallel superpositions, false for the nested one). The continuum itself is not falsified: $c = |\mathbb{R}| = 2^{\aleph_0}$ is a theorem (machine-verified), and the continuum hypothesis $c = \aleph_1$ is independent of ZFC, so neither is disprovable. The nested superposition equals $|\mathbb{R} \rightarrow \mathbb{R}|$, the function space above the continuum. The comparison shows: any phase space of size $\leq c$ — natural, real, or the full spectrum — superposes to the same continuum c (the ceiling $\kappa^{\aleph_0} = c$ for $2 \leq \kappa \leq c$); the only superposition exceeding the continuum is the all-dimension one, where the phase space itself is $2^c > c$ (Cantor) — the excess comes from the per-position space, not from the number of positions or the superposition operation.

The superposition result is the second layer of the power hierarchy, $\beth_1 = c = \aleph_0^{\aleph_0}$ (machine-verified, iso 28). The aleph hierarchy $\aleph_0 < \aleph_1 < \aleph_2 < \dots$ (ordinal-successor construction) and the beth hierarchy $\beth_0 = \aleph_0$, $\beth_1 = c$, $\beth_2 = 2^c$ (power construction) are *different constructions*: $\aleph_1$ is not reached by powers — it is the successor of $\aleph_0$ — and the superposition lands at $\beth_1$. Where $\beth_1$ sits in the aleph hierarchy, $\beth_1 = \aleph_1$ (the continuum hypothesis) or higher, is independent of ZFC — a question of independence, not of definition.

Two further counts calibrate the author’s questions. First, the superposition result $\aleph_0 < c$ is definite (Cantor), and $\aleph_1 \leq c$ always holds; there is no cardinal between $\aleph_0$ and $\aleph_1$, but this is a constructive fact about $\aleph_1$ (it is the least uncountable cardinal, a successor by construction) — recorded here only as a calibration, not as a conclusion of the superposition. If each position carries $\aleph_1$ projections (the positions are still the $\aleph_0$ natural numbers), the result is still $\aleph_1^{\aleph_0} = c$: $\aleph_1 \leq c$ always holds ($\aleph_1$ is the least uncountable cardinal), and the ceiling $c^{\aleph_0} = c$ absorbs it (machine-verified). Only if the *positions* themselves number $\aleph_1$ does one leave the continuum construction: $\aleph_1^{\aleph_1} = 2^{\aleph_1} \geq \aleph_2$ (Cantor; machine-verified), where the equality $2^{\aleph_1} = \aleph_2$ is the generalized continuum hypothesis, independent of ZFC (GCH/Easton; KNOWN). Finally, the imaginary-unit powers

$$i^{c \cdot i} = e^{-c\pi/2} \quad (c \in \mathbb{R}),$$

the case $c = 1/2$ being $i^{i/2} = e^{-\pi/4}$, real — encode a continuum-sized parameter space into the phase structure of each position (machine-verified: the parameter map is injective); even so the ceiling $c^{\aleph_0} = c$ is unchanged.

The continuum hypothesis itself remains independent of ZFC (Gödel [6], Cohen [7]); the phase-superposition reading is a *constructive generation* of the continuum (density + closure and the sequence-space count), not an intermediate-cardinality statement, and the present paper makes no claim about the continuum hypothesis.

16 Verification record

All source files, the hash ledger (`iso_hashes.md`), and the verification log are archived at the Zenodo records 10.5281/zenodo.22038154 (the framework release), and 10.5281/zenodo.22039506, 10.5281/zenodo.22044557, 10.5281/zenodo.22044634 (the successive release packages of this exercise; the Zenodo records carry the version-2 title of the first upload, and the later records contain the successive revision packages, the last one including the isolation files iso 28–31); the complete package of the present revision, including the odd/even pairing (iso 41), Layer 14 (`PreciseMigration.lean`), the second pass of Part II (`zeta_argument_iso.lean` and its companions), and the revised manuscript, is available at the GitHub repository listed in the cover letter. The journal’s submission system accepts the manuscript PDF only; the complete Lean verification package is retrievable from these records. Environment: Lean 4.32.2 [10], mathlib revision 905b95818e [3]. The development comprises 231 declarations in `proof.lean` (Layers 1–12), plus the isolation files `zeta_continuation_iso.lean` (Layer 13 Blocks A–B, 12 declarations), `zeta_odd_even_continuity_iso.lean` (iso 41: the odd/even pairing continuity of this revision, 4 declarations), `continuum_projection_iso.lean` (iso 28: the continuum by phase superposition, 35 theorems), `symmetry_cancel_phase_iso.lean` (iso 29: what symmetry cancels and the single-phase number field, 18 theorems), `path_problem_iso.lean` (iso 30: the path problem, 8 theorems), and `phase_succ_iso.lean` (iso 31: the successor as phase superposition and only-0-and-1, 9 theorems), and

`PreciseMigration.lean` (Layer 14: precise base-point migration, 2 definitions and 18 theorems/lemmas, 20 declarations), and `flip_sum_chain_iso.lean` (the Layer 12 counting chain: telescoping identity, layer-sum conservation, the lift-log bridge, and the π -jump per odd-order zero, 5 theorems); the representation bridges `known_objects_bridges_iso.lean` (the coordinate models of Layers 1 and 3 identified with the known objects: $\langle a, b \rangle \mapsto a + bi$ into mathlib’s `Complex` and $a + bj \mapsto (a + b, a - b)$ into the diagonal of $\mathbb{R} \times \mathbb{R}$, 16 theorems, 0 errors, 0 `sorry`); the fold frame of Part IV (`EulerFoldImagAxis.lean`: the fold coordinates, the four bridges, the symmetry elements, the fold arc homeomorphism, and the two-reading triangle pair of §14.3, 89 declarations; and `FoldObservation.lean`: the observation protocol — the phase ratio, the observation plane, the analytic restatement, the reconstructed components of ζ , the symmetry liftings (conjugation swap, reflection multiplier, inversion swap plus scaling), the fold modulus, the joint observation, the four-direction components, the irrational-axis (p-side) eight components, the odd/even value-locus analysis, the scale/Euler-product access, the odd/even difference structure (η), the eight-point observation signature, and the winding equality (η carries the count), 90 + 25 declarations in `FoldObservation.lean` and `ValueBridge.lean` — the 4 additions of the final revision being the value-domain anchors (`zeta_fold_sum_re`, `zeta_fold_diff_im`, `zeta_fold_dual_proj_zero_of_zero`) and the weak assertion `projection_zero_line_complete` of §14.8; and `NativeAxisEquiv.lean`: the native four-axis equivalences of the fold coordinates (constant $\sqrt{2}/2 = 2^{-1/2}$; native recovery, uniqueness, modulus preservation, surjectivity with the human-coordinate read-back exemption, 11 declarations)); each isolation file compiles independently with 0 errors and 0 `sorry`. The library theorems used by the Euler–Maclaurin continuation blocks of version 1 (all confirmed present in mathlib):

```
HasDerivAt.cpow_const,      ContinuousAt.cpow,      differentiableOn_tsum_of_summable_norm,
TendstoLocallyUniformlyOn.differentiableOn, AnalyticOnNhd.eqOn_zero_of_preconnected_of_frequently_eq_zero,
Convex.isPreconnected,      intervalIntegral.integral_ofReal,      Complex.ofReal_cpow,
ZetaAsymptotics.term_nonneg, ZetaAsymptotics.zeta_limit_aux1.
```

17 Conclusion and outlook

Final claim. Pat is a representation tool for infinite dimensions: through phase superposition and locking it represents structures from the countable ($\aleph_0$) to the continuum ($\mathfrak{c}$) and beyond (up to $2^{\mathfrak{c}}$, including infinite-dimensional function spaces); it eliminates the ambiguity of numbers — the phase-superposition ambiguity of the representatives, removed by the locking and the quotient (iso 29) — and restores the structure lost by the projection — the layers (the winding counts) carried in the lift (iso 30); both supported by the empirical studies of the framework [2]. The claim is machine-verified in known mathematics (iso 28–31: 70 theorems, 0 errors, 0 `sorry`; every object — $\mathbb{R}$, S^1 , $\mathbb{N}$, the quotient, the kernel of $\exp$ — is standard). Honest boundaries are recorded throughout: the tool is one of several possible representations (no uniqueness or novelty of the objects is claimed); the numbers themselves are unambiguous (Peano); and the continuum hypothesis remains independent of ZFC. In the same spirit, the Riemann hypothesis itself is not claimed: the chain surrounding it is machine-verified, the hypothesis is only restated in an equivalent form (Theorem 6.4), and everything beyond this chain — the zero-free regions, the moments, the distributions — remains exactly where the classical theory left it.

Within the observation frame, the final revision adds one machine-verified closure of its own: the projection value bridge in its weak form (§14.8, `projection_zero_line_complete`). The observation plane of Part IV drops axis 1 (the position along the critical line) and keeps axis 2 ($d = \sqrt{2}(\Re s - \frac{1}{2})$); the weak assertion states that this remainder — together with the value-domain anchors $U + V = \sqrt{2} \Re \zeta$, $U - V = \sqrt{2} \Im \zeta$, both machine-verified from the native four-axis equivalences of `NativeAxisEquiv.lean` — is *complete* for the judgement “the zero is on the critical line”: every input of that judgement lies in the projection quantities, and the dropped information is precisely the information the judgement does not use. The Riemann hypothesis itself remains open: the judgement it requires is $d(\rho) = 0$ for every zero ρ of the strip, and the frame in which this judgement is posed is now verified complete.

Along the path of known mathematics only, the Riemann direction is machine-verified through fourteen layers: from the complex-axis representation and the circle geometry, through the false-real-axis framework and the critical-line phase layer, the covering-space phase locking, the predictor residuals, and the argument-principle rectangle closure with its boundary estimates. The negativity $\zeta(x) < 0$ on $(0, 1)$ is closed by the odd/even pairing of this revision: the continuity of ζ on $(0, 1)$ is machine-verified (iso 41, §10.3), and the sign follows directly from the machine-verified positivity of the pairing terms and the classical doubling identity

(§10.4), completing the zero-free bottom edge of the rectangle of Layer 12.

From the boundary to the new frame. Part III makes the boundary of the chain precise: the equivalence reduction leaves exactly one task, the construction of an in-strip rigid projection of an off-line zero, and that construction is equivalent to RH (Theorem 13.1); the known generator family fails at the boundary structurally, not for lack of effort (§13.4). Part IV changes the observation frame in which the boundary is read: the fold coordinates of the Euler circle present the imaginary axis as a 90° fold with all-real coordinates (§14.5), conjugation as the coordinate swap, the critical line as a horizontal line, and the projection asymmetry of the standard frame as an explicit geometry (§14.2); the four coordinate bridges are machine-verified (0 sorry). The current work is the odd/even observation route at the false-real-axis base point $e^{\pi/4}$ (§14.8): its infrastructure is machine-verified, its observation step is open, and its status is recorded honestly — the frame is a presentation of the standard mathematics, it does not change the open status of RH, and no claim of unprovability or independence is made.

A Complete declaration inventory (231 + 12 + 4 + 20 + 5 + 12 declarations, by layer)

The declarations of the isolation files iso 28–31 (35 + 18 + 8 + 9 theorems; together with 2 + 3 + 3 + 3 definition declarations inside those files) are listed in their files (see §16), not here; the following inventory covers the fourteen layers. The second-pass theorems of Part II (16 declarations: `zeta_argument_iso.lean`, 3 theorems, and `zeta_zero_order_iso.lean`, 11 theorems and 2 definitions) and the representation bridges of Layers 1 and 3 (`known_objects_bridges_iso.lean`, 16 theorems) are likewise listed in their source files (§16) and are not part of the 284-declaration count counted here. Declarations shared between layers (e.g. `j`, `normSq`, `hardyZ_real`, `chi_ne_zero_on_line`, `zeta_im_zero_on_real`) are listed under each layer that uses them; the layer totals below therefore count with repetition, and the unique total of `proof.lean` is the 231 declarations of §16.

1. **L1 (15):** `add, neg, zero, one, J, mul, sub, lift, norm, primeCircle, criticalCircle; card_units_zmod_prime, fermat_little, fermat_little_complete, infinitely_many_primes.`
2. **L2 (7):** `orthogonal_frame_crt, factor_observable_zero, distinct_primes_observable, prime_unit_on_other_circle, zeta_zero_reflection, zeta_zero_strip_reflection, zeta_zero_offline_pair.`
3. **L3 (19):** `j, normSq, j_sq, normSq_mul, imaginary_radius_circle, real_radius_circle, orthogonal_radII, circles_disjoint, odd_prime_is_split_norm, critical_line_pseudonorm, threshold_timelike, threshold_spacelike, zero_observation_spacelike, recip_critical_line_hyperbola, imagAxis_inter_criticalCircle, realAxis_inter_primeCircle, imagAxis_inter_primeCircle, prime3Circle_inter_criticalCircle, primeCircle_inter_criticalCircle_ge5.`
4. **L4 (25):** `j, normSq, unitCircle, imaginaryCircle, transpose, transpose_involutive, transpose_maps_one_to_i, transpose_origin_fixed, transpose_maps_unit_to_imaginary, transpose_maps_imaginary_to_unit, lattice_unique_unit, lattice_unique_imaginary, realAxis, imaginaryAxis, naturalImaginaryAxis, realAxis_inter_imaginaryAxis, realAxis_inter_unitCircle, realAxis_inter_imaginaryCircle, imaginaryAxis_inter_unitCircle, imaginaryAxis_inter_imaginaryCircle, origin_natural_coefficients, origin_imaginary_part, transpose_realAxis_eq_imaginaryAxis, transpose_natural_imaginary_is_natural_real, complex_imag_axis_coefficient.`
5. **L5 (10):** `complex_mul_imag_leaks, complex_mul_cross_stays_imag, split_mul_imag_leaks, leak_sign_is_unit_square, unit_1_on_real_circle, unit_j_on_imag_circle, real_circle_is_direction_1, imag_circle_is_direction_j, directions_orthogonal, euler_plane.`
6. **L6 (10):** `complex_zeta_convergence_region, cosh_lower_bound, cosh_unbounded, log_nat_unbounded, split_zeta_term_unbounded, zeta_term_norm_split, period_pair_reduces, divergence_pair_reduces, term_conj_symmetry, functional_equation_bridges.`
7. **L7 (17):** `prime_factor_zero, trivial_zero_condition, critical_line_observation, i_pow_i_eq_exp_neg_pi_div_two, i_pow_i_ne_i, i_pow_i_pos_lt_one, axis_tick_pow, prime_no_power_decomposition, xi_symmetry0, xi_symmetry, xi_entire, zeta_series_form,`

- zeta_functional_equation, alternating_series_converges, alternating_partial_sum_bounded, eta_pair_form.
8. **L8 (39):** zeta_conj_of_one_lt_re, arg_two_ne_pi, arg_pi_ne_pi, conj_two_cpow, conj_pi_cpow, conj_sin_half, conj_gamma_one_sub, chi_conj, zeta_conj_of_re_lt_zero, conj_cpow_of_pos, conj_f_modif_zero, conj_mellin_f_modif, conj_completedRiemannZeta₀, zeta_conj_of_critical_strip, zeta_re_conj_symm, zeta_im_conj_antisymm, zeta_re_even_on_line, zeta_im_odd_on_line, zeta_eq_zero_iff_re_im, riemannZeta_ne_zero_of_re_lt_zero, riemannZeta_ne_zero_of_re_eq_zero, nontrivial_zero_in_critical_strip, zeta_eq_odd_add_even, zeta_eq_zero_iff_p_split, p_split_mul_zero_iff, odd_part_extra_zero_on_imag_axis, zero_split_normSq, zero_not_on_euler_circles, affine_image_critical_line_is_line, on_line_iff_equidistant_base_one, critical_line_equidistant_basepoint_i, recip_basepoint_i_on_unit_circle, hardyZ_real, critical_line_positional, orbit_center_half, orbit_degenerate_iff_on_line, recip_on_critical_circle_iff, zero_orbit_four.
 9. **L9 (31):** chi_mul_chi_one_sub, conj_cpow_inv_half_of_unit, hardyZ_real, chi_unit_on_line, chi_ne_zero_on_line, chi_mem_slitPlane_on_line, zero_orbit_distinct_of_offline, zero_orbit_degenerate_on_line, zero_orbit_counting, zero_left_right_bijection, cpow_two_on_line_explicit, cpow_pi_on_line_explicit, sin_pi_quarter_add_mul_I, sin_pi_half_add_mul_I, gamma_abs_sq_on_line, term_on_line_explicit, cpow_two_half_minus_im, gamma_conj_arg_angle, gamma_doubling_arg_angle, gamma_reflection_arg_angle, gamma_quarter_phase_combine, zeta_neg_nat_ne_zero_of_odd, rh_iff_all_zeros_on_line_in_strips, offline_zero_gives_half_strip, zeta_two_arg_eq_arg_chi, chi_translation_two, chi_arg_translation_two, chi_translation_four, chi_arg_translation_four, zeta_eq_chi_mul_conj_on_line, zeta_unit_sq_eq_chi.
 10. **L10 (31):** zetaZeroSet, isClosed_zetaZeroSet, isDiscrete_zetaZeroSet, zeta_zeros_finite_in_strip, zetaUnit, continuousOn_zetaUnit, chi, continuousOn_chi_line, chi_ne_zero_on_line, chiPath, theta0, exp_theta0, thetaLift, thetaLift_lifts, thetaLift_zero, uPath, uLift0, exp_uLift0, uLift, uLift_lifts, uLift_zero, expKernel, expKernel_mem_iff, expKernel_dist_ge_two_pi, phase_align_two_zeta_lift_eq_chi_lift, zeta_unit_sq_eq_chi_short, zeta_lift_half_chi_lift_pi_int.
 11. **L11 (5):** polyError, poly_error_ratio, error_seq_predicts, poly_error_monotone, error_precision_bound.
 12. **L12 (14):** flip_chi_circle_bridge, net_flip_from_chi_lift, flip_count_from_theta_lifts, completedZeta₀_log_reflection, zeta_zero_iff_completedZeta₀_one_over_sum, zeta_line_two_bounded, Gamma_one_add_it_norm_le_one, zeta_ne_zero_on_bottom_edge, zeta_im_zero_on_real, zeta_neg_one_re_neg, bottom_edge_arg_jump, zeta_partial_sum_lt_integral_bound.
 13. **L13 (12, isolation file):** zetaEmR_N, zetaEmG_N, zetaEmG_N_eq_smul, zetaEmR_N_differentiableOn, zetaEmG_N_differentiableOn, zetaTermC, zetaTermC_fund_integral, zetaTermC_closed_form, zeta_cpow_ofReal_continuousAt (plus zeta_uIcc_ge, cpow_ofReal_hasDerivAt, cpow_ofReal_neg_hasDerivAt).
 14. **L14 (20, PreciseMigration.lean):** T, TInv (the exact migration operator and its inverse); T_shift_recip (compression + shift decomposition); T_inv_right, T_inv_left (invertible change of coordinates); T_normSq_ratio (distance ratio of the two base points); T_normSq_diff ($|z - 1 - i|^2 - |z - i|^2 = 1 - 2\Re z$); T_normSq_eq_one_iff, T_normSq_lt_one_iff, T_normSq_gt_one_iff (three-tier decision: line / left / right half-strip vs. modulus 1); T_reflect (the functional-equation reflection as the unit-circle involution $w \rightarrow 1/\overline{w}$); T_on_critical_line; center_phase, log_I_phase (phase centre $\log i = i\pi/2$, $e^{i\pi/2} = i$); i_pow_i ($i^i = e^{-\pi/2}$, the false-real-axis tick); i_pow_half_inv_two_i ($i^{1/(2i)} = e^{\pi/4}$); half_succ_position (half-point = $e^{\pi/4}/2$); pair_inverse, pair_square, imag_scale_mul (paired positions and the scale product law).

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Author’s proposals, calibrations, and AI contribution statement

In compliance with the Annals of Mathematics policy on AI and LLM tools, we record (i) the ideas and directions proposed by the author, each with its location in this paper; (ii) the author’s calibrations of the development; (iii) the ideas contributed by AI tools; and (iv) the ideas contributed by AI that the author rejected or redirected. The author (a human) takes full responsibility for the content, its correctness, and the integrity and accuracy of its citations; AI agents are not authors.

Ideas and directions proposed by the author

1. *Critical-line circle and basepoint choice.* The author directed the study of how the critical-line circle arises in the Lean claims; the choice of basepoints for translation and inversion so that they cancel; the search for a single basepoint solving both critical-line circularization and translation–inversion cancellability; and the question whether intersection on the critical-line circle suffices, with the basepoint modified so that the odd and even parts converge to circles or straight lines meeting only on the critical-line circle (§3, §5, §4.2).
2. *Four-line splitting by $n/\log n$ and phase superposition.* The author proposed that the two odd/even lines be split into four lines by the scale $n/\log n$ to find four-line intersections, viewing the odd/even lines as phase superposition states and decomposing, from the 0pat viewpoint, how many phases are superposed in the two lines (§4.2, §6).
3. *Euler-circle compactification and the normalization question.* The author proposed that infinitely diverging frequencies be finitized by observation in the two orthogonal Euler circles (compactification into circles), and posed the equivalence of normalization with compactification into a circle (§3, §15).
4. *Symmetric continuation as difference, with alternating symmetry points.* The author proposed the symmetric continuation mechanism: alternating reflection about symmetry points (essentially a difference scheme), the non-closure of a single round of differences (the construction $1 \mapsto 3$ about 2, $3 \mapsto 0$ about 1.5, $0 \mapsto 4$ about 2, $4 \mapsto -1$ about 1.5, and so on, with a one-directional variant), the exchange of symmetry points with continuation points (each symmetry point is a continuation point of another), the consequence that any finite domain with sufficiently many elements extends to infinity by its own symmetry, and the symmetric continuation of arguments (numbers replaced by angles) (§15, §7, §10).
5. *Trigonometric cumulative continuation and the integer layers.* The author proposed that the integer layers of the argument (“mod π , drop π ; the count is the lifted integer”) be obtained by taking trigonometric functions and continuing by cumulative values — a Fourier phase calibration — and observed the equivalence of the count with the lifted integer, and of the lifted integer with the Riemann hypothesis itself (§7, §9).
6. *Phase modulation and phase alignment as the common mechanism.* The author proposed that the various bridges are all phase modulation and phase alignment (square-wave type), that the continuous argument is essentially phase, that continuous functions are projections of infinitely many phase-discrete components, and that a circle’s diameter can serve as an axis on which infinitely discrete arc projections become continuous segments (§6, §7, §15).

7. *Finite-infinite and period-divergence conversions as $e^{i\pi}$ transforms.* The author proposed that the finite-to-infinite and infinite-to-finite conversions, and the period-divergence conversions, are all $e^{i\pi}$ transforms tied to the basepoint position and to phase modulation, with fast paths; the decomposition $e^{i\pi} + 1 = 0$ (i : orthogonal symmetry direction / dimensional lift; π : divergence-period conversion; e : finite-infinite conversion); the role of the transcendence of π in constructing the infinite of the real domain; and the recasting of framework-dependent arguments as observation in the two orthogonal Euler circles, i.e. a geometric representation of the same structure (left formally undefined) (§1.3, §15).
8. *Polyphase numbers and the necessity of single-phase numbers.* The author identified that the three bridges (flip, Stirling phase, growth) exhibit natural numbers as polyphase counts — the winding numbers of the covering map $t \mapsto e^{it}$ ($\pi_1(S^1) \cong \mathbb{Z}$) — and that the precise definition of a number requires phase locking to obtain a single-phase number, which motivates 0pat and its application to the Riemann direction (§1.3, §15).
9. *The projection value bridge in its weak form.* The author specified the strength of the remaining bridge exactly: no full value bridge is sought, only the weak assertion that, after the projection drops axis 1 (the position along the critical line), the remaining information is *complete* for the judgement “the zero is on the critical line” — no counting, winding, or flip correlation (§14.8, §17).
10. *The native four-axis geometry and the base-point criterion.* The author established that the axes are the two orthogonal axes obtained by rotating the two orthogonal real axes by 45° : the negative side of the fold is the 90° human-defined imaginary axis (machine-verified as the diagonal $u + v = 0$), and the positive side carries the human-defined irrationality $\sqrt{2}$. The author initially identified these axes with the framework’s *four log axes*; this identification was a cognitive error — the axes carry no log-2 unit, and the $2\pi/\ln 2$ lattice of §14.8 is the factor’s single-prime structure, separate from the axes — and it was withdrawn during the development (see the AI-contribution statement). The base-point criterion that stands: any construction using i or $\pi/2$ -type transforms is a self-referential defective definition (the human coordinate presentation absorbs the imaginary information into the base-point i), so the equivalence of the four axes with the complex plane must be established natively — from the constant $\sqrt{2}/2 = 2^{-1/2}$ and real algebra alone — with i allowed only as the human-coordinate read-back layer (§14.8).
11. *Deterministic prediction of the counting remainder.* The author proposed that the growth control $\log T$ is a deterministic prediction of the counting remainder, already solved by the pat residual method and requiring only 0pat formalization, with $\log$ as e ’s finite-to-infinite; and that a mapping phase relation in phase modulation can replace iterative approximation (the residual has a definite lower bound and stable direction), since iterative approximation is itself finite-reaching-infinite (§8).
12. *Sign flips as the symmetry process; reversal docking as the period-divergence bridge.* The author proposed that oscillation is the sign-flip process of the framework (symmetry itself: a sign change with the position fixed, realized as iteration of i), that reversal docking is the period-divergence bridge, together with the continuous-discrete inverse bridge, and that the expansion of the sign-change mechanism joins the existing fast paths (§9.1, §1.3, §7).
13. *Essence of pat: cancellation of phases.* The author proposed that pat defines a single-phase number, that all numbers are phase-ambiguous multi-phase superpositions, that the method must cancel the phases by cancellation operations, and posed the question whether a symmetry group can be found that, after completion, exactly cancels the phases and leaves the theorem (§1.3, §15).
14. *From known mathematics only.* The author directed that the Riemann direction be proved from known mathematics only, without invoking unverified results of the pat framework, after the framework itself had been published at DOI 10.5281/zenodo.22038154 (§1.3).
15. *The continuum by phase superposition.* The author proposed that the continuum arises by phase superposition: countably many phase layers (each grid point carrying the $\aleph_0$ representatives $\theta + 2\pi n$, “ $\aleph_0$ layers of $\aleph_0$ points”) close to the continuum $[0, 2\pi]$, the strict form of which is the machine-verified continuum closure theorem of the framework basepaper [2] (R150–R151); the author further identified

the superposition with the sequence space $\mathbb{N} \rightarrow \mathbb{N}$, so that the count is $\aleph_0^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c} = |\mathbb{R}|$ (machine iso 28), with the calibration that the union of layers is $\aleph_0 \cdot \aleph_0 = \aleph_0$ (absorption) while the superposition itself is the power $\aleph_0^{\aleph_0}$, that each position carrying $\aleph_1$ projections still gives $\aleph_1^{\aleph_0} = \mathfrak{c}$, that $\aleph_1$ positions give $2^{\aleph_1} \geq \aleph_2$ (GCH independent), and that no intermediate-cardinality claim is made (§15).

16. *The Euler-circle fold: the imaginary axis as a 90° fold.* The author proposed that the imaginary axis be constructed, from the negative side of the Euler circle, as two orthogonal 45° rays — a 90° fold whose endpoints are the imaginary units — so that the imaginary axis is the fold (§14.5, Theorem 14.1), together with its straightening (the fold restores to $\mathbb{R}$ by a homeomorphism with arc-length preservation and unit speed) and the fold-coordinate bridges (§14.6). The author directed that the fold be assembled from net elements only, and that its equivalence with the imaginary axis be proved before any use (§14.5); the construction is a presentation-layer statement, zero new axioms, machine-verified.
17. *Net-element methodology: induction and the root net elements.* The author directed the empirical induction of the net-element methodology — the 761-compilation-log / 2042-error record distilled into the degree-lowering net elements of §11.1 — and the archaeological induction of the root net elements: the historical contradiction proofs (Euclid, Cantor, Sturm, Cauchy, Perron) and the framework’s N1–N7 family induced into four roots *R1–R4* (local-to-global lifting, invariant conservation, algebraic degree reduction, symmetric reflection), with their L1 combinations and L2 pipelines (§11.1, §13.2). The induction is over known mathematics only, and is claimed as such.

A closing remark. In the author’s view, Pat, extended to unknown domains, will necessarily present logical incompleteness. Regarding the presentation framework, this paper claims only the necessity of a new way of presenting, and the reasons why Pat is, for now, a suitable one. Should a reader find the presentation here presumptuous, or mistaken in this or that detail — the author would be profoundly grateful, and would sincerely hope that Pat is torn down completely and rebuilt into a better presentation framework. This is only an attempt — the author’s attempt to see a little more clearly.

Author’s calibrations of the development

1. *No wheel-reinvention: bridge and cancel.* Asymptotic and phase arguments must be built from symmetry cancellation and exponential bridges, not from hand-derived classical expansions (§1.3, §6, §10.2).
2. *Use known curves* (§9.2).
3. *The false real axis is proved, not KNOWN* (§5, Theorem 9.4).
4. *Epistemic calibration: Lean no sorry* (§1.5).
5. *Search the library first* (mathlib `ZetaAsymptotics`; §1.6, §10.2).
6. *Symmetric continuation* of the $s > 1$ identity to $(0, 1)$ (§10.4).
7. *Polyphase numbers* (§1.3, §15).
8. *Cardinality claim calibrated.* The union of phase layers is $\aleph_0 \cdot \aleph_0 = \aleph_0$ (absorption), while the superposition (one projection per position) is the power $\aleph_0^{\aleph_0} = \mathfrak{c}$; $\kappa^{\aleph_0} = \mathfrak{c}$ for every $2 \leq \kappa \leq \mathfrak{c}$ (so per-position choice space is irrelevant); $\aleph_1^{\aleph_0} = \mathfrak{c}$ and $\aleph_1^{\aleph_1} = 2^{\aleph_1} \geq \aleph_2$; CH/GCH independence KNOWN ([7]); no intermediate-cardinality claim (§15).
9. *Nature of the $(0, 1)$ -gap* (infrastructure, not mathematics; §10.4).
10. *Continuity covered by the odd/even splitting* (§10.3, §10).
11. *Writing discipline* (complete-logic organization, symbol discipline, precise intuition, legitimate presentation; throughout).

12. *The net-element family is an induction from known mathematics, and nothing more.* The order family of §11.1 contains no new mathematics: it is an induction over known mathematics, and the induction method itself is also known mathematics. The author’s role was personal generalization experience: after reproducing perfect generalization on a transformer, the elements were derived — there is no new theoretical discovery.

Ideas contributed by AI tools (each with its location)

1. The reduction of the $(0, 1)$ sign to the Euler–Maclaurin kernel with the identity theorem on the convex strip (technical assembly of the author’s symmetric-continuation direction) (§10, §10.4).
2. The complex closed form of the Euler–Maclaurin term via integration by parts and power-law assembly (§10.2).
3. The use of mathlib’s `ZetaAsymptotics` identity `zeta_limit_aux1` as the continuation seed (§1.6, §10.2).
4. The phase-lift construction via covering maps (`thetaLift`, `uLift`, `expKernel`) (§7).
5. The Weierstrass/identity-theorem toolkit choices (`differentiableOn_tsum_of_summable_norm` and `eqOn_zero_of_preconnected_of_frequently_eq_zero`) (§10.4, §16).
6. The mechanical Lean 4 code for all 284 declarations and the typesetting of this manuscript (§16, Appendix A).
7. The Lean implementation of the author’s native four-axis and weak-assertion directives of the final revision: the native equivalences of the fold frame (`NativeAxisEquiv.lean`, 11 declarations: recovery, uniqueness, modulus preservation, surjectivity with the human-coordinate read-back exemption), the value-domain anchors $U + V = \sqrt{2}\Re\zeta$ and $U - V = \sqrt{2}\Im\zeta$, and the weak assertion `projection_zero_line_complete` (§14.8).
8. The AI corrected the author’s identification of the four (45°) axes with the framework’s log axes: a cognitive error. The axes carry no $\log\text{-}2$ unit; the $2\pi/\ln 2$ lattice of §14.8 is the factor’s single-prime structure, separate from the axes. The correction renamed the object the *four axes*, removed the log-scale assertions from the abstract, the observation frame, the verification record, and the author’s proposal item, and recorded the withdrawal (§14.8).

Ideas contributed by AI that were rejected or redirected by the author

1. *Euler–Maclaurin continuation as the proof of continuity on $(0, 1)$.* The AI proposed to establish the $(0, 1)$ -segment by a full Euler–Maclaurin continuation argument, including the continuity of ζ on the interval. The author rejected this: the continuity is already covered by the odd/even pairing bridge of §10.3 (both sides are locally uniform limits of discrete series — continuity by discrete superposition), and the sign is likewise covered by the positivity of the pairing terms together with the doubling identity (§10.4), so the Euler–Maclaurin kernel is not needed at all.
2. *Hand-derived asymptotic expansions (Binet-type).* The AI proposed deriving Stirling-type asymptotics by classical expansions; the author rejected this as wheel-reinvention and redirected the argument to exponential bridges and known curves (§1.3, §10.2).
3. *Marking the bottom-edge real-valuedness as KNOWN.* The AI proposed citing the real-axis behavior of ζ as classical; the author corrected that the false-real-axis framework is already machine-verified and must be assembled, not cited (§5, Theorem 9.4).
4. *Difference-chain treatment of continuation.* The AI treated the symmetric continuation by difference chains; the author redirected: under `Op` at a difference chain is a symmetric operation, and the continuation must be built from symmetry (oscillation about symmetry points, exchange of symmetry and continuation points) (§15).

5. *Normalization.* The AI used normalization in the geometric constructions; the author rejected it and posed instead the equivalence of normalization with compactification into a circle (§3, §15).
6. *Iterative approximation of the residual.* The AI proposed iterative approximation for the residual; the author rejected it, requiring instead a mapping phase relation with a definite lower bound and stable direction (§8).
7. *Persistence of the asymptotic route after bridging.* After the author had bridged the Stirling phase, the AI still proposed an asymptotic route; the author rejected it and required a fast path quoting known curves (“we are already bridged, why still asymptotics”) (§1.3, §9.2).
8. *Paper scope limited to the four tasks.* The AI first framed the paper around the four tasks only; the author rejected this scope and required a complete review of the whole development (all 231 declarations in twelve layers at the time, plus the 12 declarations of Blocks A–B (`zeta_continuation_iso.lean`) and the 4 declarations of the odd/even pairing of this revision; the development now comprises 284 declarations in fourteen layers) (§1, Table 1, Appendix A).
9. *Continuing the $(0,1)$ -segment before publication.* The AI proposed continuing the machine verification of Blocks C–G immediately; the author stopped this to secure publication first, and later closed the gap with the odd/even pairing short path instead of the Euler–Maclaurin infrastructure (§10.4).
10. *Reviewing the fold frame as a presentation layer.* The AI reviewed the final-revision observation frame (the fold coordinates, the observation plane, the projection, the offline signature) as a presentation-layer extension of the equivalence chain and as evidence of circular wandering. The author corrected this: the fold frame is the main line of the tracking route (native four-axis geometry, odd/even split, observation plane with axis 1 dropped and axis 2 kept, offline-zero analysis), the weak assertion of §14.8 is its machine-verified completion in the author’s specified form, and the assessment was withdrawn (recorded as the fixed anchor of the node ledger, §14.8).

Division of labor

The author (human) proposed the ideas and directions listed in the first subsection, designed the proof architecture and the layer decomposition, calibrated the development as listed in the second subsection, and takes full responsibility for the content of this submission, including its correctness and the integrity and accuracy of its citations. AI tools (the ZCode environment and the DeepSeek API) performed the mechanical proof expansion, the writing of the Lean 4 code, and the typesetting of this manuscript; all theorems were verified by the Lean 4 compiler (0 errors, 0 `sorry`). AI agents are not authors.